DUALITY IN TENSOR-TRIANGULAR GEOMETRY
LECTURE NOTES

Overview. Tensor-triangular geometry provides a framework for studying classification problems across a range of areas. Originating in chromatic homotopy theory, its influence has since been seen in commutative algebra, geometry, and representation theory. In the course, we will see how one can prove powerful duality statements from a tensor-triangular perspective. We will focus on examples in commutative algebra, but the theory developed is rather broad, so other examples may be covered depending on the audience's interests.

Contact information. Jordan Williamson, `williamson@karlin.mff.cuni.cz`

Relevant literature: main references.

(1) P. Balmer, I. Dell'Ambrogio, and B. Sanders. Grothendieck-Neeman duality and the Wirthmüller isomorphism. *Compos. Math.*, 152(8):1740–1776, 2016
(2) H. Fausk, P. Hu, and J. P. May. Isomorphisms between left and right adjoints. *Theory Appl. Categ.*, 11:No. 4, 107–131, 2003
(3) T. Peirce, J. Williamson. Duality in tensor-triangular geometry via proxy-smallness. arXiv:2510.24415, 2025

Related literature: other helpful references.

(1) H. Krause. Localization theory for triangulated categories. In Triangulated categories, volume 375 of London Math. Soc. Lecture Note Ser., pages 161–235. Cambridge Univ. Press, Cambridge, 2010
(2) H. Krause. Homological theory of representations, volume 195 of Cambridge Studies in Advanced Mathematics. Cambridge University Press, Cambridge, 2022
(3) M. Hovey, J. H. Palmieri, and N. P. Strickland. Axiomatic stable homotopy theory. Mem. Amer. Math. Soc., 128(610):x+114, 1997
(4) C. Weibel. An introduction to homological algebra. Cambridge Studies in Advanced Mathematics. 38. Cambridge: Cambridge University Press. xiv, 450 p. (1994).
(5) A. Neeman. Triangulated categories. Annals of Mathematics Studies. 148. Princeton, NJ: Princeton University Press. vii, 449 p. (2001).

Assessment. The final exam will be an oral exam. For zápočet, students are expected to participate during the course.

1. Recollections on triangulated categories

1.A. **Axioms.** Loosely speaking, a triangulated category consists of an additive category T, together with two extra pieces of data:

(1) an equivalence of categories $\Sigma\colon \mathsf{T} \xrightarrow{\sim} \mathsf{T}$ called the *shift*;
(2) a collection of *triangles* $X \to Y \to Z \to \Sigma X$ satisfying various axioms which ensure good behaviour.

Definition 1.1. Let T be an additive category and $\Sigma\colon \mathsf{T} \xrightarrow{\sim} \mathsf{T}$ be an additive equivalence of categories. A *candidate triangle* is a diagram

$$X \xrightarrow{f} Y \xrightarrow{g} Z \xrightarrow{h} \Sigma X$$

such that the composites $g \circ f$, $h \circ g$, and $\Sigma f \circ h$ are all zero. A morphism of candidate triangles is a commutative diagram

$$\begin{tikzcd} X \arrow[r, "f"] \arrow[d, "u"] & Y \arrow[r, "g"] \arrow[d, "v"] & Z \arrow[r, "h"] \arrow[d, "w"] & \Sigma X \arrow[d, "\Sigma u"] \\ X' \arrow[r, "f'"'] & Y' \arrow[r, "g'"'] & Z' \arrow[r, "h'"'] & \Sigma X'. \end{tikzcd}$$

Considering a smaller class of candidate triangles which satisfy certain properties leads to the notion of a triangulated category. The first four axioms are easy to justify, but the final axiom is harder to motivate. It is convenient to develop the theory assuming only these first four axioms, and then add in the final one once it becomes relevant. Nonetheless, we'll give both definitions now, so that we can consider an example before embarking on the abstract theory.

Definition 1.2. A *pretriangulated category* T is an additive category together with an additive equivalence of categories $\Sigma\colon \mathsf{T} \xrightarrow{\sim} \mathsf{T}$, and a subclass of candidate triangles called *distinguished triangles* which satisfy the following axioms:

(TR0) Any candidate triangle which is isomorphic to a distinguished triangle is a distinguished triangle, and for all $X \in \mathsf{T}$ the candidate triangle

$$X \xrightarrow{1} X \to 0 \to \Sigma X$$

is distinguished.

(TR1) For all $f\colon X \to Y$ in T, there exists a distinguished triangle $X \xrightarrow{f} Y \to Z \to \Sigma X$.

(TR2) Let $X \xrightarrow{f} Y \xrightarrow{g} Z \xrightarrow{h} \Sigma X$ be a candidate triangle. This is distinguished if and only if the candidate triangle

$$Y \xrightarrow{g} Z \xrightarrow{h} \Sigma X \xrightarrow{-\Sigma f} \Sigma Y$$

is distinguished.

(TR3) For any commutative diagram

$$\begin{tikzcd} X \arrow[r, "f"] \arrow[d, "u"] & Y \arrow[r, "g"] \arrow[d, "v"] & Z \arrow[r, "h"] & \Sigma X \\ X' \arrow[r, "f'"'] & Y' \arrow[r, "g'"'] & Z' \arrow[r, "h'"'] & \Sigma X'. \end{tikzcd}$$

in which the rows are triangles, there exists a map $w\colon Z \to Z'$ (which need *not* be unique) making

$$\begin{tikzcd} X \arrow[r, "f"] \arrow[d, "u"] & Y \arrow[r, "g"] \arrow[d, "v"] & Z \arrow[r, "h"] \arrow[d, "w"] & \Sigma X \arrow[d, "\Sigma u"] \\ X' \arrow[r, "f'"'] & Y' \arrow[r, "g'"'] & Z' \arrow[r, "h'"'] & \Sigma X'. \end{tikzcd}$$

commute.

Remark 1.3. It is standard to drop the adjective 'distinguished', and just refer to them as *triangles*. For candidate triangles, we will never drop the adjective.

Remark 1.4. By combining (TR2) with (TR3), one sees that there always exists fillers in the first and second column too.

Definition 1.5. A *triangulated category* T is a pretriangulated category satisfying the following additional axiom:

(TR4) Suppose that $X \xrightarrow{f} Y \xrightarrow{g} Z' \xrightarrow{h} \Sigma X$, $Y \xrightarrow{u} Z \xrightarrow{v} X' \xrightarrow{w} \Sigma Y$ and $X \xrightarrow{uf} Z \xrightarrow{a} Y' \xrightarrow{b} \Sigma X$ are distinguished triangles. Then there exists a distinguished triangle

$$Z' \xrightarrow{\alpha} Y' \xrightarrow{\beta} X' \xrightarrow{\gamma} \Sigma Z'$$

such that $v = \beta a$, $h = b\alpha$, $\gamma = \Sigma g \circ w$, $w\beta = \Sigma f \circ b$ and $\alpha g = au$.

Pictorially this axiom can be represented by the following commuting diagram:

$$\begin{tikzcd} & Y' \arrow[dr, dashed, "b"] & & \\ Z' \arrow[ur, "\alpha"] \arrow[d, dashed, "h"'] & & X' \arrow[ll, dashed, "\Sigma g \circ w"] \arrow[ul, "\beta"'] \arrow[dl, dashed] & \\ X \arrow[rr, "uf"] \arrow[dr, "f"'] & & Z \arrow[ul, "a"'] \arrow[u, "v"'] & \\ & Y \arrow[ul, "g"] \arrow[ur, "u"'] \arrow[uur, "w"'] & & \end{tikzcd}$$

The dotted maps are of degree 1 (i.e., $f\colon X \dashrightarrow Y$ represents a map $f\colon X \to \Sigma Y$), and composites of the form $\to\to\dashrightarrow$ are triangles. The red maps are the extra data, together with the condition that they form a triangle. In order to remember this, note that the primed letters are the cones of maps, and that every triangle contains an X, Y, and a Z (primed or otherwise). In light of the shape of the above diagram, (TR4) is often referred to as the *octahedral axiom*.

Alternatively, one can give the following pictorial representation.

$$\begin{tikzcd}
X \arrow[r, "f"] \arrow[d, "1"] & Y \arrow[r, "g"] \arrow[d, "u"] & Z' \arrow[r, "h"] \arrow[d, dashed, "\alpha"] & \Sigma X \arrow[d, "1"] \\
X \arrow[r, "uf"] \arrow[d, "f"] & Z \arrow[r, "a"] \arrow[d, "1"] & Y' \arrow[r, "b"] \arrow[d, dashed, "\beta"] & \Sigma X \arrow[d, "\Sigma f"] \\
Y \arrow[r, "u"] \arrow[d, "g"] & Z \arrow[r, "v"] \arrow[d, "a"] & X' \arrow[r, "w"] \arrow[d, "1"] & \Sigma Y \arrow[d, "\Sigma g"] \\
Z' \arrow[r, dashed, "\alpha"] & Y' \arrow[r, dashed, "\beta"] & X' \arrow[r, dashed, "\gamma"] & \Sigma Z'
\end{tikzcd}$$

The first three rows are the given triangles, and (TR4) then asserts the existence of the dotted arrows making the diagram commute, so that the bottom row is also a triangle.

Let's briefly discuss the axioms and provide some motivation for them. If one thinks as triangles as a generalisation of short exact sequences, then in a triangle $X \xrightarrow{f} Y \xrightarrow{g} Z \xrightarrow{h} \Sigma X$ you should think of Z as the (homotopy coherent) cokernel of f, and X as the (homotopy coherent) kernel of g. The axioms then mean the following.

(TR0) The kernel and cokernel of the identity is zero.
(TR1) Every map has a kernel and cokernel.
(TR2) Up to sign, every map is the kernel of its cokernel and vice versa.
(TR3) Kernels and cokernels are almost functorial.
(TR4) One can interpret the given triangles as saying $Z' \simeq Y/X$, $X' \simeq Z/Y$ and $Y' \simeq Z/X$, and then the axiom asserts that $X' \simeq Y'/Z'$, i.e., $(Z/X)/(Y/X) \simeq Z/Y$.

Remark 1.6. Given a triangle $X \xrightarrow{f} Y \xrightarrow{g} Z \to \Sigma X$, it is common to call Z the *cofibre* (or *cone*) of f, and X the *fibre* (or *cocone*) of g. Sometimes it is customary to write triangles as $X \to Y \to Z$ and drop the map to the shift. We will sometimes subscribe to this later on the course for brevity, but we warn the reader that it is important not to forget this map. For example, a morphism of triangles requires the square including the map $Z \to \Sigma X$ to commute.

Let us recall some key features:

(1) Products and coproducts of triangles are again triangles (provided the required (co)products exist).
(2) If $X \to Y \to Z \to \Sigma X$ is a triangle, then $Z \simeq 0$ if and only if $X \to Y$ is an isomorphism.
(3) Given a commutative diagram of pretriangles

$$\begin{tikzcd}
X \arrow[r] \arrow[d, "f"] & Y \arrow[r] \arrow[d, "g"] & Z \arrow[r] \arrow[d, "h"] & \Sigma X \arrow[d, "\Sigma f"] \\
X' \arrow[r] & Y' \arrow[r] & Z' \arrow[r] & \Sigma X'
\end{tikzcd}$$

if f and g are isomorphisms then h is also an isomorphism. Note that this gives a good way to show that a pretriangle is infact a triangle, since triangles are closed under isomorphism.

1.B. **Examples.** We briefly describe three examples of triangulated categories.

1.B.1. *The homotopy category of a ring.* Fix a ring R. Recall that a map $f\colon M \to N$ of chain complexes of (left) R-modules is null homotopic if it is chain homotopic to the zero map. We write $\mathrm{Null}(M, N)$ for the subgroup of $\mathrm{Hom}_{\mathrm{Ch}(R)}(M, N)$ consisting of the null homotopic maps.

The *homotopy category* $\mathsf{K}(R)$ is defined by having objects the chain complexes of R-modules, and morphisms given by the homotopy classes of chain maps, i.e.,

$$\mathrm{Hom}_{\mathsf{K}(R)}(M,N) = \mathrm{Hom}_{\mathrm{Ch}(R)}(M,N)/\mathrm{Null}(M,N).$$

Equivalently, the morphisms are the chain maps up to the equivalence relation of chain homotopy equivalence. The distinguished triangles in the homotopy category are the triangles which are isomorphic in $\mathsf{K}(R)$ (i.e., chain homotopic) to those of the form

$$M \xrightarrow{f} N \to C(f) \to \Sigma M$$

for some map of chain complexes $f\colon M \to N$.

1.B.2. *The derived category of a ring.* Homological algebra is built upon taking resolutions of modules. Therefore, one seeks a category which contains precisely the homological information of modules, so that objects are resolutions, and a module is isomorphic to any resolution of it. As such, we want a category which contains all the resolutions of modules, and in which quasi-isomorphisms are isomorphisms. Since injective resolutions and projective resolutions point in opposite directions, we consider all chain complexes (i.e., rather than just those bounded above or below 0). The *derived category* of a ring R is the universal category in which quasi-isomorphisms of complexes are inverted. We denote this category by $\mathsf{D}(R) := \mathsf{K}(R)[\text{quasi isos}^{-1}]$. We now give a precise construction of this.

Remark 1.7. This construction of the derived category leads to some set-theoretic discussions, namely, why are the hom sets actually sets? For the purposes of this course we ignore this, and just remark that one can give alternative constructions bypassing this issue.

The objects of $\mathsf{D}(R)$ are the same as the objects of $\mathsf{K}(R)$, that is, they are the chain complexes of R-modules. The morphisms in $\mathsf{D}(R)$ are equivalence classes of rooves, defined as follows. Let $M, N \in \mathsf{D}(R)$. A *roof* from M to N is a pair of chain maps

$$\begin{tikzcd} & Z \arrow[dl, "s", "\simeq"'] \arrow[dr, "f"] & \\ M & & N \end{tikzcd}$$

where $Z \in \mathsf{D}(R)$ and s is a quasi-isomorphism. Two rooves $(M \leftarrow Z \to N)$ and $(M \leftarrow Z' \to N)$ are equivalent if there exists another roof $(M \leftarrow W \to N)$ and maps $W \to Z$ and $W \to Z'$ such that the diagram

$$\begin{tikzcd} & Z \arrow[dl] \arrow[dr] & \\ M & W \arrow[l] \arrow[r] \arrow[u] \arrow[d] & N \\ & Z' \arrow[ul] \arrow[ur] & \end{tikzcd}$$

commutes. The hom sets of $\mathsf{D}(R)$ are rooves up to this equivalence relation.

Remark 1.8. Defining composites requires a little work: given two rooves $(L \xleftarrow{s} Z \xrightarrow{f} M)$ and $(M \xleftarrow{s'} Z' \xrightarrow{g} N)$, their composite gf is defined as follows. We have a triangle

$$Z' \xrightarrow{s'} M \xrightarrow{\alpha} C$$

and another triangle

$$F \xrightarrow{\beta} Z \xrightarrow{\alpha f} C.$$

By the existence of fillers we have a diagram

$$\begin{tikzcd} F \arrow[r, "\beta"] \arrow[d, dashed, "\gamma'"] & Z \arrow[r, "\alpha f"] \arrow[d, "f"] & C \arrow[r] \arrow[d, "\mathrm{id}"] & \Sigma F \arrow[d, dashed, "\Sigma\beta"] \\ Z' \arrow[r, "s'"'] & M \arrow[r, "\alpha"] & C \arrow[r] & \Sigma Z' \end{tikzcd}$$

Since s' is a quasi-isomorphism, C is acyclic so β is also a quasi-isomorphism. The composite is then given by the roof

$$L \xleftarrow{s\beta} F \xrightarrow{g\gamma} N.$$

A tedious check shows everything is well-defined up to equivalence of rooves.

We end by recalling some key features of the derived category:

(1) If $0 \to L \xrightarrow{f} M \xrightarrow{g} N \to 0$ is short exact sequence of R-modules, then there exists a chain map $N \to \Sigma L$ such that $L \xrightarrow{f} M \xrightarrow{g} N \to \Sigma L$ is a triangle in $\mathsf{D}(R)$.
(2) In fact, one can generalise the previous to see that short exact sequences of complexes also give triangles in the derived category.
(3) Let M and N be R-modules. Then
$$\mathrm{Hom}_{\mathsf{D}(R)}(M, \Sigma^i N) = \mathrm{Ext}^i_R(M, N).$$
(4) For any $M \in \mathsf{D}(R)$ we have $\mathrm{Hom}_{\mathsf{D}(R)}(\Sigma^i R, M) = H_i M$.

1.B.3. *The stable module category.* Let G be a finite group and let k be a field of characteristic dividing the order of G. In this setting, Maschke's theorem fails and we are in the realm of *modular* representation theory. A key category of interest in this area is the *stable module category* $\mathsf{StMod}(kG)$.

Lemma 1.9. *A kG-module M is projective if and only if it is injective.*

Proof. We first show that kG is self-injective. Define a k-linear map $\pi\colon kG \to k$ by $\pi(g) = 1$ if $g = 1$ and $\pi(g) = 0$ otherwise. One can check that the map $kG \to \mathrm{Hom}_k(kG, k)$ defined by $g \mapsto [h \mapsto \pi(g^{-1}h)]$ is an isomorphism of kG-modules. Now

$$\mathrm{Hom}_{kG}(-, kG) = \mathrm{Hom}_{kG}(-, \mathrm{Hom}_k(kG, k)) = \mathrm{Hom}_k(\mathrm{res}^{kG}_k(-), k)$$

which is exact, hence kG is an injective kG-module.

Since kG is Noetherian, direct sums of injective modules are injective. Therefore free modules are injectives, and hence summands of free modules (i.e., projectives) are also injective. Conversely, suppose that M is injective. The unit of the induction-restriction adjunction $M \to kG \otimes_k \mathrm{res}^{kG}_k M$ is injective and hence splits. Now $kG \otimes_k \mathrm{res}^{kG}_k M$ is a free module, so M is projective. □

The objects of $\mathsf{StMod}(kG)$ are the kG-modules, and the morphisms are given by the kG-module maps modulo those which factor through some projective:

$$\mathrm{Hom}_{\mathsf{StMod}(kG)}(M, N) = \frac{\mathrm{Hom}_{kG}(M, N)}{P\mathrm{Hom}_{kG}(M, N)}$$

where $P\mathrm{Hom}_{kG}(M, N)$ denotes the linear subspace of homomorphisms which factor through some projective kG-module.

The shift functor is given by taking cosyzygies: given X we define ΣX by taking an exact sequence

$$0 \to X \to E \to \Sigma X \to 0$$

where E is injective. To show that this is an autoequivalence, one constructs Σ^{-1} by taking syzygies; here is where Lemma 1.9 comes in. Any short exact sequence $0 \to X \to Y \to Z \to 0$ fits into a commutative diagram with exact rows

$$\begin{tikzcd} 0 \arrow[r] & X \arrow[r, "\alpha"] \arrow[d] & Y \arrow[r, "\beta"] \arrow[d] & Z \arrow[r] \arrow[d, "\gamma"] & 0 \\ 0 \arrow[r] & X \arrow[r] & E \arrow[r] & \Sigma X \arrow[r] & 0 \end{tikzcd}$$

where E is injective. The triangles are those isomorphic to ones of the form

$$X \xrightarrow{\alpha} Y \xrightarrow{\beta} Z \xrightarrow{\gamma} \Sigma X$$

as in the above diagram.

We end by recalling some key features of the stable module category:

(1) We have $M \cong N$ in $\mathsf{StMod}(kG)$ if and only if there exists projectives P and Q such that $M \oplus P \cong N \oplus Q$ in $\mathsf{Mod}(kG)$.
(2) The homomorphisms in the stable module category compute Ext-groups: if $i \geq 0$ then
$$\mathrm{Hom}_{\mathsf{StMod}(kG)}(M, \Sigma^i N) = \mathrm{Ext}^i_{kG}(M, N).$$
In particular, taking $M = k$ with trivial G-action, it computes group cohomology: $H^i(G; N) = \mathrm{Hom}_{\mathsf{StMod}(kG)}(k, \Sigma^i N)$. (In fact, allowing negative i too, one obtains Tate cohomology groups.) Note that here unlike in the derived category, we are representing the elements of Ext by actual module maps, rather than by using complexes.

1.B.4. *Some other examples.*

- The derived category $\mathsf{D}_{\mathrm{qc}}(X)$ of complexes of $\mathcal{O}_X$-modules with quasi-coherent cohomology for a nice enough scheme X.
- The (equivariant) stable homotopy category of spectra (representing objects for (equivariant) cohomology theories for topological spaces).
- The Kasparov category of separable G-C^*-algebras.

2. Tensor-triangulated categories

Recall that a symmetric monoidal structure on a category $\mathcal{C}$ is the data of a bifunctor

$$- \otimes - \colon \mathcal{C} \times \mathcal{C} \to \mathcal{C}$$

called the *tensor product* (or *monoidal product*) together with an unit object $\mathbb{1} \in \mathcal{C}$ satisfying:

- unitality: $\mathbb{1} \otimes X \simeq X$ for all $X \in \mathcal{C}$;
- associativity: $(X \otimes Y) \otimes Z \simeq X \otimes (Y \otimes Z)$ for all $X, Y, Z \in \mathcal{C}$;
- symmetry: $X \otimes Y \simeq Y \otimes X$ for all $X, Y \in \mathcal{C}$;

all satisfying various coherences that we won't make precise here. Such a monoidal structure is moreover called *closed*, if there is a bifunctor $\underline{\mathrm{Hom}}(-,-) \colon \mathcal{C}^{\mathrm{op}} \times \mathcal{C} \to \mathcal{C}$ called the *internal hom* such that $- \otimes X$ is left adjoint to $\underline{\mathrm{Hom}}(X, -)$. Again, this is all subject to various coherences that we will not make explicit. Note that $\underline{\mathrm{Hom}}(\mathbb{1}, X) \simeq X$ for all $X \in \mathcal{C}$.

In order to blend this data with that of a triangulated category we need one more definition:

Definition 2.1. Let T and U be triangulated categories. A *triangulated functor* is an additive functor $F \colon \mathsf{T} \to \mathsf{U}$ together with a natural isomorphism $\varphi \colon F\Sigma \xrightarrow{\sim} \Sigma F$ such that for any triangle

$X \xrightarrow{f} Y \xrightarrow{g} Z \xrightarrow{h} \Sigma X$ in T, the candidate triangle

$$FX \xrightarrow{F(f)} FY \xrightarrow{F(g)} FZ \xrightarrow{\varphi_X \circ F(h)} \Sigma FX$$

is a triangle in U.

With this, we can make the key definition:

Definition 2.2. A *tensor-triangulated category* T (tt-category for short) is a triangulated category which also has a closed symmetric monoidal structure which is compatible in the following sense: the tensor product is a triangulated functor in both variables, and the internal hom is a triangulated functor in both variables.

Remark 2.3. In fact, we only require that $\underline{\mathrm{Hom}}(-, X)$ is triangulated up to sign. We ignore this distinction, since for most purposes the signs are irrelevant.

Definition 2.4. A functor $F\colon \mathsf{T} \to \mathsf{U}$ between tt-categories is a *tt-functor* if it is triangulated and strong monoidal. (Recall that strong monoidality means that there are natural isomorphisms $F(X \otimes Y) \simeq FX \otimes FY$ and $F(\mathbb{1}_{\mathsf{T}}) \simeq \mathbb{1}_{\mathsf{U}}$ satisfying various coherences.)

2.A. **Examples.** We record two key examples.

2.A.1. *The derived category of a commutative ring.* If R is a commutative ring then $\mathsf{D}(R)$ is a tensor-triangulated category. The tensor product is given by the derived tensor product $- \otimes_R^{\mathsf{L}} -$ and the internal hom is given by the derived hom $\mathsf{RHom}_R(-,-)$. These are defined as follows. Note that for an arbitrary $M \in \mathsf{D}(R)$, the functors $M \otimes_R -$ and $\mathrm{Hom}_R(M,-)$ do not preserve quasi-isomorphisms, so cannot induce a functor on the derived category. This is the case even if M is a complex of projective R-modules:

Example 2.5. Consider $R = \mathbb{Z}/4$ and the complex

$$M = \cdots \to \mathbb{Z}/4 \xrightarrow{\cdot 2} \mathbb{Z}/4 \xrightarrow{\cdot 2} \mathbb{Z}/4 \to \cdots .$$

The quasi-isomorphism $M \xrightarrow{\sim} 0$ is not preserved by $\mathrm{Hom}_{\mathbb{Z}/4}(M,-)$.

Instead, we use *dg-projective resolutions*: a dg-projective resolution for $M \in \mathsf{D}(R)$ consists of a quasi-isomorphism $P \xrightarrow{\sim} M$ where P has the property that $\mathrm{Hom}_R(P,-)$ preserves surjective quasi-isomorphisms. The derived tensor product is then defined by $M \otimes_R^{\mathsf{L}} - = P \otimes_R -$ where P is a dg-projective resolution of M. Likewise $\mathsf{RHom}_R(M,-) = \mathrm{Hom}_R(P,-)$. There is an analogous notion of dg-injective resolution used to define $\mathsf{RHom}_R(-,M)$.

2.A.2. *The stable module category.* Let G be a finite group and k be a field of characteristic dividing the order of G. Then the stable module category $\mathsf{StMod}(kG)$ is a tensor-triangulated category. The tensor product is given by $- \otimes_k -$ with the diagonal G-action, with monoidal unit k with the trivial action. To see that this is indeed well-defined on the stable category, let us first consider what happens on $\mathsf{Mod}(kG)$. Firstly note that $M \otimes_k N \simeq N \otimes_k M$; one easily checks that this is indeed kG-linear. One can make $\mathrm{Hom}_k(M,N)$ into a kG-module via $(g \cdot f)(m) = gf(g^{-1}m)$. In particular, one sees that $g \cdot f = f$ for all $g \in G$ if and only if $f(gm) = gf(m)$ for all g, m. Therefore,

$$\mathrm{Hom}_k(M,N)^G = \mathrm{Hom}_{kG}(M,N).$$

From this, one checks that one has the tensor-hom adjunction

$$\mathrm{Hom}_{kG}(L \otimes_k M, N) = \mathrm{Hom}_{kG}(L, \mathrm{Hom}_k(M,N)).$$

From this the following is easy:

Lemma 2.6. *For any kG-module M, if P is a projective kG-module, then $P \otimes_k M$ is again projective.*

Proof. We have
$$\mathrm{Hom}_{kG}(P \otimes_k M, -) \simeq \mathrm{Hom}_{kG}(P, \mathrm{Hom}_k(M, -))$$
by the tensor-hom adjunction. The latter is an exact functor, hence $P \otimes_k M$ is projective. □

Since the tensor product on Mod_{kG} preserves projectives, it descends to a tensor product on $\mathsf{StMod}(kG)$. In a similar way one sees that $\mathrm{Hom}_k(M, -)$ preserves injectives, and hence also passes to give a functor on $\mathsf{StMod}(kG)$. Therefore, $\mathsf{StMod}(kG)$ is indeed a tt-category as claimed.

2.B. **Smallness in tt-categories.** Let us first briefly return to the setting of a triangulated category T (i.e., not necessarily with a tensor product). We study certain types of 'small' objects in triangulated categories. It is often the case that the whole triangulated category is generated by small objects, and much of the theory of triangulated categories relies upon such assumptions.

Definition 2.7. Let T be a triangulated category which has coproducts. An object $X \in \mathsf{T}$ is said to be *compact* if the natural map
$$\bigoplus \mathrm{Hom}_{\mathsf{T}}(X, Y_i) \to \mathrm{Hom}_{\mathsf{T}}(X, \bigoplus Y_i)$$
is an equivalence for every set of objects $\{Y_i\}$. We write T^{c} for the full subcategory of T consisting of the compact objects.

Before we can give some examples of compact objects, and criteria for detecting them, we must introduce some terminology.

Definition 2.8. Let T be a triangulated category which has coproducts.

(1) A full subcategory $\mathcal{S}$ of T is *thick* if it is closed under retracts and is triangulated.
(2) A full subcategory $\mathcal{S}$ of T is *localizing* if it is thick, and closed under coproducts.

Given a set of objects $\mathcal{X}$ of T, we write $\mathrm{Thick}(\mathcal{X})$ (resp., $\mathrm{Loc}(\mathcal{X})$) for the smallest thick (resp., localizing) subcategory of T containing $\mathcal{X}$. Note that these are well defined since the intersection of thick/localizing subcategories is again thick/localizing.

Example 2.9. Let $X \in \mathsf{T}$. The full subcategory $\{Y \in \mathsf{T} \mid \mathrm{Hom}_{\mathsf{T}}(\Sigma^i X, Y) \simeq 0 \text{ for all } i \in \mathbb{Z}\}$ is thick. It is localizing if X is compact.

Definition 2.10. Let $\mathcal{X}$ be a set of objects of T. We say that $\mathcal{X}$ *generates* T if $\mathrm{Loc}(\mathcal{X}) = \mathsf{T}$. If each element of $\mathcal{X}$ is compact, then we say that $\mathcal{X}$ *compactly generates* T.

Let us give an alternative characterisation of compact generation:

Proposition 2.11. *Let T be a triangulated category with coproducts, and $\mathcal{S}$ be a set of compact objects of T. Then the following are equivalent:*

(1) $\mathcal{S}$ generates T;
(2) if $\mathrm{Hom}_{\mathsf{T}}(\Sigma^i S, X) = 0$ for all $S \in \mathcal{S}$ and $i \in \mathbb{Z}$, then $X \simeq 0$.

Proof. Left as Exercise A.1. □

If T is compactly generated, then we can characterise the compact objects in terms of building operations.

Proposition 2.12. *Let* T *be a triangulated category which is compactly generated by a set* G*. Then* $\mathsf{T}^{\mathrm{c}} = \mathrm{Thick}(\mathsf{G})$*.*

Proof. The implication that $X \in \mathsf{T}^{\mathrm{c}}$ implies that $X \in \mathrm{Thick}(\mathsf{G})$ requires some work so we omit it. For the converse, consider the set

$$\{Y \in \mathsf{T} \mid \bigoplus \mathrm{Hom}_{\mathsf{T}}(Y, Z_i) \xrightarrow{\sim} \mathrm{Hom}_{\mathsf{T}}(Y, \bigoplus Z_i) \text{ for all sets } \{Z_i\}\}.$$

This is a thick subcategory of T, and contains G. Hence it contains $\mathrm{Thick}(\mathsf{G})$ since this is the *smallest* thick subcategory of T containing G. □

Example 2.13. We have $\mathsf{D}(R)^{\mathrm{c}} = \mathsf{Perf}(R)$ where the right hand side denotes the full subcategory of *perfect complexes*, that is, those complexes which are quasi-isomorphic to a bounded complex of finitely generated projectives. We leave the proof of this as Exercise A.2.

Example 2.14. We have $\mathsf{StMod}(kG)^{\mathrm{c}} = \mathsf{stmod}(kG)$ where the right hand side denotes the full subcategory of finitely generated modules. If M is compact, then writing M as a direct limit $M = \varinjlim M_i$ of finitely generated modules, we have $\mathrm{Hom}(M, M) = \varinjlim \mathrm{Hom}(M, M_i)$. In particular, the identity on M factors through M_i so that M_i is a retract of a finitely generated and hence is finitely generated. The other inclusion is left as Exercise A.3.

If T is a tt-category, one might hope that T^{c} inherits these properties. Unfortunately, this need not always be the case: the tensor product of compacts need not be compact in general, and the unit need not be compact in general.

Example 2.15. Let $\mathsf{T} = \prod_i \mathsf{T}_i$ be the countable product of tt-categories T_i with tt-structure given pointwise. Then a tuple $(X_i) \in \mathsf{T}$ is compact if and only if for almost all i we have $X_i = 0$ and $X_j \in \mathsf{T}_j^c$ for all j. In particular, since $\mathbb{1}_{\mathsf{T}} = (\mathbb{1}_{\mathsf{T}_i})$ the unit is not compact.

There is an alternative subcategory of 'small' object which behaves better with the tensor structure: Recall that the counit of the tensor-hom adjunction is the evaluation map $\mathrm{ev}\colon X \otimes \underline{\mathrm{Hom}}(X, Y) \to Y$ and the unit is the coevaluation map $\mathrm{coev}\colon Y \to \underline{\mathrm{Hom}}(X, X \otimes Y)$.

Definition 2.16. Suppose that T is a tensor-triangulated category with coproducts.

(1) An object $X \in \mathsf{T}$ is said to be *rigid* if the natural map

$$\nu_{X,Y}\colon \underline{\mathrm{Hom}}(X, \mathbb{1}) \otimes Y \to \underline{\mathrm{Hom}}(X, Y)$$

is an equivalence for all $Y \in \mathsf{T}$. To spell it out, the natural map is the composite

$$\underline{\mathrm{Hom}}(X, \mathbb{1}) \otimes Y \xrightarrow{\mathrm{coev}} \underline{\mathrm{Hom}}(X, X \otimes \underline{\mathrm{Hom}}(X, \mathbb{1}) \otimes Y) \xrightarrow{\underline{\mathrm{Hom}}(X, \mathrm{ev})} \underline{\mathrm{Hom}}(X, Y).$$

(2) An object $X \in \mathsf{T}$ is said to be *internally compact* if the natural map

$$\bigoplus \underline{\mathrm{Hom}}(X, Y_i) \to \underline{\mathrm{Hom}}(X, \bigoplus Y_i)$$

is an equivalence for every set $\{Y_i\}$.

Example 2.17. For any tt-category T, the tensor unit $\mathbb{1}$ is both rigid and internally compact.

In order to study the properties of rigid objects it is helpful to consider an alternative definition (which we will show is in fact equivalent). We write $D\colon \mathsf{T}^{\mathrm{op}} \to \mathsf{T}$ for the functor $D = \underline{\mathrm{Hom}}(-, \mathbb{1})$. This is called the *functional dual* (or sometimes Spanier-Whitehead dual).

Definition 2.18. Let T be a tensor-triangulated category, and $X \in \mathsf{T}$. The object X is *dualizable* if and only if there exists a map $\eta\colon \mathbb{1} \to X \otimes DX$ such that the diagram

$$\begin{tikzcd} \mathbb{1} \arrow[r, "\eta"] \arrow[dr, "\mathrm{coev}"'] & X \otimes DX \arrow[d, "\nu_{X,X}"] \\ & \underline{\mathrm{Hom}}(X, X) \end{tikzcd}$$

commutes.

Firstly, note that there is a natural map $\rho\colon X \to D^2X$ given by the composite

$$X \xrightarrow{\mathrm{coev}} \underline{\mathrm{Hom}}(DX, X \otimes DX) \xrightarrow{\underline{\mathrm{Hom}}(DX,\mathrm{ev})} \underline{\mathrm{Hom}}(DX, \mathbb{1})$$

for all $X \in \mathsf{T}$. Many of the proofs of the following statements follow by tedious diagram chases, so we provide only an outline of the proofs.

Lemma 2.19. *Let T be a tensor-triangulated category and $X \in \mathsf{T}$. If X is dualizable then DX is dualizable.*

Proof. We define a map $\mathbb{1} \to DX \otimes D^2X$ via the composite

$$\mathbb{1} \xrightarrow{\eta} X \otimes DX \xrightarrow{\rho \otimes DX} D^2X \otimes DX.$$

It remains to check that the required diagram commutes. That is, we want to check that the outer square in the diagram

$$\begin{tikzcd} \mathbb{1} \arrow[rr, "\eta"] \arrow[dr, "\mathrm{coev}"] \arrow[dd, "\mathrm{coev}"'] & & X \otimes DX \arrow[dl, "\nu_{X,X}"'] \arrow[dd, "\rho \otimes DX"] \\ & \underline{\mathrm{Hom}}(X, X) \arrow[dl, "\alpha"'] & \\ \underline{\mathrm{Hom}}(DX, DX) & & D^2X \otimes DX \arrow[ll, "\nu_{DX,DX}"] \end{tikzcd}$$

commutes, where α is defined to be the composite

$$\underline{\mathrm{Hom}}(X, X) \xrightarrow{\underline{\mathrm{Hom}}(X,\rho)} \underline{\mathrm{Hom}}(X, D^2X) \simeq \underline{\mathrm{Hom}}(X \otimes DX, \mathbb{1}) \simeq \underline{\mathrm{Hom}}(DX, DX).$$

The top triangle commutes since X is dualizable, so it suffices to prove that the other subdiagrams commute. This can be checked by unravelling the definitions of α and ρ; we leave this as a (tedious) exercise for the reader. □

Lemma 2.20. *Let T be a tensor-triangulated category, and suppose that $X \in \mathsf{T}$ is dualizable. Then the natural map $\rho\colon X \to D^2X$ is an equivalence.*

Proof. We claim that the composite

$$D^2X \simeq \mathbb{1} \otimes D^2X \xrightarrow{\eta \otimes D^2X} X \otimes DX \otimes D^2X \xrightarrow{X \otimes \mathrm{ev}} X$$

is an inverse to ρ. This can be checked by diagram chasing. □

Proposition 2.21. *Let T be a tensor-triangulated category, and $X \in \mathsf{T}$. Then X is rigid if and only if it is dualizable.*

Proof. The forward direction is clear since we may take $\eta = \nu_{X,X}^{-1} \circ \mathrm{coev}$. For the reverse direction, we claim that the composite

$$\underline{\mathrm{Hom}}(X, Y) \simeq \underline{\mathrm{Hom}}(X, Y) \otimes \mathbb{1} \xrightarrow{\underline{\mathrm{Hom}}(X,Y) \otimes \eta} \underline{\mathrm{Hom}}(X, Y) \otimes X \otimes DX \xrightarrow{\mathrm{ev} \otimes DX} Y \otimes DX$$

is an inverse to $\nu_{X,Y}$. Again, one may verify this by diagram chasing. □

Remark 2.22. By Lemma 2.20, one deduces that if X is a rigid object, then we have adjunctions $X \otimes - \dashv DX \otimes - \dashv X \otimes -$.

Lemma 2.23. *Let $F\colon \mathsf{T} \to \mathsf{U}$ be a tt-functor between tt-categories. Then F preserves rigid (=dualizable) objects. Moreover, if X is rigid, then $F(DX) \simeq D(FX)$.*

Proof. For any $X \in \mathsf{T}$, we can construct a map $\varphi\colon F(DX) \to D(FX)$ as the adjunct to

$$FX \otimes F(DX) \simeq F(X \otimes DX) \xrightarrow{F(\mathrm{ev})} F(\mathbb{1}_{\mathsf{T}}) \simeq \mathbb{1}_{\mathsf{U}}.$$

If X is dualisable, then we have a map $\eta\colon \mathbb{1}_{\mathsf{T}} \to X \otimes DX$ and hence we have a map $\eta'\colon \mathbb{1}_{\mathsf{U}} \to FX \otimes D(FX)$ defined as the composite

$$\mathbb{1}_{\mathsf{U}} \simeq F(\mathbb{1}_{\mathsf{T}}) \xrightarrow{F(\eta)} F(X \otimes DX) \simeq FX \otimes F(DX) \xrightarrow{FX \otimes \varphi} FX \otimes D(FX).$$

Using this one can verify that FX is dualisable.

To see that the map $\varphi\colon F(DX) \to D(FX)$ constructed above is an isomorphism, one can construct an inverse as follows. Since FX is dualisable, $D(FX)$ is also dualisable (Lemma 2.19) so by adjunction maps $D(FX) \to F(DX)$ correspond to maps $\mathbb{1}_{\mathsf{U}} \to FX \otimes F(DX)$. We can construct such a map as the composite

$$\mathbb{1}_{\mathsf{U}} \simeq F(\mathbb{1}_{\mathsf{T}}) \xrightarrow{F(\eta)} F(X \otimes DX) \simeq FX \otimes F(DX).$$

One can check directly that this gives an inverse to φ. □

Is it natural to ask how the different notions of 'smallness' in Definition 2.7 and Definition 2.16 are related. The following two results answer this. We also note the following relation, which does not rely on a set of rigid generators.

Proposition 2.24. *Let T be a tensor-triangulated category. If $\mathbb{1}$ is compact, then any rigid object is compact.*

Proof. Let X be rigid. Then $\mathrm{Hom}(X, -) \simeq \mathrm{Hom}(\mathbb{1}, DX \otimes -)$. Since $\mathbb{1}$ is compact this commutes with coproducts showing that X is compact. □

Proposition 2.25. *Let T be a tensor-triangulated category, and suppose that T has a set of rigid generators G.*

(1) For $X \in \mathsf{T}$ we have

$$X \in \mathrm{Thick}(\mathsf{G}) \implies X \text{ is rigid} \iff X \text{ is internally compact.}$$

(2) If the elements of G are compact, then for $X \in \mathsf{T}$ we have

$$X \text{ is compact} \iff X \in \mathrm{Thick}(\mathsf{G}) \implies X \text{ is rigid} \iff X \text{ is internally compact.}$$

(3) If the elements of G are compact, and the unit $\mathbb{1}$ of T is compact, then for $X \in \mathsf{T}$ we have

$$X \text{ is compact} \iff X \in \mathrm{Thick}(\mathsf{G}) \iff X \text{ is rigid} \iff X \text{ is internally compact.}$$

Proof. For (1), firstly note that the set of rigid objects in T is thick. By assumption it contains G, so we have $\mathrm{Thick}(\mathsf{G}) \subseteq \{\text{rigid objects}\}$, which proves the first implication. That rigid objects are internally compact is an immediate consequence of the definitions. For the remaining implication, suppose X is internally compact, and consider the set

$$\mathcal{L} = \{Y \in \mathsf{T} \mid DX \otimes Y \xrightarrow{\sim} \underline{\mathrm{Hom}}(X, Y)\}.$$

The set $\mathcal{L}$ is localizing as X is internally compact, and contains G by Exercise A.4. Therefore $\mathcal{L} = \mathsf{T}$, and hence X is rigid.

For (2), it suffices to prove that X is compact if and only if $X \in \mathrm{Thick}(\mathsf{G})$, which was the content of Proposition 2.12. (3) follows from Proposition 2.24. □

Definition 2.26. Let T be a compactly generated tt-category. We say that T is *rigidly-compactly generated* if the compact and rigid objects in T coincide. Note that by Proposition 2.25(3) this is equivalent to T having a set of rigid and compact generators and the tensor unit being compact.

Example 2.27. Let T be a tt-category which is compactly generated by its tensor unit. Then T is rigidly-compactly generated. For example, $\mathsf{D}(R)$ is rigidly-compactly generated.

Example 2.28. The stable module category $\mathsf{StMod}(kG)$ is rigidly-compactly generated. Since the compacts are the finitely generated modules it is clear that it is compactly generated. It is easy to see that the compacts coincide with the rigid objects (since if M is finitely generated over kG then it has finite rank over k).

Example 2.29. Let us revisit Example 2.15: we consider $\mathsf{T} = \prod_i \mathsf{T}_i$ where each T_i is rigidly-compactly generated by G_i. The set $\mathsf{G} = \{\delta_i(G_i) \mid G_i \in \mathsf{G}_i\}$ of "characteristic objects" given by $(\delta_i X)_i = X_i$ and $(\delta_i X)_j = 0$ for $i \neq j$ is a set of rigid generators for T. Moreover these are compact. So we are in case (2) of Proposition 2.25. However, we already saw that the unit is not compact (but is rigid). So the inclusion $\mathsf{T}^{\mathrm{c}} \subset \{\text{rigids}\}$ is proper.

Finally, it is convenient to introduce terminology for the appropriate analogues of thick and localizing subcategories in the tensor-triangular setting.

Definition 2.30. Let T be a tensor-triangulated category. A thick/localizing subcategory $\mathcal{S}$ is a *$\otimes$-ideal* if for all $X \in \mathcal{S}$ and $Y \in \mathsf{T}$ we have $X \otimes Y \in \mathcal{S}$. We write $\mathrm{Thick}^{\otimes}(-)$ and $\mathrm{Loc}^{\otimes}(-)$ for the smallest thick and localizing $\otimes$-ideal generated by a set of objects.

Proposition 2.31. *Let T be a tensor-triangulated category and $X \in \mathsf{T}$ be rigid.*

(1) If Y is compact, then $X \otimes Y$ is compact.
(2) If Y is internally compact, then $X \otimes Y$ is internally compact.
(3) If Y is rigid, then $X \otimes Y$ is rigid.

Proof. For (1), we isomorphisms

$$\mathrm{Hom}(X \otimes Y, \bigoplus Z_i) = \mathrm{Hom}(X, DY \otimes \bigoplus Z_i) = \bigoplus \mathrm{Hom}(X, DY \otimes Z_i) = \bigoplus \mathrm{Hom}(X \otimes Y, Z_i)$$

showing that $X \otimes Y$ is compact. Parts (2) and (3) are similar and are left as Exercise A.5. □

The following is then immediate:

Corollary 2.32. *For any tensor-triangulated category T, the full subcategory of rigid objects forms a tensor-triangulated category.* □

3. Geometric functors

In this section we introduce the notion of a *geometric functor*. In order to demonstrate their powerful formal properties, we recall the adjoint functor theorems for triangulated categories.

Lemma 3.1. *Let $F\colon \mathsf{T} \to \mathsf{U}$ be a triangulated functor. If F has a adjoint (on either side), then this adjoint is also a triangulated functor.*

Proof. Left as Exercise A.9. □

Theorem 3.2 (Adjoint functor theorems)**.** *Let T and U be triangulated categories. Let $F\colon \mathsf{T} \to \mathsf{U}$ be a triangulated functor, and suppose that T is compactly generated.*

(1) F is coproduct preserving if and only if F has a triangulated right adjoint.
(2) F is product preserving if and only if F has a triangulated left adjoint.

Proof. This is an immediate consequence of Brown representability and Lemma 3.1. □

Proposition 3.3. *Let $F : \mathsf{T} \rightleftarrows \mathsf{U} : G$ be an adjunction between triangulated categories.*

(1) If G preserves coproducts, then F preserves compact objects.
(2) Suppose that T is compactly generated and that F preserves compacts. Then G preserves coproducts.

Proof. For (1) we let $C \in \mathsf{T}^{\mathrm{c}}$. For any set of objects $\{X_i\}$ in U, we have isomorphisms

$$\oplus\mathrm{Hom}(FC, X_i) \simeq \oplus\mathrm{Hom}(C, GX_i) \simeq \mathrm{Hom}(C, \oplus GX_i) \simeq \mathrm{Hom}(C, G(\oplus X_i)) \simeq \mathrm{Hom}(FC, \oplus X_i)$$

using in turn, the adjunction, compactness of C, the assumption that G preserves coproducts, and finally the adjunction again. For (2), by compact generation of T it suffices to prove that the canonical map $\oplus GX_i \to G(\oplus X_i)$ induces isomorphisms

$$\mathrm{Hom}(C, \oplus GX_i) \xrightarrow{\simeq} \mathrm{Hom}(C, G(\oplus X_i))$$

for all $C \in \mathsf{T}^{\mathrm{c}}$. This is then a straightforward check using the adjunction and the fact that F preserves compacts. □

We now have the necessary triangulated background to make the key definition and prove some key features of it. We will then record several examples afterwards.

Definition 3.4. Let T and U be rigidly-compactly generated tt-categories. A *geometric functor* $f^*\colon \mathsf{T} \to \mathsf{U}$ is a strong symmetric monoidal, triangulated functor which preserves coproducts.

Proposition 3.5. *Let $f^*\colon \mathsf{T} \to \mathsf{U}$ be a geometric functor. Then f^* fits into an adjoint triple*

$$\begin{tikzcd} \mathsf{T} \arrow[r, bend left=50, "f^*"] \arrow[r, bend right=50, "f^{(1)}"'] & \mathsf{U} \arrow[l, "f_*"'] \end{tikzcd}$$

where left adjoints are denoted on top.

Proof. Since f^* is coproduct preserving, it has a right adjoint f_* by Theorem 3.2. Since f^* is a strong monoidal functor it preserves rigid objects (Lemma 2.23) and hence preserves compact objects since these coincide by assumption. Therefore, f_* preserves coproducts by Proposition 3.3 and hence itself has a right adjoint $f^{(1)}$ again by Theorem 3.2. □

Let $f^*\colon \mathsf{T} \to \mathsf{U}$ be a geometric functor. There is a natural map $\pi\colon X \otimes f_*Y \to f_*(f^*X \otimes Y)$ for all $X \in \mathsf{T}$ and $Y \in \mathsf{U}$ which is constructed as the adjunct to the natural map

$$f^*(X \otimes f_*Y) \simeq f^*X \otimes f^*f_*Y \xrightarrow{f^*X \otimes \varepsilon} f^*X \otimes Y$$

where ε is the counit of the (f^*, f_*) adjunction.

Proposition 3.6 (The projection formula)**.** *Let $f^*\colon \mathsf{T} \to \mathsf{U}$ be a geometric functor. The natural map*

$$\pi\colon X \otimes f_*Y \xrightarrow{\simeq} f_*(f^*X \otimes Y)$$

is an isomorphism for all $X \in \mathsf{T}$ and $Y \in \mathsf{U}$.

Proof. Let $Z \in \mathsf{T}$ and $Y \in \mathsf{U}$ be arbitrary. We first prove that π is an isomorphism if X is rigid. In this case, we have isomorphisms

$$\begin{aligned}
\mathrm{Hom}(Z, X \otimes f_*Y) &\simeq \mathrm{Hom}(Z \otimes DX, f_*Y) && \text{since } X \text{ is rigid} \\
&\simeq \mathrm{Hom}(f^*Z \otimes f^*(DX), Y) && \text{by adjunction and monoidality of } f^* \\
&\simeq \mathrm{Hom}(f^*Z \otimes D(f^*X), Y) && \text{by Lemma 2.23} \\
&\simeq \mathrm{Hom}(Z, f_*(f^*X \otimes Y)) && \text{by rigidity of } D(f^*X) \text{ and adjunction.}
\end{aligned}$$

A diagram chase verifies that this isomorphism is induced by π (we omit tedious manipulations of commutative diagrams throughout in this course). By the Yoneda lemma we therefore deduce that π is an isomorphism when X is rigid. Now the collection of $X \in \mathsf{T}$ for which π is an isomorphism (for a fixed Y) is a localising subcategory (since both f^* and f_* preserve coproducts and are triangulated). This localising subcategory contains all the rigid=compact objects, and hence by compact generation, we deduce that π is an isomorphism for all X. □

A slick trick allows one to generate new isomorphisms from these by using uniqueness of adjunctions. Recall that if $F_1 \dashv G_1$ and $F_2 \dashv G_2$, then $F_1F_2 \dashv G_2G_1$ (notice the order reversal).

Proposition 3.7. *Let $f^* \colon \mathsf{T} \to \mathsf{U}$ be a geometric functor. There are natural isomorphisms:*

$$\underline{\mathrm{Hom}}(X, f_*Y) \simeq f_*\underline{\mathrm{Hom}}(f^*X, Y) \tag{3.8}$$

$$\underline{\mathrm{Hom}}(f_*X, Y) \simeq f_*\underline{\mathrm{Hom}}(X, f^{(1)}Y) \tag{3.9}$$

$$f^{(1)}\underline{\mathrm{Hom}}(X, Y) \simeq \underline{\mathrm{Hom}}(f^*X, f^{(1)}Y). \tag{3.10}$$

Proof. We first fix X in the projection formula (Proposition 3.6). We have adjunctions:

$$X \otimes f_*(-) = (X \otimes -) \circ f_* \dashv f^{(1)} \circ \underline{\mathrm{Hom}}(X, -)$$

and

$$f_*(f^*X \otimes -) \dashv \underline{\mathrm{Hom}}(f^*X, -) \circ f^{(1)}.$$

By uniqueness of adjoints, we therefore deduce that $f^{(1)}\underline{\mathrm{Hom}}(X, Y) \simeq \underline{\mathrm{Hom}}(f^*X, f^{(1)}Y)$, giving (3.10). Fixing Y in the projection formula instead yields (3.9). Finally, for (3.8), we have an isomorphism $f^*X \otimes f^*(-) \simeq f^*(X \otimes -)$. Taking right adjoints of this gives the claim. □

Let us return to the examples and see what the above features of geometric functors correspond to in this setting.

Example 3.11. Let $f \colon R \to S$ be a map of commutative rings. Then the extension of scalars functors $f^* = S \otimes^{\mathsf{L}}_R - \colon \mathsf{D}(R) \to \mathsf{D}(S)$ is a geometric functor. Its right adjoint f_* is the restriction of scalars functor, and the double right adjoint is the coextension of scalars functor $f^{(1)} = \mathsf{RHom}_R(S, -)$. In this case, the projection formula is not exciting: it says that for $X \in \mathsf{D}(R)$ and $Y \in \mathsf{D}(S)$ there is an isomorphism $X \otimes^{\mathsf{L}}_R Y \simeq (S \otimes^{\mathsf{L}}_R X) \otimes^{\mathsf{L}}_S Y$. The other isomorphisms are similarly "obvious" in this setting. The main selling point of the framework is that these are general isomorphisms which hold in all settings, and moreover admit uniform proofs based purely on formal manipulations.

Example 3.12. Let G be a group and H be a subgroup. There is a restriction functor $f^* = \mathrm{res}^G_H \colon \mathsf{StMod}(kG) \to \mathsf{StMod}(kH)$ given by restricting the G-action to a H-action, and this is a geometric functor. This has a right adjoint given by coinduction $f_* = \mathrm{Hom}_{kH}(kG, -)$ which has a further right adjoint given by restriction again. In this example, we see some extra features happening: f^* also has a left adjoint (the induction $kG \otimes_{kH} -$) and this left adjoint is isomorphic to the coinduction. Therefore we obtain an infinite tower of adjunctions. This is an instance of the *Wirthmüller isomorphism* which we will see later on.

Remark 3.13. We see that the extra features visible in Example 3.12 are not uniform for all geometric functors. Indeed, they do not occur in the setting of Example 3.11 in general. However, one can completely characterise the existence of extra adjoints and when these are isomorphic in general tt-terms. This is the point of much of the rest of this course.

4. Grothendieck–Neeman duality

Let $f^*\colon \mathsf{T} \to \mathsf{U}$ be a geometric functor. In this section, we explain when f^* and its double right adjoint $f^{(1)}$ agree up to a twist. Such an isomorphism is termed *Grothendieck duality*. Let us first give the background to make this precise.

Definition 4.1. Let $f^*\colon \mathsf{T} \to \mathsf{U}$ be a geometric functor. We set $\omega_f \coloneqq f^{(1)}(\mathbb{1}_{\mathsf{T}})$ and call this the *relative dualising object*.

Remark 4.2. For any $X, Y \in \mathsf{T}$ there is a natural map

$$f^{(1)}Y \otimes f^*X \to f^{(1)}(Y \otimes X)$$

constructed as the adjoint to $f_*(f^{(1)}Y \otimes f^*X) \simeq f_*f^{(1)}Y \otimes X \xrightarrow{\varepsilon \otimes X} Y \otimes X$ where the isomorphism is the projection formula (Proposition 3.6) and ε is the counit map. Taking $Y = \mathbb{1}_{\mathsf{T}}$ we obtain a natural map $\omega_f \otimes f^*X \to f^{(1)}X$. When this map is an isomorphism, we say that *Grothendieck duality* holds for X.

We first show that all rigid objects have Grothendieck duality:

Lemma 4.3. *Let $f^*\colon \mathsf{T} \to \mathsf{U}$ be a geometric functor. If $X \in \mathsf{T}$ is rigid, then the natural map*

$$f^{(1)}Y \otimes f^*X \to f^{(1)}(Y \otimes X)$$

is an isomorphism for all $Y \in \mathsf{T}$.

Proof. Let $Z \in \mathsf{T}$ be arbitrary. We have natural isomorphisms:

$$\begin{aligned}
\mathrm{Hom}(Z, f^{(1)}Y \otimes f^*X) &\simeq \mathrm{Hom}(Z \otimes f^*(DX), f^{(1)}Y) && \text{by Lemma 2.23}\\
&\simeq \mathrm{Hom}(f_*Z \otimes DX, Y) && \text{by adjunction and Proposition 3.6}\\
&\simeq \mathrm{Hom}(f_*Z, X \otimes Y) && \text{as } X \text{ is rigid}\\
&\simeq \mathrm{Hom}(Z, f^{(1)}(X \otimes Y)) && \text{by adjunction.}
\end{aligned}$$

A diagram chase verifies this is induced by the natural map $f^{(1)}Y \otimes f^*X \to f^{(1)}(Y \otimes X)$ and hence by the Yoneda lemma we are done. □

The goal is to generalise this in two directions:

(1) We identify conditions when every object of T has Grothendieck duality.
(2) Considering some examples in commutative algebra, one sees that the conditions we will identify are rather strong in some cases, so we will give an alternative condition which yields Grothendieck duality on an enlargement of the subcategory of rigid objects.

In this section we deal with (1). The next few sections treat (2).

Remark 4.4. Notice that if every object has Grothendieck duality, then $f^{(1)}$ preserves coproducts. Conversely, if $f^{(1)}$ preserves coproducts then every object has Grothendieck duality by a localising subcategory argument. We now identify further equivalent conditions in terms of the more familiar functors f^* and f_* (where more familiar means in the sense of explicit examples).

We first prove a general result about triangulated categories which provides a way to determine properties of a compact-preserving functor by restriction to compacts.

Lemma 4.5. *Let $F : \mathsf{T} \rightleftarrows \mathsf{U} : G$ be an adjunction of triangulated functors between triangulated categories, where T is compactly generated. Suppose that F preserves compacts. If the restriction $F|\colon \mathsf{T}^c \to \mathsf{U}^c$ has a right adjoint G_0, then G preserves compacts. Moreover, the restriction of G to compacts is G_0.*

Proof. We have isomorphisms

$$\mathrm{Hom}_{\mathsf{T}^c}(X, G_0(Y)) = \mathrm{Hom}_{\mathsf{U}^c}(F|(X), Y) = \mathrm{Hom}_{\mathsf{U}}(FX, Y) = \mathrm{Hom}_{\mathsf{T}}(X, GY)$$

for all $X \in \mathsf{T}^c$ and $Y \in \mathsf{U}^c$. Therefore taking $X = G_0(Y)$, we obtain a natural map $g\colon G_0(Y) \to GY$ corresponding to the identity on $G_0(Y)$. One checks that the above isomorphism of Hom's is induced by $\mathrm{Hom}_{\mathsf{T}}(-, g)$ and hence by compact generation of T we deduce that g is an isomorphism which gives the claims. □

Lemma 4.6. *Let $F : \mathsf{T} \rightleftarrows \mathsf{U} : G$ be an adjunction of triangulated functors between triangulated categories, where both T and U are compactly generated. Suppose that F preserves compacts. If the restriction $F|\colon \mathsf{T}^c \to \mathsf{U}^c$ has a left adjoint E_0, then F preserves products.*

Proof. Let $X \in \mathsf{U}^c$ and $Y \in \mathsf{T}$. There is a natural map $\alpha_{X,Y}\colon \mathrm{Hom}(E_0(X), Y) \to \mathrm{Hom}(X, FY)$ defined by the composite

$$\mathrm{Hom}(E_0(X), Y) \xrightarrow{F} \mathrm{Hom}(FE_0(X), FY) \xrightarrow{\mathrm{Hom}(\eta, FY)} \mathrm{Hom}(X, FY)$$

which is an isomorphism if Y is compact. Notice that for a fixed $X \in \mathsf{U}^c$, the collection of $Y \in \mathsf{T}$ for which $\alpha_{X,Y}$ is an isomorphism is a localising subcategory by the 5-lemma and the fact that F preserves coproducts. Therefore by compact generation of T, we have that $\alpha_{X,Y}$ is an isomorphism for all $X \in \mathsf{U}^c$ and $Y \in \mathsf{T}$.

Let $X \in \mathsf{U}^c$ and $\{Y_i\} \subset \mathsf{T}$. We then have isomorphisms

$$\begin{aligned}
\mathrm{Hom}(X, F(\textstyle\prod Y_i)) &= \mathrm{Hom}(E_0(X), \textstyle\prod Y_i) \\
&= \textstyle\prod \mathrm{Hom}(E_0(X), Y_i) \\
&= \textstyle\prod \mathrm{Hom}(X, FY_i) \\
&= \mathrm{Hom}(X, \textstyle\prod (FY_i))
\end{aligned}$$

using α in the first and third isomorphism. One can check that this is induced by the canonical comparison map $F(\prod Y_i) \to \prod FY_i$ and hence by compact generation of U we deduce that F preserves products. □

Lemma 4.7. *Let $f^*\colon \mathsf{T} \to \mathsf{U}$ be a geometric functor. If f^* has a left adjoint $f_{(1)}$ then the following hold:*

(1) f_* preserves compact objects;
(2) if $X \in \mathsf{U}^c$, then there is a natural isomorphism $D(f_{(1)}(DX)) \simeq f_*X$.

Proof. Since f^* preserves coproducts, its left adjoint $f_{(1)}$ preserves compacts by Proposition 3.3. Now by Lemma 2.23 we have a commutative diagram

$$\begin{tikzcd}
(\mathsf{T}^c)^{\mathrm{op}} \arrow[r, "D"] \arrow[d, "(f^*)^{\mathrm{op}}"'] & \mathsf{T}^c \arrow[d, "f^*"] \\
(\mathsf{U}^c)^{\mathrm{op}} \arrow[r, "D"'] & \mathsf{U}^c.
\end{tikzcd}$$

Now notice that the *right* adjoint of $(f^*)^{\mathrm{op}}$ is $(f_{(1)})^{\mathrm{op}}$. As such (since D is an equivalence of categories) we see that $D(f_{(1)}^{\mathrm{op}}(D(-)))$ is right adjoint to f^*. Therefore by Lemma 4.5 we see that f_* preserves compacts and the restriction of f_* to compacts is $D(f_{(1)}(D(-)))$. □

Theorem 4.8. *Let $f^*\colon \mathsf{T} \to \mathsf{U}$ be a geometric functor. The following conditions are equivalent:*

(1) f^* has a left adjoint $f_{(1)}$;
(2) f^* preserves products;
(3) $f^{(1)}$ has a right adjoint $f_{(-1)}$;
(4) $f^{(1)}$ preserves coproducts;
(5) f_* preserves compacts.

If any of the above equivalent conditions hold we say that f^* satisfies Grothendieck–Neeman duality*.*

Proof. We have that (1) ⇔ (2) and (3) ⇔ (4) by Theorem 3.2. We also know that (4) ⇔ (5) by Proposition 3.3. We have (1) ⇒ (5) by Lemma 4.7. Finally, (5) ⇒ (2) is left as Exercise A.10. □

Remark 4.9. If f^* has Grothendieck–Neeman duality, then we have an adjoint quintuple

$$f_{(1)} \dashv f^* \dashv f_* \dashv f^{(1)} \dashv f_{(-1)}.$$

Grothendieck–Neeman duality provides a plethora of natural isomorphisms relating all of the functors in the associated adjoint quintuple:

Corollary 4.10. *Let $f^*\colon \mathsf{T} \to \mathsf{U}$ be a geometric functor which satisfies Grothendieck–Neeman duality. Then there are natural isomorphisms on all objects:*

$$f^*X \otimes f^{(1)}Y \simeq f^{(1)}(X \otimes Y) \tag{4.11}$$
$$\omega_f \otimes f^*X \simeq f^{(1)}X \tag{4.12}$$
$$\underline{\mathrm{Hom}}(X, f_{(-1)}Y) \simeq f_*\underline{\mathrm{Hom}}(f^{(1)}X, Y) \tag{4.13}$$
$$\underline{\mathrm{Hom}}(X, f_{(-1)}Y) \simeq f_{(-1)}\underline{\mathrm{Hom}}(f^*X, Y) \tag{4.14}$$
$$f_{(-1)} \simeq f_*\underline{\mathrm{Hom}}(\omega_f, -) \tag{4.15}$$
$$f_{(1)} \simeq f_*(\omega_f \otimes -) \tag{4.16}$$
$$f^* \simeq \underline{\mathrm{Hom}}(\omega_f, f^{(1)}(-)) \tag{4.17}$$
$$\mathbb{1}_{\mathsf{U}} \simeq \underline{\mathrm{Hom}}(\omega_f, \omega_f) \tag{4.18}$$
$$X \otimes f_{(1)}Y \simeq f_{(1)}(f^*X \otimes Y) \tag{4.19}$$
$$f^*\underline{\mathrm{Hom}}(X, Y) \simeq \underline{\mathrm{Hom}}(f^*X, f^*Y) \tag{4.20}$$
$$\underline{\mathrm{Hom}}(f_{(1)}X, Y) \simeq f_*\underline{\mathrm{Hom}}(X, f^*Y) \tag{4.21}$$

Remark 4.22. Before we give the proof, let us name the most important of the above isomorphisms. We already coined (4.12) Grothendieck duality. (4.16) is called the ur-Wirthmüller isomorphism, and (4.19) is called the left projection formula.

Proof. We prove these in order:

(4.11) Fix an arbitrary $Y \in \mathsf{T}$. Notice that the subcategory

$$\{X \in \mathsf{T} \mid f^{(1)}Y \otimes f^*X \xrightarrow{\sim} f^{(1)}(Y \otimes X) \text{ is an isomorphism}\}$$

is a localising subcategory since $f^{(1)}$ preserves coproducts by Grothendieck–Neeman duality (and f^* does since it is geometric). By Lemma 4.3 this subcategory contains all rigid=compact objects, and hence is the whole of T.

(4.12) This is the case $Y = \mathbb{1}_{\mathsf{T}}$ of (4.11).

(4.13) We fix Y in (4.11) and notice that we have adjunctions

$$f^*(-) \otimes f^{(1)}Y \dashv f_*\underline{\mathrm{Hom}}(f^{(1)}Y, -) \quad \text{and} \quad f^{(1)}(Y \otimes -) \dashv \underline{\mathrm{Hom}}(-, f_{(-1)}Y)$$

so we obtain the claim from (4.11) by taking right adjoints after fixing Y.

(4.14) This follows from (4.11) by taking right adjoints after fixing X.

(4.15) This is the case $X = \mathbb{1}_{\mathsf{T}}$ of (4.13).

(4.16) We first construct the natural map. By adjunction, maps $f_*(\omega_f \otimes X) \to f_{(1)}X$ are in bijection with maps $\omega_f \otimes X \to f^{(1)}f_{(1)}X$ and we have such a natural map by taking the composite

$$\omega_f \otimes X \xrightarrow{\omega_f \otimes \eta} \omega_f \otimes f^* f_{(1)}X \xrightarrow{\sim} f^{(1)}f_{(1)}X$$

where the isomorphism is Grothendieck duality (4.12). We note that the collection of X for which the natural map $f_*(\omega_f \otimes X) \to f_{(1)}X$ is an isomorphism is a localising subcategory (since all the functors involved preserve coproducts). Therefore it suffices to prove it is an isomorphism for X rigid. So let $X \in \mathsf{U}$ be rigid and $Z \in \mathsf{T}$ be arbitrary. We then have isomorphisms

$$\begin{aligned}
\mathrm{Hom}(Z, f_*(\omega_f \otimes X)) &\simeq \mathrm{Hom}(f^*Z \otimes DX, \omega_f) && \text{by adjunction and rigidity of } X \\
&\simeq \mathrm{Hom}(f_*(f^*Z \otimes DX), \mathbb{1}_{\mathsf{T}}) && \text{by adjunction as } \omega_f = f^{(1)}(\mathbb{1}_{\mathsf{T}}) \\
&\simeq \mathrm{Hom}(Z \otimes f_*(DX), \mathbb{1}_{\mathsf{T}}) && \text{by projection formula (3.6)} \\
&\simeq \mathrm{Hom}(Z \otimes D(f_{(1)}X), \mathbb{1}_{\mathsf{T}}) && \text{by Lemma 4.7} \\
&\simeq \mathrm{Hom}(Z, f_{(1)}X) && \text{as } f_{(1)}X \text{ is rigid.}
\end{aligned}$$

A (rather involved) diagram chase shows that this isomorphism is induced by the natural map $f_*(\omega_f \otimes X) \to f_{(1)}X$ and hence by the Yoneda lemma we are done.

(4.17) This is the right adjoint of (4.16).

(4.18) This follows by taking $\mathbb{1}_{\mathsf{T}}$ in (4.17) as f^* is strong monoidal.

(4.19) We have isomorphisms

$$X \otimes f_{(1)}Y \simeq X \otimes f_*(\omega_f \otimes Y) \simeq f_*(f^*X \otimes \omega_f \otimes Y) \simeq f_{(1)}(f^*X \otimes Y)$$

by using the ur-Wirthmüller isomorphism (4.16), the projection formula Proposition 3.6 and again the ur-Wirthmüller isomorphism in turn.

(4.20) Fix X in (4.19) and take right adjoints.

(4.21) Fix Y in (4.19) and take right adjoints.

□

Remark 4.23. We have proved that Grothendieck–Neeman duality implies a whole host of isomorphisms as listed in Corollary 4.10. However, we have some converses too. For example, (4.12) implies Grothendieck–Neeman duality since it ensures that $f^{(1)}$ preserves coproducts. We leave it as an exercise (Exercise A.11) for the reader to work out which other isomorphisms listed there imply Grothendieck–Neeman duality.

Remark 4.24. One may wonder why the rather odd name of the *ur*-Wirthmüller isomorphism. The prefix "ur-" means primitive or original, and indeed, later on we will see the Wirthmüller isomorphism. The difference boils down to the fact, that in the "ur-" version, we realise $f_{(1)}$ as a twisted form of f_* where in the full blown Wirthmüller isomorphism, we can also realise f_* as a twisted form of $f_{(1)}$. However, the existence of a full Wirthmüller isomorphism requires a stronger setup than Grothendieck–Neeman duality.

Let us return to our running examples. Let's deal with modular representation theory first since there is less to say here:

Example 4.25. Consider the geometric functor $f^* = \mathrm{res}^G_H \colon \mathsf{StMod}(kG) \to \mathsf{StMod}(kH)$. It is easy to see that this preserves products, and hence has Grothendieck–Neeman duality. In this case, the ur-Wirthmüller isomorphism is the well-known fact that induction $kG \otimes_{kH} -$ and coinduction $\mathrm{Hom}_{kH}(kG, -)$ are isomorphic. (It follows in particular that $\omega_f = \mathbb{1}$.)

Remark 4.26. The previous example highlights an important point: the general framework here does not tell you how to work out what ω_f is (although there are some partial results in this direction which we may revisit if there is time). Often giving an explicit description of ω_f relies on some pre-existing knowledge of the case at hand.

Let us now turn to commutative algebra where we already see several interesting features. We first make the following trivial observation:

Lemma 4.27. *Let $f^* \colon \mathsf{T} \to \mathsf{U}$ be a geometric functor. If U is compactly generated by its tensor unit (i.e., $\mathsf{U} = \mathrm{Loc}(\mathbb{1}_{\mathsf{U}})$) then f^* has Grothendieck–Neeman duality if and only if $f_*(\mathbb{1}_{\mathsf{U}})$ is compact.*

Proof. Recall from Proposition 2.12 that $\mathsf{U}^c = \mathrm{thick}(\mathbb{1}_{\mathsf{U}})$. Notice that $\{X \in \mathsf{U} \mid f_*X \text{ is compact}\}$ is a thick subcategory of U. So if $f_*\mathbb{1}_{\mathsf{U}}$ is compact, then f_* preserves compacts. □

Example 4.28. Let $(R, \mathfrak{m}, k)$ be a commutative Noetherian local ring (i.e., $\mathfrak{m}$ is the maximal ideal, and $k = R/\mathfrak{m}$ is the residue field). Let $f^* = k \otimes_R - \colon \mathsf{D}(R) \to \mathsf{D}(k)$ be the geometric functor given by extension of scalars along the quotient map $R \to k$. Recall that the right adjoint to this is the restriction of scalars functor. This satisfies Grothendieck–Neeman duality if and only if $k \in \mathsf{D}(R)^c$. By Example 2.13, this is equivalent to k having finite projective dimension over R. It is a standard fact in commutative algebra that this is equivalent to R being a regular local ring (i.e., if $\mathfrak{m} = (x_1, \cdots, x_n)$ with n minimal, then $\dim(R) = n$). We immediately see that this is rather restrictive from a commutative algebraist's perspective: regular local rings form a very small class of the rings we are interested in. As an example to keep in mind for later, $R = k[x]/x^2$ is not regular: it has Krull dimension 0, but the maximal ideal can't be generated by 0 elements! (In fact, fields are the only regular local rings of Krull dimension 0.) This in particular motivates a generalisation of the above results which weaken the requirement that k is compact in $\mathsf{D}(R)$.

5. Proxy-small objects

In this section we introduce a substantial weakening of the notion of compact objects. It is motivated by the observation that for a commutative Noetherian local ring $(R, \mathfrak{m}, k)$ the residue field k is compact if and only if R is regular, which is a very restrictive condition. On the other hand, we will see that k is always proxy-small in $\mathsf{D}(R)$.

5.A. Proxy-small objects.

We saw in Theorem 4.8 that f^* has Grothendieck–Neeman duality if and only if f_* preserves compact. By Example 4.28 this is a rather restrictive assumption in general. However, it is not so rare that $f_*(\mathbb{1}_{\mathsf{U}})$ is "almost" compact in the following sense:

Definition 5.1. Let T be a rigidly-compactly generated tt-category. We say that a set of objects $\mathcal{K}$ of T is *proxy-small* if there exists a set of compact objects $\mathcal{W}$ such that $\mathcal{W} \subseteq \mathrm{thick}_{\otimes}(\mathcal{K})$ and $\mathcal{K} \subseteq \mathrm{Loc}_{\otimes}(\mathcal{W})$. We call $\mathcal{W}$ the *witness* for the proxy-smallness of $\mathcal{K}$, and note this implies $\mathrm{Loc}_{\otimes}(\mathcal{K}) = \mathrm{Loc}_{\otimes}(\mathcal{W})$.

Remark 5.2. It is helpful to introduce some terminology to make discussing proxy-small objects easier. If $X \in \mathrm{thick}_{\otimes}(Y)$, we say that Y *finitely builds* X. If $X \in \mathrm{Loc}_{\otimes}(Y)$ we say that Y *builds* X.

A priori, the definition of proxy-smallness depends on the choice of *some* witness, but one can give an equivalent definition in terms which are intrinsic to $\mathcal{K}$:

Lemma 5.3. *Let $\mathcal{K}$ be a set of objects in a rigidly-compactly generated tt-category T. Then $\mathcal{K}$ is proxy-small if and only if* $\mathrm{Loc}_{\otimes}(\mathcal{K}) = \mathrm{Loc}_{\otimes}(\mathrm{thick}_{\otimes}(\mathcal{K}) \cap \mathsf{T}^c)$.

Proof. By definition, $\mathrm{thick}_{\otimes}(\mathcal{K}) \cap \mathsf{T}^c$ is a set of compact objects contained in $\mathrm{thick}_{\otimes}(\mathcal{K})$. The reverse implication is then immediate since we may take $\mathrm{thick}_{\otimes}(\mathcal{K}) \cap \mathsf{T}^c$ as the witness. For the forward implication, we have a witness $\mathcal{W}$ set which must satisfy $\mathcal{W} \subseteq \mathrm{thick}_{\otimes}(\mathcal{K}) \cap \mathsf{T}^c$. Since $\mathcal{K} \subseteq \mathrm{Loc}_{\otimes}(\mathcal{W})$ it follows that $\mathcal{K} \subseteq \mathrm{Loc}_{\otimes}(\mathrm{thick}_{\otimes}(\mathcal{K}) \cap \mathsf{T}^c)$. The other inclusion required is trivial. □

Remark 5.4. Despite the definition being independent of the choice of witness, we note that choosing a witness is often convenient. This is illustrated in particular by some of the following examples.

Example 5.5. Proxy-small objects are a weakening of the notation of compact objects (an unfortunate terminology clash gives rise to the small vs compact: some authors call compact objects small). If $\mathcal{K}$ is a set of compact objects of T, then $\mathcal{K}$ is proxy-small by taking $\mathcal{W} = \mathcal{K}$.

Example 5.6. Let k be a field. Any object M in $\mathsf{D}(k)$ is proxy-small with k as a witness. Indeed, since any object of $\mathsf{D}(k)$ may be written as a sum of shifted copies of k we have $M \simeq \oplus_{i \in I} \Sigma^{a_i} k$. The claim then follows clearly from the definition.

Example 5.7. Let $(R, \mathfrak{m}, k)$ be a commutative Noetherian local ring $(R, \mathfrak{m}, k)$. We claim that $k = R/\mathfrak{m}$ is proxy-small with witness the Koszul complex on the generators of $\mathfrak{m}$. In contrast, recall that the residue field k is compact if and only if R is regular. Proving this requires us setting up a bit of background which we do below, see Corollary 5.17.

Example 5.8. We give one more example from commutative algebra. Recall that $(R, \mathfrak{m}, k)$ is a complete intersection if its completion is a regular local ring quotiented by an ideal generated by a regular sequence. Write $\mathsf{D}^f(R)$ for the full subcategory of $\mathsf{D}(R)$ on those complexes whose homology is bounded and degreewise finitely generated. It is a recent theorem of Pollitz (building on work of Dwyer–Greenlees–Iyengar) that a commutative local Noetherian ring $(R, \mathfrak{m}, k)$ is a complete intersection if and only if every object of $\mathsf{D}^f(R)$ is proxy-small.

We leave various closure properties of proxy-small objects as an exercise, see Exercise A.12.

Let us also give a non-example.

Example 5.9. Let $R = k[x_1, x_2, \ldots]$ be the infinite polynomial ring. Then the residue field k is not proxy-small in $\mathsf{D}(R)$. Suppose that k is proxy-small with witness $\mathcal{W}$ and argue by contradiction. Recall that $Dk \simeq 0$ in $\mathsf{D}(R)$. Since $\mathcal{W} \subseteq \mathrm{thick}_{\otimes}(k)$ it follows that $D\mathcal{W} \subseteq \mathrm{thick}_{\otimes}(Dk)$ and hence $DW = 0$ for all $W \in \mathcal{W}$. Since $\mathcal{W}$ is a set of compacts (hence rigids), we have $W \simeq D^2 W$ (see Lemma 2.20), and hence $W = 0$ too. Note that this argument applies more generally in any rigidly-compactly generated tt-category and to any object k with $Dk \simeq 0$. For example, it also applies to show that $\mathbb{F}_p$ is not proxy-small in the stable homotopy category of spectra.

5.B. **Proxy-smallness of residue fields.** We now return to Example 5.7 and justify it. In order to do this, we need to make some recollections on homotopy colimits:

Definition 5.10. Let T be a triangulated category with coproducts, and let $X_0 \xrightarrow{f_0} X_1 \xrightarrow{f_1} X_2 \to \cdots$ be a system of maps in T. The *homotopy colimit* of this system, denoted $\mathrm{hocolim} X_i$, is the cone of the map

$$\bigoplus X_i \xrightarrow{1-f} \bigoplus X_i$$

which on the nth component is given by

$$X_n \xrightarrow{x \mapsto (x, -f_n(x))} X_n \oplus X_{n+1} \xrightarrow{\mathrm{inc}} \bigoplus X_i.$$

Lemma 5.11. *Let T be a triangulated category with coproducts, and let $X_0 \xrightarrow{f_0} X_1 \xrightarrow{f_1} X_2 \to \cdots$ be a system of maps in T. Let $Y \in \mathsf{T}$, and suppose that there exists maps $g_n \colon X_n \to Y$ such that $g_n = g_{n+1} f_n$ for all n. Then there exists a map $\overline{g} \colon \operatorname{hocolim} X_i \to Y$ such that $g = \overline{g} q$ where q is the induced map $q \colon \bigoplus X_i \to \operatorname{hocolim} X_i$.*

Proof. By universal property of the coproduct, we obtain a map $g \colon \bigoplus X_i \to Y$ by assembling the g_i. Note that $g(1-f) = 0$ since on the nth component we have $g(1-f)(x) = g(x_n, -f_n(x_n)) = g_n(x_n) - g_{n+1} f_n(x_n) = 0$. So consider the diagram

$$\begin{tikzcd}
\bigoplus X_i \arrow[r, "1-f"] \arrow[d] & \bigoplus X_i \arrow[r, "q"] \arrow[d, "g"] & \operatorname{hocolim} X_i \arrow[r] & \Sigma \bigoplus X_i \arrow[d] \\
0 \arrow[r] & Y \arrow[r, "\mathrm{id}"'] & Y \arrow[r] & 0.
\end{tikzcd}$$

The top row is a triangle by definition of the homotopy colimit, and the bottom row is a triangle by (TR0). The left hand square commutes since $g(1-f) = 0$, so by (TR3) there exists a map $\overline{g} \colon \operatorname{hocolim} X_i \to Y$ making the diagram commute. □

Lemma 5.12. *Let T be a triangulated category with coproducts. Let C be a compact object and let $X_0 \xrightarrow{f_0} X_1 \xrightarrow{f_1} X_2 \to \cdots$ be a system in T. Then the canonical map*

$$\varinjlim \operatorname{Hom}_{\mathsf{T}}(C, X_i) \to \operatorname{Hom}_{\mathsf{T}}(C, \operatorname{hocolim} X_i)$$

is an isomorphism.

Proof. By definition of the homotopy colimit and since $\operatorname{Hom}(C, -)$ is a homological functor we obtain a long exact sequence,

$$\cdots \to \operatorname{Hom}_{\mathsf{T}}(C, \bigoplus X_i) \to \operatorname{Hom}_{\mathsf{T}}(C, \bigoplus X_i) \to \operatorname{Hom}_{\mathsf{T}}(C, \operatorname{hocolim} X_i) \to \operatorname{Hom}_{\mathsf{T}}(C, \bigoplus \Sigma X_i) \to \cdots .$$

We therefore have a commutative diagram

$$\begin{tikzcd}
0 \arrow[r] & \bigoplus \operatorname{Hom}_{\mathsf{T}}(C, X_i) \arrow[r] \arrow[d] & \bigoplus \operatorname{Hom}_{\mathsf{T}}(C, X_i) \arrow[r] \arrow[d] & \varinjlim \operatorname{Hom}_{\mathsf{T}}(C, X_i) \arrow[r] \arrow[d] & 0 \\
\cdots \arrow[r] & \operatorname{Hom}_{\mathsf{T}}(C, \bigoplus X_i) \arrow[r] & \operatorname{Hom}_{\mathsf{T}}(C, \bigoplus X_i) \arrow[r] & \operatorname{Hom}_{\mathsf{T}}(C, \operatorname{hocolim} X_i) \arrow[r] & \cdots
\end{tikzcd}$$

The first two columns are isomorphisms since C is compact, and so we see that the map $\operatorname{Hom}_{\mathsf{T}}(C, \bigoplus X_i) \to \operatorname{Hom}_{\mathsf{T}}(C, \bigoplus X_i)$ is injective. Similarly, we may apply the same argument to ΣX_i to deduce that $\operatorname{Hom}_{\mathsf{T}}(C, \bigoplus \Sigma X_i) \to \operatorname{Hom}_{\mathsf{T}}(C, \bigoplus \Sigma X_i)$ is also injective. Therefore the long exact sequence collapses into short exact sequences. We then have a commutative diagram

$$\begin{tikzcd}
0 \arrow[r] & \bigoplus \operatorname{Hom}_{\mathsf{T}}(C, X_i) \arrow[r] \arrow[d] & \bigoplus \operatorname{Hom}_{\mathsf{T}}(C, X_i) \arrow[r] \arrow[d] & \varinjlim \operatorname{Hom}_{\mathsf{T}}(C, X_i) \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & \operatorname{Hom}_{\mathsf{T}}(C, \bigoplus X_i) \arrow[r] & \operatorname{Hom}_{\mathsf{T}}(C, \bigoplus X_i) \arrow[r] & \operatorname{Hom}_{\mathsf{T}}(C, \operatorname{hocolim} X_i) \arrow[r] & 0
\end{tikzcd}$$

in which the first two columns are isomorphisms as C is compact. The result then follows. □

Corollary 5.13. *Let T be a compactly generated triangulated category. Let $X_0 \xrightarrow{f_0} X_1 \xrightarrow{f_1} X_2 \to \cdots$ be a system in T, and let $Y \in \mathsf{T}$ such that there exists maps $g_n \colon X_n \to Y$ with $g_n = g_{n+1} f_n$ for all n. If the canonical map*

$$\varinjlim \operatorname{Hom}_{\mathsf{T}}(C, X_i) \to \operatorname{Hom}_{\mathsf{T}}(C, Y)$$

is an isomorphism for all $C \in \mathsf{T}^{\mathrm{c}}$, then $Y \simeq \operatorname{hocolim} X_i$.

Proof. By Lemma 5.11 we obtain a map $\overline{g}\colon \operatorname{hocolim} X_i \to Y$. Since T is compactly generated, to prove this is an isomorphism it suffices to check that the induced map

$$\operatorname{Hom}_{\mathsf{T}}(C, \operatorname{hocolim} X_i) \to \operatorname{Hom}_{\mathsf{T}}(C, Y)$$

is an isomorphism for all $C \in \mathsf{T}^{\mathrm{c}}$. This is immediate from Lemma 5.12. □

We now have the necessary background on homotopy colimits to return to the example of commutative Noetherian rings.

Definition 5.14. Let R be a commutative ring and $I = (x_1, \cdots, x_n)$ be a finitely generated ideal.

(1) The *unstable Koszul complex* $K(I)$ is defined by

$$K(I) = \bigotimes_{i=1}^{n} (R \xrightarrow{\cdot x} R)$$

where the complex $R \xrightarrow{\cdot x} R$ is in degrees 0 and -1. Note that by definition, this is a compact object in $\mathsf{D}(R)$.

(2) The *stable Koszul complex* $K_\infty(I)$ is defined by

$$K_\infty(I) = \bigotimes_{i=1}^{n} (R \to R[1/x_i])$$

where the complex $R \to R[1/x]$ is in degrees 0 and -1.

For an element $x \in R$, we note that we have a map $K(x^i) \to K(x^{i+1})$ for all i, represented by the commutative diagram

$$\begin{tikzcd} R \arrow[r, "\cdot x^i"] \arrow[d, "1"'] & R \arrow[d, "\cdot x"] \\ R \arrow[r, "\cdot x^{i+1}"'] & R \end{tikzcd}$$

Therefore we have a tower $\cdots \to K(x^i) \to K(x^{i+1}) \to \cdots$ and we may take the homotopy colimit of this.

Lemma 5.15. *Let $x \in R$. Then* $\operatorname{hocolim}(K(x^i)) \simeq K_\infty(x)$.

Proof. By Corollary 5.13 it suffices to show that we have compatible maps $K(x^i) \to K_\infty(x)$ for all i, so that the induced map

$$\varinjlim \operatorname{Hom}_{\mathsf{D}(R)}(\Sigma^j R, K(x^i)) \to \operatorname{Hom}_{\mathsf{D}(R)}(\Sigma^j R, K_\infty(x))$$

is an isomorphism for all j. The existence of compatible maps is clear from the commutative diagram

$$\begin{tikzcd} R \arrow[r, "\cdot x^i"] \arrow[d] & R \arrow[d] \\ R \arrow[r] & R[1/x]. \end{tikzcd}$$

Recall that $\operatorname{Hom}_{\mathsf{D}(R)}(\Sigma^j R, -) = H_j(-)$, so it suffices to treat the cases $j = 0$ and $j = -1$. For $j = -1$, $H_{-1}(K(x^i)) = R/x^i$ and $H_{-1}(K_\infty(x)) = R[1/x]/R = R/x^\infty$. Therefore we see that the claim holds for $j = -1$. For $j = 0$, $H_0(K(x^i)) = \{m \mid x^i m = 0\}$ and $H_0(K_\infty(x)) = \{m \mid x^i m = 0 \text{ for some } i\}$. As such we see that $\varinjlim H_0(K(x^i)) = H_0(K_\infty(x))$ which completes the proof of the claim. □

Theorem 5.16. *Let R be a commutative ring and $I = (x_1, \cdots, x_n)$ be a finitely generated ideal. Then*

$$\mathrm{Loc}(R/I) = \mathrm{Loc}(K(I)) = \mathrm{Loc}(K_\infty(I)).$$

If R/I is moreover a regular ring, then $K(I) \in \mathrm{thick}(R/I)$.

Proof. We will prove that (1) R/I builds $K(I)$, (2) $K(I)$ builds $K_\infty(I)$, and (3) $K_\infty(I)$ builds R/I. Note that this suffices since it implies that

$$\mathrm{Loc}(R/I) \subseteq \mathrm{Loc}(K_\infty(I)) \subseteq \mathrm{Loc}(K(I)) \subseteq \mathrm{Loc}(R/I).$$

(1) For each $i \in \mathbb{Z}$ there is a triangle

$$H_i(K(I)) \to \tau_{\leq i} K(I) \to \tau_{\leq i-1} K(I)$$

where $\tau_{\leq i}$ denotes the smart truncation (i.e., $(\tau_{\leq i} X)_j = X_j$ for $j < i$, is 0 for $j > i$ and is $\mathrm{coker}(d_{j+1})$ for $j = i$). We now claim that the homology groups of $K(I)$ are modules over R/I. To see this, note that multiplication by x is null homotopic as a map $K(x) \to K(x)$. As such, for any $x \in I$, $x\colon K(I) \to K(I)$ is null homotopic. Therefore x acts as 0 in homology, and as such the homology groups of $K(I)$ are all R/I-modules. Therefore $H_i(K(I))$ may be built from R/I in $\mathsf{D}(R/I)$. The restriction of scalars functor $\mathsf{D}(R/I) \to \mathsf{D}(R)$ is triangulated and preserves coproducts, and hence R/I also builds $H_i(K(I))$ in $\mathsf{D}(R)$. Now we argue by induction that $\tau_{\leq i} K(I)$ is built from R/I, for each $-n \leq i \leq 0$. We start with $i = -n$ as the base case. When $i = -n$, $\tau_{\leq i-1} K(I) \simeq 0$ so the base case holds by the above triangle. Then inducting, we obtain that $\tau_{\leq i} K(I)$ is built from R/I for each $-n \leq i \leq 0$. When $i = 0$, $\tau_{\leq i} K(I) = K(I)$ so this gives the claim as desired. If R/I is moreover regular, then note that $H_i(K(I))$ is finitely built from R/I in $\mathsf{D}(R/I)$, so the same proof then shows that R/I finitely builds $K(I)$ in $\mathsf{D}(R)$.

(2) One can check that there is a triangle $K(x^{i-1}) \to K(x^i) \to K(x)$, and so by induction we have that $K(x^i) \in \mathrm{thick}(K(x))$ for all i. Since $K_\infty(x)$ is the homotopy colimit of the $K(x^i)$ by Lemma 5.15, it follows that $K_\infty(x)$ lies in the localizing subcategory generated by $K(x)$. The general case follows from this by tensoring.

(3) The stable Koszul complex $K_\infty(I)$ is a dg-flat complex, and hence $R/I \otimes^{\mathsf{L}}_R K_\infty(I) = R/I \otimes_R K_\infty(I)$. As $R/I \otimes_R R[1/x_i] \simeq 0$, one sees that $R/I \otimes^{\mathsf{L}}_R K_\infty(I) \simeq R/I$. As R builds any M in $\mathsf{D}(R)$, $K_\infty(I) \simeq K_\infty(I) \otimes^{\mathsf{L}}_R R$ builds $K_\infty(I) \otimes^{\mathsf{L}}_R M$ for all $M \in \mathsf{D}(R)$. Therefore $K_\infty(I)$ builds $R/I \otimes^{\mathsf{L}}_R K_\infty(I) \simeq R/I$ as required.

Therefore we have the above chain of inclusions of localising subcategories, and hence they are all equal. □

Corollary 5.17. *Let $(R, \mathfrak{m}, k)$ be a commutative Noetherian local ring. Then k is proxy-small in $\mathsf{D}(R)$.*

Proof. Applying Theorem 5.16 to $I = \mathfrak{m}$ we see that $K(\mathfrak{m}) \in \mathrm{thick}(k)$ and $k \in \mathrm{Loc}(K(\mathfrak{m}))$. Since $K(\mathfrak{m})$ is also compact by definition it follows that k is proxy-small in $\mathsf{D}(R)$ with witness $K(\mathfrak{m})$. □

6. Torsion and completion

6.A. **Localisation theory.** In this subsection we set up the basic theory of localisation in triangulated and tt-categories. We start off in just the triangulated setting.

Definition 6.1. A triangulated functor $L\colon \mathsf{T} \to \mathcal{S}$ is a *localization* if it has a fully faithful right adjoint. Note then that $\mathcal{S} \simeq L\mathsf{T}$, the essential image of L. Frequently we write L for the composite of L followed by the inclusion. A *colocalization* is the dual concept.

Remark 6.2. Some authors might call the above a *Bousfield localization* to distinguish it from other types of localizations. In this course, we will only see Bousfield localizations.

Remark 6.3. An alternative more explicit definition of a localization is as follows. A localization is an exact functor $L\colon \mathsf{T} \to \mathsf{T}$ together with a natural transformation $\eta\colon \mathrm{id} \to L$ such that $L\eta\colon L \to L^2$ is an isomorphism, and $L\eta = \eta L$. Note the difference between the codomains in the definitions.

Definition 6.4. Let L be a localization on T. An object $X \in \mathsf{T}$ is said to be L-local if the unit map $X \to LX$ is an isomorphism, and is said to be L-acyclic if $LX \simeq 0$.

With the definitions set, we may now turn to proving various properties of (co)localizations. Firstly, we show that localizations and colocalizations always come in pairs.

Proposition 6.5. *Let T be a triangulated category. There is a one-to-one correspondence between localizations of T and colocalizations of T, in which L and Γ correspond if and only if $\Gamma X \to X \to LX \to \Sigma\Gamma X$ is a triangle for all $X \in \mathsf{T}$.*

Proof. Suppose that L is a localization of T. By (TR1) we may extend the unit map $X \to LX$ to give a triangle

$$\Gamma X \to X \to LX \to \Sigma\Gamma X.$$

By applying the triangulated functor L and using that $L \xrightarrow{\sim} L^2$ is an isomorphism we see that $L\Gamma X \simeq 0$, so ΓX is L-acyclic. Conversely, suppose that Z is L-acyclic. Then $\mathrm{Hom}_{\mathsf{T}}(Z, LX) = 0$ since $\mathrm{Hom}_{\mathsf{T}}(Z, LX) = \mathrm{Hom}_{\mathsf{T}}(LZ, LX) = 0$ by adjunction. Therefore, applying $\mathrm{Hom}_{\mathsf{T}}(Z, -)$ to the above triangle, we see that $\mathrm{Hom}_{\mathsf{T}}(Z, \Gamma X) = \mathrm{Hom}_{\mathsf{T}}(Z, X)$. Therefore ΓX has a universal property: if Z is L-acyclic and there is a map $f\colon Z \to X$, then f factors as $Z \to \Gamma X \to X$.

Normally applying (TR3) to construct maps is not functorial, but the above universal property of ΓX allows us to make Γ functorial as we now describe. Let $f\colon A \to X$ be a morphism in T. By (TR3) we obtain a map $\Gamma(f)$ in the commutative diagram

$$\begin{tikzcd}
\Sigma^{-1}LA \arrow[r] \arrow[d] & \Gamma A \arrow[r] \arrow[d, "\Gamma(f)"] & A \arrow[r] \arrow[d] & LA \arrow[d] \\
\Sigma^{-1}LX \arrow[r] & \Gamma X \arrow[r] & X \arrow[r] & LX
\end{tikzcd}$$

So to see that Γ is functorial, it suffices to show that $\Gamma(f)$ is unique which is immediate from the universal property. Therefore, given a localization L, we have constructed a functor $\Gamma\colon \mathsf{T} \to \mathsf{T}$.

It remains to prove that Γ is triangulated and is a colocalization. Firstly, we consider the diagram

$$\begin{tikzcd}
LX \arrow[r] \arrow[d, "\sim"] & \Sigma\Gamma X \arrow[r] \arrow[d, dashed] & \Sigma X \arrow[r] \arrow[d, "1"] & \Sigma LX \arrow[d, "\sim"] \\
\Sigma^{-1}L\Sigma X \arrow[r] & \Gamma\Sigma X \arrow[r] & \Sigma X \arrow[r] & L\Sigma X
\end{tikzcd}$$

in which the top row is the triangle obtained by applying Σ to the standard triangle, and the bottom row is the standard triangle on ΣX. Using (TR3) and the 5-lemma, we deduce that $\Gamma\Sigma \simeq \Sigma\Gamma$. We use this isomorphism implicitly in what follows.

Consider a triangle $X \xrightarrow{f} Y \xrightarrow{g} Z \xrightarrow{h} \Sigma X$. We may complete the map Γf to a triangle by (TR1). Consider the diagram

$$\begin{tikzcd}
\Gamma X \arrow[r, "\Gamma f"] \arrow[d, "\epsilon_X"] & \Gamma Y \arrow[r, "\alpha"] \arrow[d, "\epsilon_Y"] & C \arrow[r, "\beta"] \arrow[d, dashed, "\epsilon"] & \Sigma\Gamma X \arrow[d, "\Sigma\epsilon_X"] \\
X \arrow[r, "f"'] & Y \arrow[r, "g"'] & Z \arrow[r, "h"'] & \Sigma X
\end{tikzcd}$$

in which the dashed map exists by (TR3). Suppose that W is L-acyclic. By applying $\mathrm{Hom}_{\mathsf{T}}(W,-)$ and the 5-lemma, we see that $\mathrm{Hom}_{\mathsf{T}}(W,C) \to \mathrm{Hom}_{\mathsf{T}}(W,Z)$ is an isomorphism. Therefore C is terminal amongst L-acyclic objects mapping to Z. We proved above that ΓZ admits the same universal property, and hence $C \simeq \Gamma Z$. As in the proof of functoriality above, one then identifies α and β with Γg and Γh. Therefore Γ is triangulated as required.

It remains to prove that Γ is a colocalization. Consider the inclusion of the L-acyclic objects into T. We have seen that for any L-acyclic Z, we have that $\mathrm{Hom}_{\mathsf{T}}(Z,\Gamma X) \to \mathrm{Hom}_{\mathsf{T}}(Z,X)$ is an isomorphism. Therefore Γ is right adjoint to this inclusion functor. To see that the inclusion is fully faithful, we just need to see that the counit is an isomorphism, i.e., that the map $\epsilon\colon \Gamma Z \to Z$ is an isomorphism if Z is L-acyclic. This is immediate by the standard triangle. □

Next, we give various equivalent characterisations of local objects. We leave the dual colocal version to the reader. We use the following notation: for any subcategory $\mathcal{X}$,

$$\mathcal{X}^{\perp} = \{Y \in \mathsf{T} \mid \mathrm{Hom}_{\mathsf{T}}(X,Y) = 0 \text{ for all } X \in \mathcal{X}\}.$$

Proposition 6.6. *Let T be a triangulated category and L be a localization. Write Γ for corresponding colocalization as in Proposition 6.5. The following are equivalent for an object $X \in \mathsf{T}$:*

(1) X is L-local;
(2) $X \simeq LY$ for some $Y \in \mathsf{T}$;
(3) $\mathrm{Hom}_{\mathsf{T}}(Z,X) \simeq 0$ for all L-acyclic objects Z;
(4) X is Γ-acyclic.

In particular, $\mathrm{im}(L) = \ker(L)^{\perp}$.

Proof. The implication (1) ⇒ (2) is trivial. For (2) ⇒ (3) we have $\mathrm{Hom}_{\mathsf{T}}(Z,X) = \mathrm{Hom}_{\mathsf{T}}(Z,LY) = \mathrm{Hom}_{\mathsf{T}}(LZ,LY)$ since L is a localization. Now this is zero for all L-acyclic Z as required. For (3) ⇒ (4), as $L\Gamma \simeq 0$ as proved in Proposition 6.5, the map $\epsilon\colon \Gamma X \to X$ is zero by assumption, and hence $\Gamma\epsilon = 0$. However, $\Gamma\epsilon\colon \Gamma^2 X \to \Gamma X$ is an isomorphism. Therefore $\Gamma X \simeq 0$ as required. Finally, the implication (4) ⇒ (1) follows immediately from the defining triangle $\Gamma X \to X \to LX \to \Sigma\Gamma X$. □

Corollary 6.7. *Let L be a localization. Then the full subcategory $\ker(L)$ of L-acyclics is localizing, and the full subcategory $\mathrm{im}(L)$ of L-local objects is colocalizing.*

Proof. Since L is a triangulated functor, $\ker(L)$ is thick. Now $X \in \ker(L)$ if and only if $\mathrm{Hom}_{\mathsf{T}}(X,Y) = 0$ for all L-local objects Y by the dual form of Proposition 6.6. Therefore $\ker(L)$ is closed under coproducts, and hence is localizing. The argument that $\mathrm{im}(L)$ is colocalizing is similar. □

We now turn to consider localization functors on tensor-triangulated categories. In order to ensure good behaviour with the tensor product, it is necessary to refine the notion of localization slightly.

Definition 6.8. A localization functor $L\colon \mathsf{T} \to \mathsf{T}$ is said to be *monoidal* if $LX \simeq 0$ implies that $L(X \otimes Y) \simeq 0$ for all $Y \in \mathsf{T}$. A colocalization functor $\Gamma\colon \mathsf{T} \to \mathsf{T}$ is *monoidal* if $\Gamma X \simeq 0$ implies that $\Gamma\underline{\mathrm{Hom}}(Y,X) \simeq 0$ for all $Y \in \mathsf{T}$.

Remark 6.9. Notice that if Γ is monoidal, then $\mathrm{im}(\Gamma)$ is a localising $\otimes$-ideal. If $X \in \mathrm{im}(\Gamma) = \ker(L)$ and $Y \in \mathsf{T}$, then $L(X \otimes Y) \simeq 0$ since L is monoidal (see Exercise A.13), so $X \otimes Y \in \mathrm{im}(\Gamma)$ as claimed. We leave it to the reader to state and prove the analogous statement for L.

Lemma 6.10. *Let $L\colon \mathsf{T} \to \mathsf{T}$ be a monoidal localization. There is a natural map $\alpha_X\colon L\mathbb{1} \otimes X \to LX$ for all $X \in \mathsf{T}$.*

Proof. Applying L to the defining triangle we have a triangle

$$L(\Gamma\mathbb{1} \otimes X) \to LX \to L(L\mathbb{1} \otimes X) \to \Sigma L(\Gamma\mathbb{1} \otimes X).$$

The first entry is 0 since L is monoidal, and hence $LX \to L(L\mathbb{1} \otimes X)$ is an isomorphism. Therefore we have a natural map $L\mathbb{1} \otimes X \to L(L\mathbb{1} \otimes X) \simeq LX$ as required. □

Proposition 6.11. *Let T be a tensor-triangulated category, and let L be a monoidal localization. Write $\mathcal{S}$ for the full subcategory of L-local objects. The following are equivalent:*

(i) The natural map $\alpha_X \colon L\mathbb{1} \otimes X \to LX$ from Lemma 6.10 is an equivalence for all $X \in \mathsf{T}$.
(ii) $i \colon \mathcal{S} \to \mathsf{T}$ preserves coproducts (equivalently, $L \colon \mathsf{T} \to \mathsf{T}$ preserves coproducts).
(iii) $\mathcal{S}$ is a localizing subcategory of T.

If any of these equivalent conditions hold, we say that L is smashing.

Proof. This is left as Exercise A.15. □

Proposition 6.12. *Let $L \colon \mathsf{T} \to \mathsf{T}$ be a monoidal localisation functor. Then the category $L\mathsf{T}$ of L-local objects is a tensor-triangulated category with:*

- *triangles are those $X \to Y \to Z$ which are triangles in T;*
- *monoidal product $X \otimes_L Y \coloneqq L(X \otimes Y)$;*
- *tensor unit $\mathbb{1}_{L\mathsf{T}} = L\mathbb{1}_{\mathsf{T}}$;*
- *internal hom $\underline{\mathrm{Hom}}_L(X, Y) \coloneqq \underline{\mathrm{Hom}}(X, Y)$.*

The localisation functor $L \colon \mathsf{T} \to L\mathsf{T}$ is then a coproduct preserving tt-functor.

Proof. Since $L\mathsf{T}$ is a thick subcategory of T, the above makes $L\mathsf{T}$ into a triangulated category. It is clear that the monoidal product and tensor unit defined above are L-local. Notice that the internal hom is automatically L-local: by Proposition 6.6 it suffices to show that $\mathrm{Hom}(Z, \underline{\mathrm{Hom}}(X, Y)) = 0$ if $LZ \simeq 0$. By adjunction we have

$$\mathrm{Hom}(Z, \underline{\mathrm{Hom}}(X, Y)) \simeq \mathrm{Hom}(Z \otimes X, Y) \simeq 0$$

where the final isomorphism holds by definition of a monoidal localisation. It is then a straightforward check that the tensor product and internal hom are adjoint triangulated functors. □

Remark 6.13. Notice that if L is a smashing localisation then it is not necessary to apply L to obtain the correct tensor product since $L(X \otimes Y) = L\mathbb{1} \otimes X \otimes Y = X \otimes Y$ since X (or/and Y) is L-local.

Remark 6.14. Notice that if T is rigidly-compactly generated, then $L\mathsf{T}$ need not be rigidly-compactly generated. It is left as an exercise (see Exercise A.16) to prove that $L\mathsf{T}$ is rigidly-compactly generated if and only if L is smashing.

We now have the required background to prove the existence of (co)localisation functors based on sets of compact objects.

Theorem 6.15. *Let T be a tt-category which is compactly generated by rigid objects and let $\mathcal{K} \subseteq \mathsf{T}^c$. Define*

$$\mathsf{T}^{\mathcal{K}-\mathrm{tors}} \coloneqq \mathrm{Loc}_{\otimes}(\mathcal{K}), \qquad \mathsf{T}^{\mathcal{K}-\mathrm{loc}} \coloneqq (\mathsf{T}^{\mathcal{K}-\mathrm{tors}})^{\perp}, \qquad \mathsf{T}^{\mathcal{K}-\mathrm{comp}} \coloneqq (\mathsf{T}^{\mathcal{K}-\mathrm{loc}})^{\perp}.$$

There are monoidal localization and colocalization functors:

$$\Gamma \colon \mathsf{T} \to \mathsf{T}^{\mathcal{K}\text{-tors}}, \quad L \colon \mathsf{T} \to \mathsf{T}^{\mathcal{K}\text{-loc}}, \quad V \colon \mathsf{T} \to \mathsf{T}^{\mathcal{K}\text{-loc}}, \quad \Lambda \colon \mathsf{T} \to \mathsf{T}^{\mathcal{K}\text{-comp}}$$

fitting into triangles $\Gamma X \to X \to LX$ and $VX \to X \to \Lambda X$. These satisfy the following properties:

(1) We have $\ker(L) = \mathrm{im}(\Gamma) = \mathsf{T}^{\mathcal{K}\text{-tors}}$;

(2) We have $\ker(\Gamma) = \operatorname{im}(L) = \operatorname{im}(V) = \ker(\Lambda) = \mathsf{T}^{\mathcal{K}\text{-loc}}$*;*
(3) We have $\ker(V) = \operatorname{im}(\Lambda) = \mathsf{T}^{\mathcal{K}\text{-comp}}$*.*
(4) for any $X \in \mathsf{T}$*,* $LX \simeq L\mathbb{1} \otimes X$ *and* $\Gamma X \simeq \Gamma\mathbb{1} \otimes X$*;*
(5) for all $X \in \mathsf{T}$*, the natural maps* $\Gamma X \to \Gamma\Lambda X$ *and* $\Lambda\Gamma X \to \Lambda X$ *are isomorphisms;*
(6) the functors Γ *and* Λ *give an equivalence of categories* $\mathsf{T}^{\mathcal{K}\text{-tors}} \simeq \mathsf{T}^{\mathcal{K}\text{-comp}}$*.*

Proof. Writing $\mathcal{G}$ for a set of compact-rigid generators for T, since $\operatorname{Loc}_{\otimes}(\mathcal{K}) = \operatorname{Loc}(\mathcal{K} \otimes \mathcal{G})$, the category $\mathsf{T}^{\mathcal{K}-\text{tors}}$ is compactly generated. Therefore by Theorem 3.2, the inclusion $\mathsf{T}^{\mathcal{K}-\text{tors}} \hookrightarrow \mathsf{T}$ has a right adjoint Γ. This has a corresponding localisation functor L. We define $V = \underline{\operatorname{Hom}}(L\mathbb{1}, -)$ and $\Lambda = \underline{\operatorname{Hom}}(\Gamma\mathbb{1}, -)$. These are triangulated functors. We will verify that these are (co)localizations later on.

Proof of (1): By construction, $\operatorname{im}(\Gamma) = \mathsf{T}^{\mathcal{K}-\text{tors}}$. Note that $\ker(L)$ is localizing by Corollary 6.7 and contains $\mathcal{K} \otimes \mathcal{G}$. Therefore $\mathsf{Loc}^{\otimes}(\mathcal{K}) \subseteq \ker(L)$. Conversely, if $LX \simeq 0$, then $X \simeq \Gamma X \in \mathsf{Loc}^{\otimes}(\mathcal{K})$.

Proof of (4): Note that $\Gamma X \simeq 0$ if and only if $\operatorname{Hom}_{\mathsf{T}}(Z, X) = 0$ for all $Z \in \operatorname{Loc}_{\otimes}(\mathcal{K})$ as $\operatorname{Hom}_{\mathsf{T}}(Z, X) = \operatorname{Hom}_{\mathsf{T}}(Z, \Gamma X)$. So if $\Gamma X \simeq 0$, then $\operatorname{Hom}_{\mathsf{T}}(Z, \underline{\operatorname{Hom}}(Y, X)) = \operatorname{Hom}_{\mathsf{T}}(Y, \underline{\operatorname{Hom}}(Z, X)) = 0$ and hence $\Gamma\underline{\operatorname{Hom}}(Y, X) \simeq 0$ for all $Y \in \mathsf{T}$ so that Γ is monoidal. By Exercise A.13, it follows that L is monoidal. Now X is L-local if and only if $\operatorname{Hom}_{\mathsf{T}}(Z, X) = 0$ for all $Z \in \operatorname{thick}_{\otimes}(\mathcal{K})$. Since any such Z is compact, we see that the category of L-local objects is closed under coproducts, and hence L is smashing by Proposition 6.11. By the 5 lemma, it follows that $\Gamma X \simeq \Gamma\mathbb{1} \otimes X$ for all $X \in \mathsf{T}$.

Proof of (2): Notice that (4) shows that V and Λ are idempotent functors. We have $\operatorname{im}(L) = \ker(L)^{\perp} = \ker(\Gamma) = \mathsf{T}^{\mathcal{K}\text{-loc}}$ by Proposition 6.6 together with (1). By the idempotence of V together with the associated triangle $VX \to X \to \Lambda X$, we also see that $\ker(\Lambda) = \operatorname{im}(V)$. So it remains to prove that $\ker(\Lambda) = \mathsf{T}^{\mathcal{K}\text{-loc}}$.

If $X \in \mathsf{T}^{\mathcal{K}\text{-loc}}$ then $\operatorname{Hom}(\Gamma\mathbb{1} \otimes G, X) \simeq 0$ for all $G \in \mathcal{G}$ by (1). Therefore $\Lambda\mathbb{1} \simeq \underline{\operatorname{Hom}}(\Gamma\mathbb{1}, X) \simeq 0$. Conversely, suppose that $\Lambda X \simeq 0$ and $Y \in \mathsf{Loc}^{\otimes}(\mathcal{K})$. Then $\operatorname{Hom}_{\mathsf{T}}(Y, X) = \operatorname{Hom}_{\mathsf{T}}(\Gamma Y, X)$ by (1). Since Γ is smashing, this is equal to $\operatorname{Hom}_{\mathsf{T}}(\Gamma\mathbb{1} \otimes Y, X) \simeq \operatorname{Hom}_{\mathsf{T}}(Y, \Lambda X) \simeq 0$.

Proof of (3): This follows the same strategy as the second paragraph of (2).

Proof of (5): We have $\Lambda LX \simeq 0$ for all X by (2). Therefore by applying Λ to the triangle $\Gamma X \to X \to LX$ we conclude that $\Lambda\Gamma X \to \Lambda X$ is an isomorphism. The other isomorphism is similar.

Proof of (6): This is immediate from (1), (3), and (5).

Finally we argue that Λ is a localization functor. To do this, it suffices to show that the canonical map $\operatorname{Hom}_{\mathsf{T}}(\Lambda X, \Lambda Y) \to \operatorname{Hom}_{\mathsf{T}}(X, \Lambda Y)$ is an equivalence. This is equivalent to $\operatorname{Hom}_{\mathsf{T}}(VX, \Lambda Y) \simeq 0$. Now by (2), $VX \in \mathsf{T}^{\mathcal{K}\text{-loc}}$, and by (3) $\Lambda X \in \mathsf{T}^{\mathcal{K}\text{-comp}}$ so the claim follows. Finally, we need that show that Λ is moreover a monoidal localization. To see this, if $\Lambda X \simeq 0$, then $\Gamma X \simeq 0$ by (5). Again by (5) together with (4), $\Lambda(X \otimes Y) \simeq \Lambda\Gamma(X \otimes Y) \simeq \Lambda(\Gamma X \otimes Y) \simeq 0$ as required. □

Remark 6.16. If $\mathcal{K}$ is proxy-small in T, then the previous theorem applies too, since we have a set of compact objects $\mathcal{W}$ such that $\operatorname{Loc}_{\otimes}(\mathcal{K}) = \operatorname{Loc}_{\otimes}(\mathcal{W})$.

Remark 6.17. If confusion may occur, we record the subscript of the set of compact objects on the associated functors, e.g., $\Gamma_{\mathcal{K}}$.

Corollary 6.18 (Greenlees-May duality)**.** *Let* T *be a tt-category which is compactly generated by rigid objects and let* $\mathcal{K} \subseteq \mathsf{T}^c$*. Then*

$$\Gamma\colon \mathsf{T} \rightleftarrows \mathsf{T}\colon \Lambda$$

form an adjoint pair which internalises, i.e., for all $X, Y \in \mathsf{T}$*,*

$$\underline{\operatorname{Hom}}(\Gamma X, Y) \simeq \underline{\operatorname{Hom}}(X, \Lambda Y).$$

Proof. By definition of Λ as in the proof of Theorem 6.15 we have $\Lambda = \underline{\mathrm{Hom}}(\Gamma\mathbb{1}, -)$. Therefore,

$$\underline{\mathrm{Hom}}(\Gamma X, Y) \simeq \underline{\mathrm{Hom}}(\Gamma\mathbb{1} \otimes X, Y) \simeq \underline{\mathrm{Hom}}(X, \underline{\mathrm{Hom}}(\Gamma\mathbb{1}, Y)) \simeq \underline{\mathrm{Hom}}(X, \Lambda Y)$$

since Γ is smashing. Applying $\mathrm{Hom}_{\mathsf{T}}(\mathbb{1}, -)$ to the internalised adjunction yields the external form of the adjunction $\mathrm{Hom}_{\mathsf{T}}(\Gamma X, Y) \simeq \mathrm{Hom}_{\mathsf{T}}(X, \Lambda Y)$. □

Remark 6.19. By Proposition 6.12, the category $\mathsf{T}^{\mathcal{K}\text{-comp}}$ is a tt-category with tensor product given by $X \widehat{\otimes} Y := \Lambda(X \otimes Y)$, tensor unit $\Lambda\mathbb{1}_{\mathsf{T}}$ and internal hom the restriction of the internal hom in T. Using the MGM equivalence, this induces a tt-structure on $\mathsf{T}^{\mathcal{K}\text{-tors}}$ with tensor product given by the underlying tensor product in T, tensor unit $\Gamma\mathbb{1}_{\mathsf{T}}$ and internal hom $\Gamma\underline{\mathrm{Hom}}(-,-)$. The functors $\Gamma\colon \mathsf{T} \to \mathsf{T}^{\mathcal{K}\text{-tors}}$ and $\Lambda\colon \mathsf{T} \to \mathsf{T}^{\mathcal{K}\text{-comp}}$ are both strong symmetric monoidal functors. Moreover the equivalence of categories $\mathsf{T}^{\mathcal{K}\text{-tors}} \simeq \mathsf{T}^{\mathcal{K}\text{-comp}}$ is an equivalence of tt-categories.

6.B. **Relative proxy-smallness.** We now mix the notion of proxy-smallness with the definition of geometric functors:

Definition 6.20. Let $f^*\colon \mathsf{T} \to \mathsf{U}$ be a geometric functor. We say that f^* is *proxy-small* if there exists a set of compact objects $\mathcal{W} \subseteq \mathsf{T}^c$ such that:

(1) $f_*\mathbb{1}_{\mathsf{U}}$ is proxy-small with witness $\mathcal{W}$;
(2) $\mathbb{1}_{\mathsf{U}}$ is proxy-small with witness $f^*\mathcal{W}$.

Notice that given (1), since f^* preserves compact objects, condition (2) is just the statement that $\mathbb{1}_{\mathsf{U}} \in \mathrm{Loc}_{\otimes}(f^*\mathcal{W})$.

Remark 6.21. Condition (1) is the most important: it turns out that if f^* is *enhanced* then condition (2) is automatic. We will see later why we impose the compatibility condition (2): it ensures that the colocalisation $\Gamma\colon \mathsf{T} \to \mathrm{Loc}_{\otimes}(\mathcal{K})$ behaves well with respect to f^*, namely $f^* \simeq f^*\Gamma$.

Example 6.22. Let $(R, \mathfrak{m}, k)$ be a commutative Noetherian local ring and let $f^* = k \otimes^{\mathsf{L}}_R - \colon \mathsf{D}(R) \to \mathsf{D}(k)$. Then f^* is a proxy-small geometric functor by Example 5.7: that gives condition (1), and condition (2) follows since $k \otimes^{\mathsf{L}}_R K(\mathfrak{m})$ is a sum of shifted copies of k.

As with the proxy-small condition on objects (recall Lemma 5.3), the proxy-small condition on geometric functors is independent of the choice of witness.

Lemma 6.23. *Let $f^*\colon \mathsf{T} \to \mathsf{U}$ be a geometric functor where $f_*\mathbb{1}_{\mathsf{U}}$ is proxy-small in T. Then f^* is proxy-small if and only if $\mathrm{Loc}_{\otimes}(f^*f_*\mathbb{1}_{\mathsf{U}}) = \mathsf{U}$.*

Proof. For the reverse direction, as noted above it suffices to show that $\mathbb{1}_{\mathsf{U}} \in \mathrm{Loc}_{\otimes}(f^*\mathcal{W})$. Applying f^* to the definition of $f_*\mathbb{1}_{\mathsf{U}}$ being proxy-small we have that $f^*f_*\mathbb{1}_{\mathsf{U}} \in \mathrm{Loc}_{\otimes}(f^*\mathcal{W})$. By assumption, $f^*f_*\mathbb{1}_{\mathsf{U}}$ builds $\mathbb{1}_{\mathsf{U}}$ so we obtain that $\mathbb{1}_{\mathsf{U}} \in \mathrm{Loc}_{\otimes}(f^*\mathcal{W})$ as required. For the forward implication, notice that we have $\mathbb{1}_{\mathsf{U}} \in \mathrm{Loc}_{\otimes}(f^*\mathcal{W})$ and $f^*\mathcal{W} \in \mathrm{thick}_{\otimes}(f^*f_*\mathbb{1}_{\mathsf{U}})$. So $\mathbb{1}_{\mathsf{U}} \in \mathrm{Loc}_{\otimes}(f^*f_*\mathbb{1}_{\mathsf{U}})$, from which it follows that $\mathrm{Loc}_{\otimes}(f^*f_*\mathbb{1}_{\mathsf{U}}) = \mathsf{U}$. □

Definition 6.24. Let f^* be a proxy-small geometric functor. By applying Theorem 6.15 to T and $\mathcal{W}$ (a witness for the proxy-smallness of $f_*\mathbb{1}_{\mathsf{U}}$), we obtain torsion, localisation and completion functors which we denote by Γ_f, L_f, and Λ_f respectively. Likewise we denote the associated (co)localised categories by $\Gamma_f\mathsf{T}$ and so on.

The relevance of the proxy-small condition on geometric functors is that it ensures good compatibility between f^* and the associated torsion and completion functors Γ_f and Λ_f.

Proposition 6.25. *Let $f^*\colon \mathsf{T} \to \mathsf{U}$ be a proxy-small geometric functor. There are natural isomorphisms:*

(1) $\Gamma_f f_* Y \simeq f_* Y$ *for all* $Y \in \mathsf{U}$*;*
(2) $f^* \Gamma_f X \simeq f^* X$ *for all* $X \in \mathsf{T}$*;*
(3) $f^{(1)} \Lambda_f X \simeq f^{(1)} X$ *for all* $X \in \mathsf{T}$*;*
(4) $\Lambda_f f_* Y \simeq f_* Y$ *for all* $Y \in \mathsf{U}$*.*

Proof. For (1), by considering the triangle $\Gamma_f X \to X \to L_f X$ it suffices to show that $f_* Y \in \mathrm{Loc}_\otimes(f_* \mathbb{1}_\mathsf{U})$ as $\ker(L_f) = \mathrm{Loc}_\otimes(f_* \mathbb{1}_\mathsf{U})$. We have $Y \in \mathsf{U} = \mathrm{Loc}_\otimes(f^* f_* \mathbb{1}_\mathsf{U}) = \mathrm{Loc}(f^* f_* \mathbb{1}_\mathsf{U} \otimes u \mid u \in \mathsf{U})$ by proxy-smallness (see Lemma 6.23). By applying the coproduct-preserving triangulated functor f_* we therefore have

$$f_* Y \in \mathrm{Loc}(f_*(f^* f_* \mathbb{1}_\mathsf{U} \otimes u) \mid u \in \mathsf{U}) = \mathrm{Loc}(f_* \mathbb{1}_\mathsf{U} \otimes f_* u \mid u \in \mathsf{U}) \subseteq \mathrm{Loc}_\otimes(f_* \mathbb{1}_\mathsf{U})$$

where the equality uses the projection formula (Proposition 3.6).

For (2), by the Yoneda lemma it suffices to show that the canonical map

$$\mathrm{Hom}(Z, \alpha) \colon \mathrm{Hom}(Z, f^* \Gamma_f X) \to \mathrm{Hom}(Z, f^* X)$$

is an isomorphism for all $Z \in \mathsf{U}$. By proxy-smallness, we have a set $\mathcal{W} \subseteq \mathsf{T}^c$ such that $\mathsf{U} = \mathrm{Loc}_\otimes(f^* \mathcal{W})$. Since $\mathrm{Loc}_\otimes(f^* \mathcal{W}) = \mathrm{Loc}(f^* \mathcal{W} \otimes C \mid C \in \mathsf{U}^c)$ it therefore suffices to show that $\mathrm{Hom}(f^* W \otimes C, \alpha)$ is an isomorphism for all $W \in \mathcal{W}$ and $C \in \mathsf{U}^c$. This is equivalent to $\underline{\mathrm{Hom}}(f^* W, \alpha)$ being an isomorphism for all $W \in \mathcal{W}$ which is in turn equivalent to $f^*(DW) \otimes \alpha$ being an isomorphism for all $W \in \mathcal{W}$ by Lemma 2.23. This now holds since f^* is strong monoidal.

Taking the right adjoint of (1) yields (3), and (4) is the right adjoint to (2). □

Proposition 6.26. *Let* $f^* \colon \mathsf{T} \to \mathsf{U}$ *be a proxy-small geometric functor and write* $\mathcal{W} \subseteq \mathsf{T}^c$ *for a witness. Let* α *be a map in* T*. Then the following are equivalent:*

(1) $W \otimes \alpha$ *is an isomorphism for all* $W \in \mathcal{W}$*;*
(2) $\Gamma_f \alpha$ *is an isomorphism;*
(3) $\Lambda_f \alpha$ *is an isomorphism;*
(4) $f^* \alpha$ *is an isomorphism;*
(5) $f^{(1)} \alpha$ *is an isomorphism.*

Proof. That (1) and (2) are equivalent follows from the facts that Γ_f is smashing and elements of $\mathcal{W}$ are $f_* \mathbb{1}_\mathsf{U}$-torsion. The equivalence between (2) and (3) is then immediate from Theorem 6.15. For (2) $\Rightarrow$ (4), if $\Gamma_f \alpha$ is an isomorphism, then $f^* \alpha \simeq f^* \Gamma_f \alpha$ is an isomorphism by Proposition 6.25. Similarly, (3) $\Rightarrow$ (5) holds since we have $f^! \alpha \simeq f^! \Lambda_f \alpha$ by Proposition 6.25.

To see that (4) $\Rightarrow$ (1), we have $f_* \mathbb{1}_\mathsf{U} \otimes \alpha \simeq f_* f^* \alpha$ by the projection formula (Proposition 3.6). As $\mathcal{W} \subseteq \mathrm{thick}_\otimes(f_* \mathbb{1}_\mathsf{U})$, by a thick subcategory argument it then follows that $W \otimes \alpha$ is an isomorphism for all $W \in \mathcal{W}$. Finally, for (5) $\Rightarrow$ (1), we have

$$\mathrm{Hom}(X, \underline{\mathrm{Hom}}(f_* \mathbb{1}_\mathsf{U}, \alpha)) \simeq \mathrm{Hom}(X, f_* f^! \alpha) \simeq \mathrm{Hom}(f^* X, f^! \alpha)$$

for any $X \in \mathcal{C}$, by (3.9) and adjunction. As such, $\underline{\mathrm{Hom}}(f_* \mathbb{1}_\mathsf{U}, \alpha)$ is an isomorphism. Since $\mathcal{W} \subseteq \mathrm{thick}_\otimes(f_* \mathbb{1}_\mathsf{U})$ we see that $\underline{\mathrm{Hom}}(W, \alpha)$ is an isomorphism for each $W \in \mathcal{W}$. Now since each $W \in \mathcal{W}$ is rigid, W is a retract of $W \otimes DW \otimes W$ (see Exercise A.8). Therefore $W \otimes \alpha$ is a retract of $W \otimes \underline{\mathrm{Hom}}(W, \alpha) \otimes W$, and hence is an isomorphism. □

Remark 6.27. We say that a map α in $\mathcal{C}$ is an f^**-isomorphism* if $f^* \alpha$ is an isomorphism, and similarly for $f^!$, Γ_f, and Λ_f.

6.C. **An aside on enhancements.** In this subsection, we prove the assertion of Remark 6.21; namely that an *enhanced* geometric functor $f^* \colon \mathsf{T} \to \mathsf{U}$ is proxy-small if and only if $f_* \mathbb{1}_\mathsf{U}$ is proxy-small in T.

Let us first make precise what we mean by an enhancement. The key point is that we require a "good" notion of modules. Let T is a tt-category and R is a ring object in T (i.e., we have a multiplication map $\mu\colon R \otimes R \to R$ and a unit map $\eta\colon \mathbb{1} \to R$ satisfying the obvious relations). An R-module M is then an object $M \in \mathsf{T}$ together with an action map $a\colon R \otimes M \to M$ which satisfies the expected relations with μ and η. The problem is that, the category $\mathrm{Mod}_{\mathsf{T}}(R)$ of R-modules in T does not inherit a triangulated structure from T in any obvious way. For our purposes the use of enhancements fixes this issue.

For those "in the know", an *enhanced tt-category* for us means a presentably symmetric monoidal stable ∞-category whose tensor product commutes with colimit separately in each variable. Its homotopy category is then a tt-category in the usual sense, and we say it is rigidly-compactly generated if its homotopy category is so. An *enhanced geometric functor* between rigidly-compactly generated enhanced tt-categories is a colimit-preserving symmetric monoidal functor.

For those not in the know, one can assume that we are working with tt-categories where the category of modules over a ring object is again a tt-category. One does not lose much from just pretending this for the purposes of this aside. As such the following is the key observation:

Proposition 6.28. *Let T be a rigidly-compactly generated enhanced tt-category and let R be a commutative ring object in T. Then the category $\mathrm{Mod}_{\mathsf{T}}(R)$ of R-modules is a rigidly-compactly generated enhanced tt-category.*

Proof. We take for granted that $\mathrm{Mod}_{\mathsf{T}}(R)$ is again an enhanced tt-category, so it suffices to prove that it is rigidly-compactly generated. Note that the tensor product in $\mathrm{Mod}_{\mathsf{T}}(R)$ is $- \otimes_R -$ and so has unit object R. In particular, the free functor $R \otimes -\colon \mathsf{T} \to \mathrm{Mod}_{\mathsf{T}}(R)$ is a strong symmetric monoidal functor with right adjoint the forgetful functor.

Write G for a set of compact generators for T. We claim that $R \otimes \mathsf{G}$ is a set of compact generators for T. By the free-forgetful adjunction,

$$\mathrm{Hom}_{\mathrm{Mod}_{\mathsf{T}}(R)}(R \otimes \mathsf{G}, -) = \mathrm{Hom}_{\mathsf{T}}(\mathsf{G}, -).$$

From this it is immediate that $R \otimes \mathsf{G}$ is a set of compact generators for $\mathrm{Mod}_{\mathsf{T}}(R)$. These are also rigid since $R \otimes -\colon \mathsf{T} \to \mathrm{Mod}_{\mathsf{T}}(R)$ is a strong symmetric monoidal functor, see Lemma 2.23. Therefore we have shown that $\mathrm{Mod}_{\mathsf{T}}(R)$ has a set of rigid-compact generators, and moreover the unit object $\mathbb{1}_{\mathrm{Mod}_{\mathsf{T}}(R)} = R$ is compact. Therefore, $\mathrm{Mod}_{\mathsf{T}}(R)$ is rigidly-compactly generated by Proposition 2.25. □

Recall that a functor $F\colon \mathsf{T} \to \mathsf{U}$ between triangulated categories is *conservative* if $FX \simeq 0$ implies that $X \simeq 0$.

Lemma 6.29. *Let $g^*\colon \mathsf{T} \to \mathsf{U}$ be a geometric functor. The functor $g_*\colon \mathsf{U} \to \mathsf{T}$ is conservative if and only if* $\mathrm{thick}(g^*(\mathsf{T}^c)) = \mathsf{U}^c$.

Proof. For the reverse implication, suppose that $g_*X \simeq 0$. By compact generation, $X \simeq 0$ if and only if $\mathrm{Hom}(C, X) = 0$ for all $C \in \mathsf{U}^c$. By assumption this is moreover equivalent to $\mathrm{Hom}(g^*C', X) = 0$ for all $C' \in \mathsf{T}^c$. By adjunction, $\mathrm{Hom}(g^*C', X) = \mathrm{Hom}(C', g_*X) = 0$ so $X = 0$ as required. Conversely, suppose that g_* is conservative. To see that $\mathrm{thick}(g^*(\mathsf{T}^c)) = \mathsf{U}^c$ notice that since $g^*(\mathsf{T}^c)$ is a set of compacts (by Lemma 2.23) it suffices by Proposition 2.12 to show that $g^*(\mathsf{T}^c)$ is a set of generators. As such, we need to prove that $\mathrm{Hom}(g^*C, X) = 0$ for all $C \in \mathsf{T}^c$ implies that $X = 0$. Now if $\mathrm{Hom}(g^*C, X) = 0$ for all $C \in \mathsf{T}^c$, then $\mathrm{Hom}(C, g_*X) = 0$ for all $C \in \mathsf{T}^c$ so $g_*X = 0$ by compact generation. Therefore by conservativity $X = 0$ as required. □

Lemma 6.30. *Let $g^*\colon \mathsf{T} \to \mathsf{U}$ be a geometric functor whose right adjoint g_* is conservative. Then for any $\mathcal{X} \subseteq \mathsf{U}$ we have*

$$g_*(\mathrm{thick}_\otimes(\mathcal{X})) \subseteq \mathrm{thick}_\otimes(g_*\mathcal{X}) \quad \textit{and} \quad g_*(\mathrm{Loc}_\otimes(\mathcal{X})) \subseteq \mathrm{Loc}_\otimes(g_*\mathcal{X}).$$

Proof. Note that $\mathrm{thick}_\otimes(\mathcal{X}) = \mathrm{thick}(\mathcal{X} \otimes \mathsf{U}^c) = \mathrm{thick}(\mathcal{X} \otimes g^*(\mathsf{T}^c))$ where the latter equality holds by Lemma 6.29. As g_* is an exact functor, we therefore have

$$g_* \,\mathrm{thick}(\mathcal{X} \otimes g^*(\mathsf{T}^c)) \subseteq \mathrm{thick}(g_*(\mathcal{X} \otimes g^*(\mathsf{T}^c))) = \mathrm{thick}(g_*\mathcal{X} \otimes \mathsf{T}^c) = \mathrm{thick}_\otimes(g_*\mathcal{X})$$

by the projection formula (Proposition 3.6) as claimed. The statement with localising $\otimes$-ideals follows similarly since g_* is coproduct-preserving. □

Proposition 6.31. *If $f^*\colon \mathsf{T} \to \mathsf{U}$ is an enhanced geometric functor, then $\mathrm{Loc}_\otimes(f^*f_*\mathbb{1}_\mathsf{U}) = \mathsf{U}$.*

Proof. There are maps of commutative algebras

$$\mathbb{1}_\mathsf{U} \xrightarrow{\alpha} f^*f_*\mathbb{1}_\mathsf{U} \xrightarrow{\varepsilon} \mathbb{1}_\mathsf{U}$$

with composite the identity; explicitly, $\alpha = f^*(\eta)$ where $\eta\colon \mathbb{1}_\mathsf{T} \to f_*f^*\mathbb{1}_\mathsf{T} \simeq f_*\mathbb{1}_\mathsf{U}$ is the unit map of the adjunction. The fact that the composite is the identity is then precisely the triangle identity. Write $\varepsilon^*\colon \mathrm{Mod}_\mathsf{U}(f^*f_*\mathbb{1}_\mathsf{U}) \to \mathrm{Mod}_\mathsf{U}(\mathbb{1}_\mathsf{U}) = \mathsf{U}$ for the extension of scalars functor, so that ε_* is restriction of scalars. Therefore, $\varepsilon_*\mathbb{1}_\mathsf{U}$ denotes the underlying object $\mathbb{1}_\mathsf{U}$ equipped with a $f^*f_*\mathbb{1}_\mathsf{U}$-module structure via ε. In particular, $\varepsilon_*\mathbb{1}_\mathsf{U} \in \mathrm{Loc}_\otimes(f^*f_*\mathbb{1}_\mathsf{U})$ in the category $\mathrm{Mod}_\mathsf{U}(f^*f_*\mathbb{1}_\mathcal{D})$ by Proposition 6.28. By applying Lemma 6.30, we deduce that $\mathbb{1}_\mathsf{U} = \alpha_*\varepsilon_*\mathbb{1}_\mathsf{U} \in \mathrm{Loc}_\otimes(\alpha_*f^*f_*\mathbb{1}_\mathsf{U})$ in U. Now it just remains to note that $\alpha_*f^*f_*\mathbb{1}_\mathsf{U}$ is just $f^*f_*\mathbb{1}_\mathsf{U}$ viewed as an object in U (rather than in the category of $f^*f_*\mathbb{1}_\mathsf{U}$-modules). Therefore we have proved the claim. □

Corollary 6.32. *Let $f^*\colon \mathsf{T} \to \mathsf{U}$ be an enhanced geometric functor. Then f^* is proxy-small if and only if $f_*\mathbb{1}_\mathsf{U}$ is proxy-small in T.*

Proof. Combine Lemma 6.23 with Proposition 6.31. □

7. Grothendieck duality via proxy-smallness

In this section we come to the promised statement that under the proxy-smallness assumption, Grothendieck duality holds on a certain canonical subcategory.

7.A. Classifying rigid objects in the torsion category.

In this section we give various equivalent characterisations of the subcategory of interest.

Definition 7.1. Let $f^*\colon \mathsf{T} \to \mathsf{U}$ be a geometric functor. We say that $X \in \mathsf{T}$ is *f^*-rigid* if f^*X is rigid in U.

Note that any rigid object is f^*-rigid by Lemma 2.23. However, there are many many more f^*-rigid objects in general:

Example 7.2. For any geometric functor $f^*\colon \mathsf{T} \to \mathsf{U}$ and compact $C \in \mathsf{T}^c$, the object $\Gamma_f C$ is f^*-rigid: we have $f^*\Gamma_f C \simeq f^*C$ by Proposition 6.25. For example, when $f^* = k \otimes^\mathsf{L}_R -$ for a commutative Noetherian local ring $(R, \mathfrak{m}, k)$, the stable Koszul complex $K_\infty(\mathfrak{m}) = \Gamma_f R$ is f^*-rigid, but is not rigid in general.

In order to state and prove the main theorem of this section we set some notation.

Remark 7.3. Let F be a strong symmetric monoidal functor.

(1) Consider the natural map

$$\alpha_{M,N}\colon F\underline{\mathrm{Hom}}(M,N) \to \underline{\mathrm{Hom}}(FM,FN)$$

constructed as the adjunct of

$$FM \otimes F\underline{\mathrm{Hom}}(M,N) \simeq F(M \otimes \underline{\mathrm{Hom}}(M,N)) \xrightarrow{F(\mathrm{ev}_{M,N})} FN$$

This is an isomorphism provided that M is rigid. In particular, if M is rigid, then

$$\alpha_{M,\mathbb{1}}\colon F(DM) \xrightarrow{\sim} D(FM)$$

is a natural isomorphism. (Indeed, we already saw this last claim as Lemma 2.23.)

(2) There is a natural transformation

$$\rho_{M,N}\colon DM \otimes N \to \underline{\mathrm{Hom}}(M,N)$$

constructed as the adjunct of

$$M \otimes DM \otimes N \xrightarrow{\mathrm{ev}_{M,\mathbb{1}} \otimes N} \mathbb{1}_{\mathcal{C}} \otimes N \simeq N.$$

(3) There is also a natural transformation

$$\nu_{M,X,N}\colon \underline{\mathrm{Hom}}(M,X) \otimes N \to \underline{\mathrm{Hom}}(M, X \otimes N)$$

constructed as the adjunct of

$$\mathrm{ev}_{M,X} \otimes N\colon M \otimes \underline{\mathrm{Hom}}(M,X) \otimes N \to X \otimes N.$$

Note that $\rho_{M,N} = \nu_{M,\mathbb{1},N}$.

Theorem 7.4. *Let $f^*\colon \mathsf{T} \to \mathsf{U}$ be a proxy-small geometric functor. Let $M \in \mathsf{T}$. The following are equivalent:*

(1) the map $\rho_{M,N}\colon DM \otimes N \to \underline{\mathrm{Hom}}(M,N)$ is an f^*-isomorphism for all $N \in \mathsf{T}$, i.e., $f^*(\rho_{M,N})$ is an isomorphism for all $N \in \mathsf{T}$;
(2) $\Gamma_f M \in \Gamma_f \mathsf{T}$ is rigid;
(3) $\Lambda_f M \in \Lambda_f \mathsf{T}$ is rigid;
(4) $f^* M \in \mathsf{U}$ is rigid.

Moreover, these are implied by $\Gamma_f M \in \mathrm{thick}(\Gamma_f(\mathsf{T}^c))$.

Proof. We first prove (1) $\Leftrightarrow$ (3). Write D_{Λ_f} for the functional dual in $\Lambda_f \mathsf{T}$. Note that $\Lambda_f\colon \mathsf{T} \to \Lambda_f \mathsf{T}$ is a strong symmetric monoidal functor (see Remark 6.19), so that Remark 7.3 applies. We have a diagram

$$\begin{tikzcd}
\Lambda_f(DM) \widehat{\otimes} \Lambda_f N \arrow[r, "\simeq"] \arrow[d, "\alpha_{M,\mathbb{1}} \widehat{\otimes} \Lambda_f N"'] & \Lambda_f(DM \otimes N) \arrow[r, "\Lambda_f(\rho_{M,N})"] & \Lambda_f \underline{\mathrm{Hom}}(M,N) \arrow[d, "\alpha_{M,N}"] \\
D_{\Lambda_f}(\Lambda_f M) \widehat{\otimes} \Lambda_f N \arrow[rr, "\rho_{\Lambda_f M, \Lambda_f N}"'] & & \underline{\mathrm{Hom}}(\Lambda_f M, \Lambda_f N)
\end{tikzcd}$$

for any $N \in \mathsf{T}$. This commutes by a diagram chase (omitted here). Since $\Lambda_f \underline{\mathrm{Hom}}(M,N) \simeq \underline{\mathrm{Hom}}(M, \Lambda_f N)$ by Corollary 6.18, the map $\alpha_{M,N}$ is an isomorphism for all $M, N \in \mathsf{T}$. Therefore $\Lambda_f(\rho_{M,N})$ is an isomorphism if and only if $\rho_{\Lambda_f M, \Lambda_f N}$ is an isomorphism if and only if $\Lambda_f M \in \Lambda_f \mathsf{T}$ is rigid. Now by Proposition 6.26, $\Lambda_f(\rho_{M,N})$ is an isomorphism if and only if $f^*(\rho_{M,N})$ is an isomorphism. This proves that (1) is equivalent to (3). Conditions (2) and (3) are equivalent by the MGM equivalence (see Remark 6.19).

For (2) $\Rightarrow$ (4), we have that $f^*\colon \mathsf{T} \to \mathsf{U}$ factorises as

$$\mathsf{T} \xrightarrow{\Gamma_f} \Gamma_f \mathsf{T} \xrightarrow{f^*} \mathsf{U}$$

by Proposition 6.25. Since $f^*\colon \Gamma_f\mathsf{T} \to \mathsf{U}$ is strong symmetric monoidal, and $\Gamma_f M$ is rigid by assumption, it follows that f^*M is rigid, as strong monoidal functors preserve rigid objects, see Lemma 2.23.

To conclude the proof, we prove that (4) $\Rightarrow$ (1). Recall the natural transformation

$$\nu_{M,X,N}\colon \underline{\mathrm{Hom}}(M,X)\otimes N \to \underline{\mathrm{Hom}}(M, X\otimes N)$$

as defined in Remark 7.3, and let $\mathcal{X}$ be the subcategory of $X \in \mathsf{T}$ such that $\nu_{M,X,N}$ is an equivalence for all $N \in \mathsf{T}$. Note that for $C \in \mathcal{C}$ rigid, by naturality $\nu_{M,X\otimes C,N}$ is identified under the obvious isomorphisms with $C \otimes \nu_{M,X,N}$. Thus we see that $\mathcal{X}$ is a thick $\otimes$-ideal of T.

We show that $f_*Y \in \mathcal{X}$ for all $Y \in \mathsf{U}$. To do so, we note that we have isomorphisms fitting into a diagram

$$\begin{tikzcd}
\underline{\mathrm{Hom}}(M, f_*Y)\otimes N \arrow[r, "\nu_{M,f_*Y,N}"] \arrow[d, "\simeq"', "(3.8)"] & \underline{\mathrm{Hom}}(M, f_*Y\otimes N) \arrow[d, "(3.6)"', "\simeq"] \\
f_*\underline{\mathrm{Hom}}(f^*M, Y)\otimes N \arrow[d, "\simeq"', "(3.6)"] & \underline{\mathrm{Hom}}(M, f_*(Y\otimes f^*N) \arrow[d, "(3.8)"', "\simeq"] \\
f_*(\underline{\mathrm{Hom}}(f^*M,Y)\otimes f^*N) \arrow[r, "f_*(\nu_{f^*M,Y,f^*N})"'] & f_*(\underline{\mathrm{Hom}}(f^*M, Y\otimes f^*N)
\end{tikzcd}$$

commutes. Since f^*M is rigid, ν_{f^*M,Y,f^*N} is an isomorphism for all N, and therefore by the above diagram we see that $\nu_{M,f_*Y,N}$ is also an isomorphism so $f_*Y \in \mathcal{X}$ as claimed.

Since $\mathcal{X}$ is a thick $\otimes$-ideal, it follows that $\mathcal{W} \subseteq \mathcal{X}$ where $\mathcal{W}$ is a witness for proxy-smallness as $\mathcal{W} \subseteq \mathrm{thick}_\otimes(f_*\mathbb{1}_\mathsf{T})$. Since each $W \in \mathcal{W}$ is rigid, we have that $\nu_{M,W,Y}$ corresponds under natural isomorphisms to $W \otimes \nu_{M,\mathbb{1}_\mathsf{T},Y}$. As $\nu_{M,\mathbb{1}_\mathsf{T},Y} = \rho_{M,Y}$, this proves that $W \otimes \rho_{M,Y}$ is an isomorphism for all $W \in \mathcal{W}$ and $Y \in \mathsf{T}$, which by Proposition 6.26 gives (1).

For the final statement, since $\Gamma_f\colon \mathsf{T} \to \Gamma_f\mathsf{T}$ is strong monoidal it preserves rigid objects (see Lemma 2.23). Therefore any object in $\Gamma_f(\mathsf{T}^c)$ is rigid in $\Gamma_f\mathcal{C}$, so the claim follows by a thick subcategory argument. □

Remark 7.5. We note that the condition that $\Gamma_f M \in \mathrm{thick}(\Gamma_f(\mathsf{T}^c))$ is not equivalent to the other conditions in the previous theorem in general. The counterexample comes from the map of ring spectra $L_nS^0 \to K(n)$ and is way beyond the scope of this course.

7.B. **Case study: local algebra.** We now give an explicit example of the previous theorem in action. For a commutative Noetherian local ring $(R, \mathfrak{m}, k)$, we consider the geometric functor $f^* = k \otimes^{\mathsf{L}}_R -\colon \mathsf{D}(R) \to \mathsf{D}(k)$ and write $\Gamma_\mathfrak{m} = \Gamma_f$ and similarly for $\Lambda_\mathfrak{m}$.

Remark 7.6. The functor $\Gamma_\mathfrak{m}$ computes the total right derived functor of the $\mathfrak{m}$-power torsion functor, and the functor $\Lambda_\mathfrak{m}$ computes the total left derived functor of the $\mathfrak{m}$-adic completion functor.

In this case, we can in fact show that slightly more than Theorem 7.4 claims is true. In order to do so we need the following two lemmas:

Lemma 7.7. *Let $(R, \mathfrak{m}, k)$ be a commutative Noetherian local ring, and let $M \in \Lambda_\mathfrak{m}\mathsf{D}(R)$. If $H_*(k \otimes^{\mathsf{L}}_R M)$ is bounded, then the natural map*

$$M \simeq R \otimes^{\mathsf{L}}_R M \to k \otimes^{\mathsf{L}}_R M$$

is non-zero in homology.

Proof. To prove this we need the following "black box":

If $N \in \Lambda_{\mathfrak{m}}\mathsf{D}(R)$, then $H_n(N)/\mathfrak{m}H_n(N) = 0$ implies that $H_n(N) = 0$.

The proof of the above black box is not too difficult but it would take us too far from the point so we omit it. Since $H_*(k \otimes^{\mathsf{L}}_R M)$ is bounded by assumption, it follows that $H_*(K(\mathfrak{m}) \otimes^{\mathsf{L}}_R M)$ is also bounded by Theorem 5.16. Since $H_*(K(\mathfrak{m}) \otimes^{\mathsf{L}}_R M)$ is bounded, each $x \in \mathfrak{m}$ acts as an isomorphism on $H_n(M)$ for $n \ll 0$. Therefore by the black box, we have $H_n(M) = 0$ for $n \ll 0$.

Now, write $i = \inf\{n \mid H_n(M) \neq 0\}$; this is permitted since $H_*(M)$ is bounded. One checks that $H_i(k \otimes^{\mathsf{L}}_R M) = k \otimes_R H_iM = H_iM/\mathfrak{m}H_iM$. Therefore the canonical map induces the non-zero quotient map

$$H_iM \to H_i(k \otimes^{\mathsf{L}}_R M) = H_iM/\mathfrak{m}H_iM$$

in homology. □

Lemma 7.8. *Let $(R, \mathfrak{m}, k)$ be a commutative Noetherian local ring and let $M \in \Lambda_{\mathfrak{m}}\mathsf{D}(R)$. If $\operatorname{rank}_k H_*(k \otimes^{\mathsf{L}}_R M)$ is finite then $M \in \operatorname{thick}(\Lambda_{\mathfrak{m}}R)$.*

Proof. We argue by induction on $r = \operatorname{rank}_k(H_*(k \otimes^{\mathsf{L}}_R M))$. For the base case $r = 0$, this means that $k \otimes^{\mathsf{L}}_R M \simeq 0$. Therefore by Proposition 6.26 we see that $M \simeq \Lambda_{\mathfrak{m}}M \simeq 0$ giving the base case.

Now suppose that $r \geq 1$. Since the rank is finite, we have that $H_*(k \otimes^{\mathsf{L}}_R M)$ is bounded. Therefore by Lemma 7.7, we have a non-zero map $\alpha\colon H_nM \to H_n(k \otimes^{\mathsf{L}}_R M)$ for some n. Since $H_n(-) = \operatorname{Hom}_{\mathsf{D}(R)}(\Sigma^n R, -)$, this non-zero map α is equivalent to the data of a map $\Sigma_n R \to M$ such that the induced map $\Sigma^n k \to k \otimes^{\mathsf{L}}_R M$ is non-zero.

The map α gives rise to a map $\Sigma^n \Lambda_{\mathfrak{m}} R \to M$ since $M \in \Lambda_{\mathfrak{m}}\mathsf{D}(R)$, and therefore to a triangle

$$\Sigma^n \Lambda_{\mathfrak{m}} R \to M \to N.$$

Applying $k \otimes^{\mathsf{L}}_R -$ we get a triangle

$$\Sigma^n k \to k \otimes^{\mathsf{L}}_R M \to k \otimes^{\mathsf{L}}_R N$$

in $\mathsf{D}(k)$ in which the first map is non-zero. Therefore taking the long exact sequence in homology, we see that $\operatorname{rank}_k(H_*(k \otimes^{\mathsf{L}}_R M)) = \operatorname{rank}_k(H_*(k \otimes^{\mathsf{L}}_R N)) + 1$. Therefore by inductive hypothesis we have that $N \in \operatorname{thick}(\Lambda_{\mathfrak{m}}R)$. Hence by the above triangle, we also have $M \in \operatorname{thick}(\Lambda_{\mathfrak{m}}R)$. □

As an aside, we note a corollary of the proof of the previous lemma which will become relevant later on in the course:

Corollary 7.9. *Let $(R, \mathfrak{m}, k)$ be a commutative Noetherian local ring and let $M \in \Lambda_{\mathfrak{m}}\mathsf{D}(R)$. If $k \otimes^{\mathsf{L}}_R M \simeq \Sigma^a k$ for some $a \in \mathbb{Z}$, then $M \simeq \Sigma^a \Lambda_{\mathfrak{m}} R$.*

Proof. The assumption that $k \otimes^{\mathsf{L}}_R M \simeq \Sigma^a k$ says that $\operatorname{rank}_k H_*(k \otimes^{\mathsf{L}}_R M) = 1$. The proof of the inductive step in Lemma 7.8 then gives the claim. □

Proposition 7.10. *Let $(R, \mathfrak{m}, k)$ be a commutative Noetherian local ring. The following are equivalent for $M \in \mathsf{D}(R)$:*

(1) $\Gamma_{\mathfrak{m}}M \in \Gamma_{\mathfrak{m}}\mathsf{D}(R)$ is rigid;
(2) $k \otimes^{\mathsf{L}}_R M \in \mathsf{D}(k)$ is rigid;
(3) $\operatorname{rank}_k H^*(k \otimes^{\mathsf{L}}_R M)$ is finite;
(4) $\Gamma_{\mathfrak{m}}M \in \operatorname{thick}(\Gamma_{\mathfrak{m}}R)$.

Proof. Conditions (1) and (2) are equivalent and are implied by (4) by Theorem 7.4. Condition (3) is a rephrasing of condition (2) since k is a field. So it suffices to prove that (3) implies (4). As $k \simeq \Gamma_{\mathfrak{m}} k$, we see that the canonical map

$$k \otimes^{\mathsf{L}}_R X \to k \otimes^{\mathsf{L}}_R \Lambda_{\mathfrak{m}} X$$

is an isomorphism, by the MGM equivalence and the fact that $\Gamma_{\mathfrak{m}}$ is smashing. Therefore by Lemma 7.8 we see that $\Lambda_{\mathfrak{m}} X \in \operatorname{thick}(\Lambda_{\mathfrak{m}} R)$. Now the MGM equivalence (see Theorem 6.15) restricts to an equivalence of categories

$$\operatorname{thick}(\Gamma_{\mathfrak{m}} R) \underset{\Lambda_{\mathfrak{m}}}{\overset{\Gamma_{\mathfrak{m}}}{\underset{\longrightarrow}{\overset{\longleftarrow}{\simeq}}}} \operatorname{thick}(\Lambda_{\mathfrak{m}} R)$$

from which we deduce that $\Gamma_{\mathfrak{m}} X \simeq \Gamma_{\mathfrak{m}} \Lambda_{\mathfrak{m}} X \in \operatorname{thick}(\Gamma_{\mathfrak{m}} R)$ as required. □

7.C. **Grothendieck duality.** We now move onto the consequences of Theorem 7.4, and show that f^*-rigidity of $M \in \mathsf{T}$ is usually sufficient for M to enjoy properties in relation to f^* that one would typically only expect if M itself were rigid. These properties will allow us to prove Grothendieck duality for f^*-rigid objects below.

Corollary 7.11. *Suppose $f^* \colon \mathsf{T} \to \mathsf{U}$ is a proxy-small geometric functor and let $M \in \mathsf{T}$. If f^*M is rigid, then the canonical map*

$$f^*(d_M) \colon f^*M \to f^*(D^2 M)$$

is an isomorphism.

Proof. By Proposition 6.26 it suffices to prove that $\Lambda_f(d_M)$ is an isomorphism. Since f^*M is rigid in U, by Theorem 7.4 we have that $\Lambda_f M$ is rigid in $\Lambda_f \mathsf{T}$ and hence the natural map $d'_{\Lambda_f M} \colon \Lambda_f M \to D^2_{\Lambda_f}(\Lambda_f M)$ is an isomorphism, where D_{Λ_f} denotes the functional dual in $\Lambda_f \mathsf{T}$. Recall from Proposition 6.12 that the internal hom in complete modules is computed in the ambient category and that the tensor unit is $\Lambda_f \mathbb{1}$. Since $\Lambda_f \colon \mathsf{T} \to \Lambda_f \mathsf{T}$ is closed monoidal, there is a natural isomorphism $D^2_{\Lambda_f}(\Lambda_f(-)) \simeq \Lambda_f D^2(-)$ making the diagram

$$\begin{tikzcd}
\Lambda_f M \arrow[r, "\Lambda_f(d_M)"] \arrow[dr, "d'_{\Lambda_f M}"', "\simeq"] & \Lambda_f D^2 M \arrow[d, "\simeq"] \\
& D^2_{\Lambda_f}(\Lambda_f M)
\end{tikzcd}$$

commute. We see from this diagram that $\Lambda_f(d_M)$ is an isomorphism as claimed. □

Corollary 7.12. *Suppose $f^* \colon \mathsf{T} \to \mathsf{U}$ is a proxy-small geometric functor and let $M \in \mathsf{T}$. Recall the natural map $\alpha_{M,N} \colon f^* \underline{\operatorname{Hom}}(M,N) \to \underline{\operatorname{Hom}}(f^*M, f^*N)$ from Remark 7.3. If f^*M is rigid, then for all $N \in \mathsf{T}$ there are natural isomorphisms:*

$$\alpha_{M,N} \colon f^* \underline{\operatorname{Hom}}(M,N) \xrightarrow{\sim} \underline{\operatorname{Hom}}(f^*M, f^*N) \tag{7.13}$$

$$\alpha_{M,\mathbb{1}} \colon f^*(DM) \xrightarrow{\sim} D(f^*M). \tag{7.14}$$

Proof. Recall from Proposition 6.25 that $f^* \colon \mathsf{T} \to \mathsf{U}$ admits a factorisation

$$\mathsf{T} \xrightarrow{\Gamma_f} \Gamma_f \mathsf{T} \xrightarrow{f^*_t} \mathcal{D}$$

where f_t^* is the restriction of f^* to $\Gamma_f\mathsf{T}$. We note that f_t^* is also strong symmetric monoidal. Thus, the construction of Remark 7.3 composes, i.e., $\alpha_{M,N}$ is the composition

$$f_t^*(\Gamma_f\underline{\mathrm{Hom}}_{\mathsf{T}}(M,N)) \xrightarrow{f_t^*(\gamma_{M,N})} f_t^*\underline{\mathrm{Hom}}_{\Gamma_f\mathsf{T}}(\Gamma_f M,\Gamma_f N) \xrightarrow{\beta_{\Gamma_f M,\Gamma_f N}} \underline{\mathrm{Hom}}_{\mathsf{U}}(f_t^*\Gamma_f M, f_t^*\Gamma_f N)$$

where β and γ are the natural transformations analogous to α associated to f_t^* and Γ_f. Since $\Gamma_f\colon \mathsf{T}\to\Gamma_f\mathsf{T}$ is closed monoidal, γ is a natural isomorphism, and hence $f_t^*(\gamma_{M,N})$ is an isomorphism. Therefore, to prove (7.13) it suffices to prove that $\beta_{\Gamma_f M,\Gamma_f N}$ is an isomorphism. Since f^*M being rigid implies $\Gamma_f M\in\Gamma_f\mathsf{T}$ is rigid by Theorem 7.4, this is immediate from Remark 7.3. Finally, (7.14) is the case $N=\mathbb{1}_{\mathcal{C}}$ in (7.13). □

We are now ready to prove the promised generalisation of Grothendieck duality. Recall from Lemma 4.3 that rigid objects have Grothendieck duality. The following shows that under the proxy-small hypothesis, this extends to the larger class of f^*-rigid objects:

Theorem 7.15. *Let $f^*\colon\mathsf{T}\to\mathsf{U}$ be a proxy-small geometric functor, and $M\in\mathsf{T}$. If f^*M is rigid, then there is a natural isomorphism*

$$f^!X\otimes f^*M\simeq f^!(X\otimes M) \tag{7.16}$$

for all $X\in\mathsf{T}$. In particular, if f^*M is rigid there is an isomorphism

$$\omega_f\otimes f^*M\simeq f^!M. \tag{7.17}$$

Proof. We have natural isomorphisms

$$f^!X\otimes f^*M\simeq\underline{\mathrm{Hom}}(D(f^*M),f^!X)\simeq\underline{\mathrm{Hom}}(f^*(DM),f^!X)\simeq f^!\underline{\mathrm{Hom}}(DM,X)$$

using the rigidity of f^*M, (7.14), and then (3.10). To conclude it therefore suffices to show that the canonical map

$$M\otimes X\xrightarrow{d_M\otimes X}D^2M\otimes X\xrightarrow{\rho_{DM,N}}\underline{\mathrm{Hom}}(DM,X)$$

is an $f^!$-isomorphism, where $\rho_{DM,N}$ is as in Remark 7.3. By Proposition 6.26, this is equivalent to checking that it is an f^*-isomorphism.

Note that since f^*M is rigid, it follows that $f^*(DM)$ is also rigid (since it is isomorphic to $D(f^*M)$ by (7.14)). Now consider the following diagram in which the left hand vertical is the map we want to show is an isomorphism:

$$\begin{tikzcd}
f^*(M\otimes X) \arrow[r, "\simeq"] \arrow[d, "f^*(d_M\otimes X)"'] & f^*M\otimes f^*X \arrow[d, "f^*(d_M)\otimes f^*X", "\simeq\ (7.11)"'] & \\
f^*(D^2M\otimes X) \arrow[r, "\simeq"] \arrow[d, "f^*(\rho_{DM,N})"'] & f^*(D^2M)\otimes f^*X \arrow[r, "\alpha_{DM,\mathbb{1}}\otimes f^*X", "\simeq\ (7.14)"'] & D(f^*DM)\otimes f^*X \arrow[d, "\rho_{f^*DM,f^*X}", "\simeq"'] \\
f^*\underline{\mathrm{Hom}}(DM,X) \arrow[rr, "\simeq\ (7.13)", "\alpha_{DM,X}"'] & & \underline{\mathrm{Hom}}(f^*DM,f^*X)
\end{tikzcd}$$

Note that the diagram commutes: the top square by the strong monoidality of f^*, and the bottom rectangle by a diagram chase left to the reader. Therefore, we see that the claimed map is an isomorphism as required. This completes the proof of (7.16). Taking $X=\mathbb{1}_{\mathsf{T}}$ in (7.16) gives (7.17). □

8. The Wirthmüller isomorphism

We now have a good understanding of Grothendieck duality and when it holds. Recall that it holds for all objects when f^* has Grothendieck–Neeman duality. Recall that the Grothendieck–Neeman duality setup supplies a natural isomorphism $f_{(1)}(X) \simeq f_*(\omega_f \otimes X)$ called the ur-Wirthmüller isomorphism (4.16). To "remove the ur-" we want to find an object C such that $f_* \simeq f_{(1)}(C \otimes -)$. One way to get such a C is to suppose that ω_f is invertible meaning that there exists a Y such that $\omega_f \otimes Y \simeq \mathbb{1}$. Then we can deduce the Wirthmüller isomorphism from the ur-Wirthmüller isomorphism by inputting $X = Y \otimes -$. The point of this section is to prove that the invertibility of ω_f precisely characterises when a geometric functor with Grothendieck–Neeman duality has a Wirthmüller isomorphism.

In order to state and prove this, we first introduce some terminology and notation.

8.A. Invertibility and Picard groups.

In this section, we introduce the basics of invertible objects in tt-categories. We assume that T is a tt-category.

Definition 8.1. An object $X \in \mathsf{T}$ is *invertible* if there exists an object $X^{-1} \in \mathsf{T}$ such that $X \otimes X^{-1} \simeq \mathbb{1}_{\mathsf{T}}$. Equivalently, X is invertible if $X \otimes -\colon \mathsf{T} \to \mathsf{T}$ is an equivalence of categories.

Example 8.2. The unit object $\mathbb{1}_{\mathsf{T}}$ is always invertible with inverse $\mathbb{1}_{\mathsf{T}}^{-1} = \mathbb{1}_{\mathsf{T}}$. Moreover, any shift of the unit $\Sigma^n \mathbb{1}_{\mathsf{T}}$ is also invertible with inverse $\Sigma^{-n}\mathbb{1}_{\mathsf{T}}$.

It is straightforward to check that the tensor product of two invertible objects is again invertible: if X and Y are invertible, then $X \otimes Y$ has inverse $X^{-1} \otimes Y^{-1}$ by using the symmetry of the tensor product. Therefore we can make the following definition:

Definition 8.3. The *Picard group* of T is the collection of isomorphism classes of invertible objects of T. This forms an abelian group under the tensor product operation. We denote this by $\mathrm{Pic}(\mathsf{T})$.

Remark 8.4. If $F\colon \mathsf{T} \to \mathsf{U}$ is a strong symmetric monoidal functor, then F preserves invertible objects with $F(X)^{-1} = F(X^{-1})$. This means that F induces a group homomorphism $\mathrm{Pic}(\mathsf{T}) \to \mathrm{Pic}(\mathsf{U})$.

We now prove some basic facts about invertible objects.

Lemma 8.5. *If X is invertible then $X^{-1} \simeq DX$.*

Proof. For all Z we have

$$\mathrm{Hom}(Z, X^{-1}) \simeq \mathrm{Hom}(Z \otimes X, X \otimes X^{-1}) \simeq \mathrm{Hom}(Z \otimes X, \mathbb{1}) \simeq \mathrm{Hom}(Z, DX)$$

so we conclude by Yoneda. □

Remark 8.6. Under the isomorphism of the previous lemma, the map $X \otimes X^{-1} \to \mathbb{1}$ corresponds to the evaluation map $X \otimes DX \to \mathbb{1}$. As such, an object $X \in \mathsf{T}$ is invertible if and only if the evaluation map $DX \otimes X \to \mathbb{1}$ is an isomorphism.

Lemma 8.7. *An object X is invertible if and only if it is rigid and $\underline{\mathrm{Hom}}(X, X) \simeq \mathbb{1}$.*

Proof. For the forward implication, by Lemma 8.5 we have $X^{-1} \simeq DX$. For any Y, Z we have isomorphisms

$$\mathrm{Hom}(Z, DX \otimes Y) \simeq \mathrm{Hom}(Z \otimes X, X \otimes DX \otimes Y) \simeq \mathrm{Hom}(Z \otimes X, Y) \simeq \mathrm{Hom}(Z, \underline{\mathrm{Hom}}(X, Y))$$

so we see that invertible objects are rigid by Yoneda. Therefore $\underline{\mathrm{Hom}}(X, X) \simeq DX \otimes X \simeq \mathbb{1}$. For the reverse implication it suffices to show that $X \otimes DX \simeq \mathbb{1}$ which is immediate from the assumptions. □

Example 8.8. Let R be a commutative Noetherian local ring. Then if $X \in \mathsf{D}(R)$ is invertible we have $X \simeq \Sigma^n R$ for some $n \in \mathbb{Z}$. We will prove this in the next subsection.

The previous shows that there are no 'exotic' invertible objects in $\mathsf{D}(R)$. We codify this behaviour.

Definition 8.9. We say that T has *trivial Picard group* if the group homomorphism $\mathbb{Z} \to \mathrm{Pic}(\mathsf{T})$ given by $n \mapsto \Sigma^n \mathbb{1}$ is surjective.

Remark 8.10. The terminology may appear contradictory at first, since $\mathrm{Pic}(\mathsf{T})$ need not be the trivial group for T to have trivial Picard group! The point however is that it is very rare for $\mathrm{Pic}(\mathsf{T})$ to be actually isomorphic to the trivial group, since it requires that $\Sigma\mathbb{1} \simeq \mathbb{1}$. This also highlights why we only require surjectivity for the map $\mathbb{Z} \to \mathrm{Pic}(\mathsf{T})$ rather than it being an isomorphism. For instance, consider the graded field $R = k[x, x^{-1}]$ where x is in degree $2n$. The derived category of dg-R-modules has Picard group $\mathbb{Z}/2n$ since $\Sigma^{2n} R \simeq R$.

8.B. **An aside on derived categories.** We now prove the assertion of Example 8.8. This will require a little bit of work, so we split it up into some lemmas. We do not give a complete proof since it would take too long, although it does not use any significantly new ideas beyond what we have already discussed.

The key point is the following which we do not prove (but we do indicate a line of proof and an alternative approach below):

Lemma 8.11. *Let R be a commutative Noetherian ring. Write $\kappa(\mathfrak{p}) = R_{\mathfrak{p}}/\mathfrak{p}R_{\mathfrak{p}}$ for the residue field at $\mathfrak{p} \in \mathrm{Spec}(R)$. The collection $\{\kappa(\mathfrak{p}) \otimes^{\mathsf{L}}_R - \colon \mathsf{D}(R) \to \mathsf{D}(\kappa(\mathfrak{p}))\}_{\mathfrak{p} \in \mathrm{Spec}(R)}$ is jointly conservative (i.e., detects isomorphisms).*

Proof. Suppose that $\kappa(\mathfrak{p}) \otimes_R X \simeq 0$ for all $\mathfrak{p} \in \mathrm{Spec}(R)$. By a localising subcategory argument, we therefore have $Y \otimes_R X \simeq 0$ for all $Y \in \mathrm{Loc}(\kappa(\mathfrak{p}) \mid \mathfrak{p} \in \mathrm{Spec}(R))$. Therefore it suffices to prove that $R \in \mathrm{Loc}(\kappa(\mathfrak{p}) \mid \mathfrak{p} \in \mathrm{Spec}(R))$. We omit the proof of this. □

There is an alternative version of the previous lemma which is easier to prove (although we only give a proof in certain cases) and would actually suffice for our purposes:

Lemma 8.12. *Let $(R, \mathfrak{m}, k)$ be a commutative Noetherian local ring. Then $k \otimes_R - \colon \mathsf{D}(R)^c \to \mathsf{D}(k)^c$ is conservative. (Note the restriction to compact objects!)*

Proof. This is a consequence of Nakayama's lemma. For instance, if R is a PID, then $k \otimes_R H_n(X) \to H_n(k \otimes_R X)$ is injective. Therefore by assumption, we have $k \otimes_R H_n(X) = 0$ for all n. As X is compact (hence perfect), $H_n(X)$ is finitely generated. Therefore by Nakayama's lemma, $H_n(X) = 0$ for all n, and hence $X = 0$ as required. When R is not a PID, there is a spectral sequence argument which does the job. □

We now turn to classifying the invertible objects.

Lemma 8.13. *Let $(R, \mathfrak{m}, k)$ be a local commutative Noetherian ring. If M is an invertible R-module (i.e., in the abelian category of R-modules rather than the derived category), then M is isomorphic to R.*

Proof. If M is invertible it is projective and also finitely generated (since $\mathrm{Hom}_R(M, -)$ commutes with colimits). Since projective modules over local rings are free, this implies that $M = R^n$ for some n. Since $k \otimes_R -$ is a strong monoidal functor, $k \otimes_R M \simeq k^n$ is also invertible as a k-module. Since k is a field, we see that $n = 1$ so $M = R$. □

Proposition 8.14. *Let $(R, \mathfrak{m}, k)$ be a commutative Noetherian local ring. If $X \in \mathsf{D}(R)$ is invertible, then $X \simeq \Sigma^n R$ for some $n \in \mathbb{Z}$.*

Proof. Since X is invertible, it is rigid and hence perfect. As such, X has bounded homology. Set $i(X) = \inf\{i \mid H_i X \neq 0\}$. It is a standard fact that for any bounded below X, Y we have

$$H_{i(X)} X \otimes_R H_{i(Y)} Y = H_{i(X)+i(Y)}(X \otimes^{\mathsf{L}}_R Y). \tag{8.15}$$

Taking Y to be the inverse of X, we therefore see that $H_{i(X)} X \otimes_R H_{i(Y)} Y = R$. Hence $H_{i(X)} X$ is an invertible R-module, so by Lemma 8.13 we have $H_{i(X)} X = R$. As $\mathrm{Hom}_{\mathsf{D}(R)}(\Sigma^{i(X)} R, -) = H_{i(X)}(-)$, we obtain a map $f \colon \Sigma^{i(X)} R \to X$ corresponding to the identity element of R. By Lemma 8.11 it suffices to show that $\kappa(\mathfrak{p}) \otimes^{\mathsf{L}}_R f$ is an isomorphism for all $\mathfrak{p} \in \mathrm{Spec}(R)$. (Or alternatively one could use Lemma 8.12 here.) Now $\kappa(\mathfrak{p}) \otimes^{\mathsf{L}}_R X$ is invertible, and hence $\kappa(\mathfrak{p}) \otimes^{\mathsf{L}}_R X \simeq \Sigma^{n_{\mathfrak{p}}} k$ for some integer $n_{\mathfrak{p}}$ as k is a field. By (8.15) we see that $n_{\mathfrak{p}} = i(X)$ for all $\mathfrak{p}$. It follows that $\kappa(\mathfrak{p}) \otimes^{\mathsf{L}}_R f$ is an isomorphism for all $\mathfrak{p} \in \mathrm{Spec}(R)$. Hence f is an isomorphism, so $X \simeq \Sigma^{i(X)} R$ as required. □

8.C. **The Wirthmüller isomorphism.** We now have a pretty good understanding of Grothendieck duality and when it holds. Recall from Theorem 4.8 that the following are equivalent for a geometric functor $f^* \colon \mathsf{T} \to \mathsf{U}$:

(1) Grothendieck duality holds for all $X \in \mathsf{T}$, i.e., $f^* X \otimes \omega_f \simeq f^{(1)} X$ for all $X \in \mathsf{T}$;
(2) f_* preserves compacts;
(3) there is an adjoint quintuple $f_{(1)} \dashv f^* \dashv f_* \dashv f^{(1)} \dashv f_{(-1)}$.

When these hold we say that f^* has *Grothendieck–Neeman duality*. In this case, we moreover have the ur-Wirthmüller isomorphism $f_{(1)} \simeq f_*(\omega_f \otimes -)$.

There are two ways to motivate where we are going next. Firstly, one can ask the natural question, when can the adjoint quintuple in (3) above be extended? Alternatively, one can ask when the ur-Wirthmüller isomorphism can be improved to give a description of f_* in terms of $f_{(1)}$ instead. In turns out these two are equivalent.

We are ready to state and prove the main result of this section.

Theorem 8.16. *Let $f^* \colon \mathsf{T} \to \mathsf{U}$ be a geometric functor which has Grothendieck–Neeman duality. The following are equivalent:*

(1) the functor $f_{(1)}$ has a left adjoint (or equivalently it preserves products);
(2) the functor $f_{(-1)}$ has a right adjoint (or equivalently it preserves coproducts);
(3) the relative dualising object ω_f is compact;
(4) the relative dualising object ω_f is invertible;
(5) there is a Wirthmüller isomorphism, i.e., there exists an invertible object ω such that $f_{(1)} \simeq f_*(\omega \otimes -)$ or equivalently such that $f_* \simeq f_{(1)}(\omega^{-1} \otimes)$;
(6) there is an infinite tower of adjunctions $\cdots \dashv f^{(-n)} \dashv f_{(n)} \dashv f^{(-n+1)} \dashv f_{(n-1)} \dashv \cdots$ for $n \in \mathbb{Z}$ with the convention that $f^* = f^{(0)}$ and $f_* = f_{(0)}$.

Moreover when these equivalent conditions hold the adjoints are related by the formulas

$$f^{(n)} \simeq \omega_f^{\otimes n} \otimes f^*(-) \quad \text{and} \quad f_{(n)} \simeq f_*(\omega_f^{\otimes n} \otimes -). \tag{8.17}$$

Proof. (1) ⇒ (2): We have six functors fitting into adjunctions

$$f^{(-1)} \dashv f_{(1)} \dashv f^* \dashv f_* \dashv f^{(1)} \dashv f_{(-1)}$$

and we want to show that $f_{(-1)}$ preserves coproducts. Equivalently by Proposition 3.3 we want to show that $f^{(1)}$ preserves compacts. By restricting to compact objects we have adjunctions

$$f^{(-1)}|_{\mathsf{T}^c} \dashv f_{(1)}|_{\mathsf{U}^c} \dashv f^*|_{\mathsf{T}^c} \dashv f_*|_{\mathsf{U}^c}.$$

Since f^* is strong monoidal we have a commutative diagram

$$\begin{tikzcd} (\mathsf{T}^c)^{\mathrm{op}} \arrow[r, "D"] \arrow[d, "(f^*)^{\mathrm{op}}"'] & \mathsf{T}^c \arrow[d, "f^*"] \\ (\mathsf{U}^c)^{\mathrm{op}} \arrow[r, "D"'] & \mathsf{U}^c. \end{tikzcd}$$

Since the right adjoint of $(f^*)^{\mathrm{op}}$ is $(f_{(1)})^{\mathrm{op}}$ we obtain that $f_*|_{\mathsf{U}^c} = D \circ f_{(1)}|_{\mathsf{U}^c} \circ D$. We see that this has right adjoint given by $D \circ f^{(-1)}|^{\mathrm{op}}_{\mathsf{T}^c} \circ D$. Therefore we conclude by applying Lemma 4.5.

(2) $\Rightarrow$ (3): If $f_{(-1)}$ preserves coproducts, then its left adjoint $f^{(1)}$ preserves compacts by Proposition 3.3. Therefore $\omega_f = f^{(1)}(\mathbb{1}_{\mathsf{T}})$ is compact.

(3) $\Leftrightarrow$ (4): Recall from (4.18) that $\underline{\mathrm{Hom}}(\omega_f, \omega_f) \simeq \mathbb{1}_{\mathsf{U}}$. Therefore this follows from Lemma 8.7.

(4) $\Rightarrow$ (5): This is immediate by the ur-Wirthmüller isomorphism (4.16).

(5) $\Rightarrow$ (6): Define functors as in (8.17) using ω in place of ω_f. So it is enough to check that for all n we have $\omega^{\otimes(-n)} \otimes f^*(-) \dashv f_*(\omega^{\otimes n} \otimes -) \dashv \omega^{\otimes(-n+1)} \otimes f^*(-)$. For the first of these, we have

$$\mathrm{Hom}(\omega^{\otimes(-n)} \otimes f^*X, Y) = \mathrm{Hom}(f^*X, \omega^{\otimes n} \otimes Y) = \mathrm{Hom}(X, f_*(\omega^{\otimes n} \otimes Y)).$$

The second is similar but uses the Wirthmüller isomorphism in the first step:

$$\begin{aligned} \mathrm{Hom}(f_*(\omega^{\otimes n} \otimes X), Y) &= \mathrm{Hom}(f_{(1)}(\omega^{\otimes(n-1)} \otimes X), Y) \\ &= \mathrm{Hom}(\omega^{\otimes(n-1)} \otimes X, f^*Y) \\ &= \mathrm{Hom}(X, \omega^{\otimes(-n+1)} \otimes f^*Y). \end{aligned}$$

(6) $\Rightarrow$ (1): This is immediate. □

Definition 8.18. We say that f^* is a *Wirthmüller context* if the equivalent conditions of Theorem 8.16 hold.

As an immediate corollary we have the following trichotomy theorem:

Theorem 8.19. *Let $f^*\colon \mathsf{T} \to \mathsf{U}$ be a geometric functor. Then exactly one of the following must hold:*

(1) there is an adjoint triple $f^* \dashv f_* \dashv f^{(1)}$ with no further adjoints;
(2) there is an adjoint quintuple $f_{(1)} \dashv f^* \dashv f_* \dashv f^{(1)} \dashv f_{(-1)}$ with no further adjoints;
(3) there is an infinite tower of adjunctions in both directions.

Proof. By Proposition 3.5 we have at least 3 functors in the adjunction. If there is a fourth on either side, then there must also be a fifth by Theorem 4.8. If there are any more adjoints then there are in fact infinitely many by Theorem 8.16. □

Example 8.20. Let $(R, \mathfrak{m}, k)$ be a local Noetherian commutative ring, and let $f^* = k \otimes^{\mathsf{L}}_R -$ be the extension of scalars functor. We saw in Example 4.28 that f^* has Grothendieck–Neeman duality if and only if R is a regular local ring. Since $\mathrm{Pic}(\mathsf{D}(k)) = \mathbb{Z}$, we see that f^* is a Wirthmüller context if and only if $\omega_f = \mathsf{R}\mathrm{Hom}_R(k, R) \simeq \Sigma^a k$ for some $a \in \mathbb{Z}$. Taking homology this is equivalent to the condition that $\mathrm{Ext}^i_R(k, R)$ is zero in all but one degree. This is a characterisation of Gorenstein rings, and since any regular ring is Gorenstein, we see that Grothendieck–Neeman duality automatically gives the Wirthmüller context in this setting.

Remark 8.21. However, there are many more Gorenstein rings than the regular rings. As such, we want to capture the phenomena one sees for Gorenstein rings without assuming regularity. Said differently, what happens if one assumes that ω_f is invertible, without assuming that f^* has Grothendieck duality? This is the subject of the next few sections of the course.

We end this section by discussing how to verify the existence of a Wirthmüller context.

Proposition 8.22. *Let $f^*\colon \mathsf{T} \to \mathsf{U}$ be a geometric functor satisfying Grothendieck–Neeman duality. Then f^* is a Wirthmüller context if any of the following conditions hold:*

(1) f_* is faithful;
(2) for all $X \in \mathsf{U}$, the counit map $f^*f_*X \to X$ is a split epimorphism;
(3) f^* is surjective up to summands, i.e., for all $X \in \mathsf{U}$ there exists $Y \in \mathsf{T}$ such that X is a retract of f^*Y;
(4) f_* detects compacts, i.e., if $f_*X \in \mathsf{T}^c$ then $X \in \mathsf{U}^c$;
(5) if $f_*(X \otimes Y) \in \mathsf{T}^c$ for all $Y \in \mathsf{U}^c$ then $X \in \mathsf{U}^c$.

Proof. We will first show that (1) $\Leftrightarrow$ (2) $\Leftrightarrow$ (3) $\Rightarrow$ (4) $\Rightarrow$ (5).

(1) $\Rightarrow$ (2): Extend the counit map on X to a triangle

$$f^*f_*X \xrightarrow{\varepsilon_X} X \xrightarrow{\alpha} C.$$

Applying f_* to this triangle, the first map in the resulting triangle $f_*(\varepsilon_X)$ is a split epimorphism by the triangle identities. Moreover, since it is a triangle we have $f_*(\alpha) \circ f_*(\varepsilon_X) = 0$, so $f_*(\alpha) = 0$ as $f_*(\varepsilon_X)$ is an epimorphism. Therefore as f_* is faithful we have $\alpha = 0$. Standard triangulated category theory then tells us that $f^*f_*X \to X$ is a split epimorphism.

(2) $\Rightarrow$ (3): By assumption X is a retract of f^*f_*X.

(3) $\Rightarrow$ (1): By assumption, for each $X \in \mathsf{U}$ there exists $X' \in \mathsf{U}$ and $Y \in \mathsf{T}$ such that $X \oplus X' \simeq f^*Y$. Therefore $\varepsilon_{f^*Y} = \varepsilon_X \oplus \varepsilon_{X'}$. By the triangle identities, ε_{f^*Y} is a split epimorphism and hence ε_X is also an epimorphism. Standard category theory then implies that f_* is faithful.

(2) $\Rightarrow$ (4): if $f_*X \in \mathsf{T}^c$ then $f^*f_*X \in \mathsf{U}^c$. But X is a retract of f^*f_*X by assumption and hence is also compact.

(4) $\Rightarrow$ (5): take $Y = \mathbb{1}_{\mathsf{U}}$.

With these implications proved, it is therefore sufficient to prove that (5) implies that f^* is a Wirthmüller context. We show that it implies that ω_f is compact.

For all $Y \in \mathsf{U}^c$ we have that $f_*(\omega_f \otimes Y)$ is compact since $f_*(\omega_f \otimes Y) \simeq f_{(1)}Y$ by the ur-Wirthmüller isomorphism (4.16). Therefore by condition (5), it follows that ω_f is compact as required. □

The procedure for getting a Wirthmüller isomorphism above is purely formal; one starts with a geometric functor and isolates the isomorphism in terms of features of ω_f. In examples, this is not how we usually proceed. For example, consider the geometric functor given by restriction to a subgroup $\mathrm{res}^G_H\colon \mathsf{StMod}(kG) \to \mathsf{StMod}(kH)$. Here it is standard that res^G_H has a left adjoint (we do not need formal nonsense to get this), and it is basic representation theory that this left adjoint is isomorphic to the right adjoint of res^G_H. Let us also relate these approaches. Said differently, if we know what the correct twist is, we can determine ω_f formally.

Proposition 8.23. *Let $f^*\colon \mathsf{T} \to \mathsf{U}$ be a geometric functor. Suppose that f^* has a left adjoint $f_{(1)}$ and there exists an object $C \in \mathsf{U}$ such that there is a natural Wirthmüller isomorphism $f_* \simeq f_{(1)}(C \otimes -)$. Then $\omega_f \simeq DC$. In particular, if C is compact then f^* is a Wirthmüller context.*

Proof. Since f^* has a left adjoint we have Grothendieck–Neeman duality (see Theorem 4.8). Taking right adjoints of the assumed natural isomorphism we have

$$f^{(1)} \simeq \underline{\mathrm{Hom}}(C, f^*(-)).$$

Evaluating at $\mathbb{1}_{\mathsf{T}}$ yields that $\omega_f \simeq DC$ from which the claim follows. □

Remark 8.24. Hypothetically, the previous result says that one can have Grothendieck–Neeman duality and something looking like the Wirthmüller isomorphism without fitting into the Wirthmüller context. However in practice, this C is compact. For example, for the restriction functor between stable module categories we have $C = k$ (the tensor unit), and for the restriction functor between categories of equivariant spectra, we have $C = S^{-L}$ where L is the tangent H-representation of G/H at the identity coset.

Remark 8.25. In the previous proposition, it would in fact be enough to just assume that $f_*(\mathbb{1}_{\mathsf{U}}) \simeq f_{(1)}(C)$ since from this one can deduce the existence of the Wirthmüller isomorphism. We don't go into details here, and refer the interested reader to Fausk–Hu–May "Isomorphisms between left and right adjoints" for further details.

9. Matlis duality

9.A. **Matlis duality and Gorenstein geometric functors.** Recall that for a commutative Noetherian local ring $(R, \mathfrak{m}, k)$, the geometric functor $f^* = k \otimes_R -$ has $\omega_f = \mathbb{R}\mathrm{Hom}_R(k, R)$. This is invertible if and only if $\mathbb{R}\mathrm{Hom}_R(k, R) \simeq \Sigma^a k$ for some $a \in \mathbb{Z}$, equivalently if R is a *Gorenstein* ring.

This motivates the following terminology:

Definition 9.1. Let $f^* \colon \mathsf{T} \to \mathsf{U}$ be a geometric functor. Then f^* is *Gorenstein* if $\omega_f = f^{(1)}(\mathbb{1}_{\mathsf{T}})$ is invertible in U, i.e., if $\omega_f \in \mathrm{Pic}(\mathsf{U})$.

Remark 9.2. We can now make precise the statement that f^* is a Wirthmüller context if and only if f^* has Grothendieck–Neeman duality and is Gorenstein. We saw in Remark 8.21 that in our guiding example, this collapses, since Grothendieck–Neeman duality already implies Gorenstein. The point of the rest of the course, is what happens if f^* is Gorenstein but does not have Grothendieck–Neeman duality.

Example 9.3. The discussion above shows that for $f^* = k \otimes_R - \colon \mathsf{D}(R) \to \mathsf{D}(k)$ we have that f^* is Gorenstein if and only if $(R, \mathfrak{m}, k)$ is a Gorenstein local ring.

Example 9.4. There is a map of commutative ring spectra $KO \to KU$ given by complexification, and Wood's theorem says that there is a triangle

$$\Sigma KO \to KO \to KU$$

of KO-modules. It follows that the extension of scalars functor $f^* = KU \otimes_{KO} - \colon \mathrm{Mod}(KO) \to \mathrm{Mod}(KU)$ is Gorenstein. Indeed, we have $f^{(1)} = \mathrm{Hom}_{KO}(KU, -)$, so applying this to the triangle above, we obtain a triangle

$$\mathrm{Hom}_{KO}(KU, KO) \to KO \to \Sigma^{-1} KO.$$

Therefore by shifting and rotating, we obtain a triangle

$$\Sigma KO \to KO \to \Sigma^2 \mathrm{Hom}_{KO}(KU, KO).$$

Hence $f^{(1)}(KO) \simeq \Sigma^{-2} KU$ which is clearly invertible.

Example 9.5. Let G be a finite group and consider the commutative ring spectrum $C^*(BG; k) := \underline{\mathrm{Hom}}(\Sigma^\infty BG_+, Hk)$ of cochains where k is a field. This is a model for the cochains in the usual

sense, since

$$\pi_n C^*(BG;k) = \mathrm{Hom}(S^n, \underline{\mathrm{Hom}}(\Sigma^\infty BG_+, Hk)) = \mathrm{Hom}(\Sigma^n BG_+, Hk) = H^{-n}(BG;k).$$

That is, its homotopy computes the group cohomology ring of G with coefficients in k. The geometric functor $f^* = k \otimes_{C^*(BG;k)} -\colon \mathrm{Mod}(C^*(BG;k)) \to \mathrm{Mod}(k)$ is a Gorenstein geometric functor. We may prove this later depending on time and interest of the audience.

Although ω_f has a distinguished role (e.g., through Grothendieck duality), it is in fact convenient to consider a more general form of the previous definition:

Definition 9.6. Let $f^*\colon \mathsf{T} \to \mathsf{U}$ be a geometric functor and $M \in \mathsf{T}$. We say that M is a *Matlis dualising object* if $f^{(1)}(M) \in \mathrm{Pic}(\mathsf{U})$. (Note that f^* is Gorenstein if and only if $\mathbb{1}_{\mathsf{T}}$ is a Matlis dualising object.)

Example 9.7. Let $(R, \mathfrak{m}, k)$ be a commutative Noetherian local ring and $f^* = k \otimes_R -\colon \mathsf{D}(R) \to \mathsf{D}(k)$. Then the injective hull of k, denoted $E(k)$, is a Matlis dualising object. Indeed, as $E(k)$ is injective we have $f^{(1)}(E(k)) = \mathbb{R}\mathrm{Hom}_R(k, E(k)) = \mathrm{Hom}_R(k, E(k)) \simeq k$.

We now justify the terminology of Matlis dualising objects, showing that they give rise to an abstract form of Matlis duality. Let $f^*\colon \mathsf{T} \to \mathsf{U}$ be a geometric functor and $M \in \mathsf{T}$. We write $D_M \coloneqq \underline{\mathrm{Hom}}(-, M)$ for the M-dual, and note that there is a natural transformation

$$\varphi_X^M \colon X \to D_M^2(X)$$

for all $X \in \mathsf{T}$, defined as the adjunct of the evaluation map $\mathrm{ev}_{X,M}\colon X \otimes \underline{\mathrm{Hom}}(X, M) \to M$. Likewise, we define $\psi_Y^M \colon Y \to D^2_{f^{(1)}M} Y$ for all $Y \in \mathsf{U}$. These fit into the following commutative diagram:

$$\begin{tikzcd} f_*Y \arrow[r, "f_*\psi_Y^M"] \arrow[d, "\varphi^M_{f_*Y}"'] & f_* D^2_{f^!M}(Y) \arrow[d, "\simeq"] \\ D_M^2(f_*Y) \arrow[r, "\simeq"'] & D_M f_* D_{f^!M}(Y) \end{tikzcd} \tag{9.8}$$

where the indicated natural isomorphisms come from (3.9).

Proposition 9.9. *Let $f^*\colon \mathsf{T} \to \mathsf{U}$ be a geometric functor, and let $M \in \mathsf{T}$ be Matlis dualising. Then the natural map $\varphi_X^M \colon X \to D_M^2(X)$ is an isomorphism for all $X \in \mathrm{thick}_\otimes(f_*(\mathsf{U}^c))$. Moreover there is an equivalence of categories*

$$D_M\colon \mathrm{thick}_\otimes(f_*(\mathsf{U}^c))^{\mathrm{op}} \xrightarrow{\simeq} \mathrm{thick}_\otimes(f_*(\mathsf{U}^c)).$$

Proof. Let $Y \in \mathsf{U}^c$. By considering the commutative diagram (9.8), we see that $\varphi^M_{f_*Y}$ is a natural isomorphism if and only if $f_*\psi_Y^M$ is an isomorphism. Since Y is rigid and $f^!M$ is invertible, the latter is an isomorphism, and hence $\varphi^M_{f_*Y}$ is an isomorphism. If $C \in \mathsf{T}^c$, then for all $X \in \mathsf{T}$, the map $\varphi^M_{C\otimes X}$ may be identified (up to natural isomorphism) with $C \otimes \varphi_X^M$. As such, the collection of X for which φ_X^M is an isomorphism is a thick $\otimes$-ideal, and hence φ_X^M is an isomorphism for all $X \in \mathrm{thick}_\otimes(f_*(\mathsf{U}^c))$.

To conclude the equivalence of categories it suffices to show that if $X \in \mathrm{thick}_\otimes(f_*(\mathsf{U}^c))$ then $D_M(X) \in \mathrm{thick}_\otimes(f_*(\mathsf{U}^c))$. By a thick subcategory argument we reduce to showing that if $Y \in \mathsf{U}^c$ then $D_M(f_*Y) \in \mathrm{thick}_\otimes(f_*(\mathsf{U}^c))$. By (3.9) we have

$$\underline{\mathrm{Hom}}(f_*Y, M) \simeq f_*\underline{\mathrm{Hom}}(Y, f^!M) \simeq f_*(DY \otimes f^!M).$$

Since M is Matlis dualising, $f^!M$ is rigid, so we conclude. $\square$

Example 9.10. Let $(R, \mathfrak{m}, k)$ be a commutative Noetherian local ring, and let $f^*\colon \mathsf{D}(R) \to \mathsf{D}(k)$ be the extension of scalars $k \otimes^{\mathsf{L}}_R -$ along $R \to k$. Then the injective hull $E(k)$ is a Matlis dualising object and the previous result generalises classical Matlis duality. Recall that an R-module M has finite length if and only if it has a finite composition series, i.e., a filtration

$$0 = M_0 \subsetneq M_1 \subsetneq M_2 \subsetneq \cdots \subsetneq M_n = M$$

where each quotient M_{i+1}/M_i is simple (that is, has no non-zero submodules). The simple modules in this context are just the residue field k. To see this, suppose that N is simple. Then for any non-zero $n \in N$, the map $f\colon R \to N$ given by $f(r) = rn$ is surjective by simplicity (since the image is the cyclic submodule nR of N). Therefore $N = R/\ker(f)$. We claim that $\ker(f)$ must be maximal. If not, then $\ker(f) \subsetneq \mathfrak{m}$ and so $\mathfrak{m}/\ker(f) \subseteq R/\ker(f)$ which would contradict simplicity of $R/\ker(f)$. Putting this all together we see that an R-module M has finite length if and only if it has a finite filtration with subquotients isomorphic to k. This means that M is in the thick subcategory of $\mathsf{D}(R)$ generated by k. By the previous result, $D_{E(k)}\colon \operatorname{thick}(k)^{\mathrm{op}} \to \operatorname{thick}(k)$ is an equivalence of categories. In particular, $M \simeq D^2_{E(k)}M$ if M is a finite length module. (This is the classical statement of Matlis duality.)

9.B. **Classifying Matlis dualising objects in terms of Matlis duality.** We can actually make the relationship between the existence of Matlis duality and Matlis dualising objects precise. To formulate this we make the following definition:

Definition 9.11. Let $f^*\colon \mathsf{T} \to \mathsf{U}$ be a geometric functor. We say that $M \in \mathsf{T}$ is an *external dualising object* if $D_M\colon \operatorname{thick}_\otimes(f_*(\mathsf{U}^c)) \to \operatorname{thick}_\otimes(f_*(\mathsf{U}^c))$ is an equivalence of categories. Equivalently, if φ^M_Y is an isomorphism for all $Y \in \operatorname{thick}_\otimes(\mathsf{U}^c)$.

Proposition 9.9 shows that Matlis dualising objects are external dualising objects. We now characterise when the converse holds, at least when f_* is conservative.

Proposition 9.12. *Let $f^*\colon \mathsf{T} \to \mathsf{U}$ be a geometric functor and suppose that f_* is conservative. The following are equivalent for $M \in \mathsf{T}$:*

(1) M is a Matlis dualising object;
(2) M is an external dualising object and $f^{(1)}M$ is rigid;
(3) $\psi^M_{\mathbb{1}_\mathsf{U}}\colon \mathbb{1}_\mathsf{U} \to \underline{\mathrm{Hom}}(f^{(1)}M, f^{(1)}M)$ is an isomorphism and $f^{(1)}M$ is rigid.

Proof. (1) $\Rightarrow$ (2): this follows from Proposition 9.9.

(2) $\Rightarrow$ (3): Since M is an external dualising object, $\varphi^M_X\colon X \to D^2_M X$ is an isomorphism for all $X \in \operatorname{thick}_\otimes(f_*(\mathsf{U}^c))$. Taking $X = f_*\mathbb{1}_\mathsf{U}$ and using the commutative diagram (9.8), this gives that $f_*\mathbb{1}_\mathsf{U} \to f_*D^2_{f^{(1)}M}(\mathbb{1}_\mathsf{U}) = f_*\underline{\mathrm{Hom}}(f^{(1)}M, f^{(1)}M)$ is an isomorphism. Since f_* is conservative, we conclude.

(3) $\Rightarrow$ (1): this is immediate from the definitions (e.g. see Lemma 8.7). □

As an immediate consequence we obtain the following for $M = \mathbb{1}_\mathsf{T}$:

Proposition 9.13. *Let $f^*\colon \mathsf{T} \to \mathsf{U}$ be a geometric functor and suppose that f_* is conservative. Then f^* is Gorenstein if and only if ω_f is compact and $f_*\mathbb{1}_\mathsf{U}$ is reflexive (i.e., $f_*\mathbb{1}_\mathsf{U} \to D^2(f_*\mathbb{1}_\mathsf{U})$ is an isomorphism).*

Remark 9.14. It is worth comparing this to the setting with Grothendieck–Neeman duality. In that context, $f_*\mathbb{1}_\mathsf{U}$ is compact since $f^{(1)}$ preserves coproducts, and hence is automatically reflexive. As such, the previous result says that if f^* has Grothendieck–Neeman duality, then it is a Wirthmüller context if and only if ω_f is compact.

By restricting the kinds of U we consider, one can also get sharper results.

Definition 9.15. A compactly generated triangulated category U is *pure semisimple* if every object of U is a coproduct of compact objects. This implies that U^c is a Krull-Schmidt category. Therefore any compact object of U may be written as a finite coproduct of indecomposable compact objects in a unique way up to permutation of factors.

The feature of pure semisimple triangulated categories we require are the following two lemmas:

Lemma 9.16. *Let U be a pure semisimple triangulated category, and $C \in \mathsf{U}^c$. There exists an $N \geq 0$, such that if C can be written as a sum of n non-zero compact objects, then $n \leq N$.*

Proof. We can write $C = \oplus_{i=1}^N A_i$ for indecomposable compacts A_i by the Krull-Schmidt property. If $C = \oplus_{j=1}^n B_j$ for some non-zero compacts B_j then again by the Krull-Schmidt property we have $C = \oplus_{j=1}^n \oplus_{k=1}^{\alpha_j} E_k$ for indecomposable compacts E_k. Since decompositions into indecomposable compacts are unique up to permutation, we must have $N = \sum_{j=1}^n \alpha_j$. In particular, $n \leq N$. □

Lemma 9.17. *Let U be a pure semisimple triangulated category and $X \in \mathsf{U}$. If DX is compact then X is compact.*

Proof. Since U is pure semisimple we may write $X = \bigoplus_{i \in I} C_i$ for some non-zero indecomposable compacts C_i. We wish to show I is finite, so suppose not for contradiction. We have $DX = \prod_{i \in I} D(C_i)$, and since I is infinite, for any n we can choose a partition $I = \coprod_{r=1}^n I_r$ of I into non-empty subsets I_r, so that $DX = \bigoplus_{r=1}^n \prod_{i \in I_r} D(C_i)$. This contradicts Lemma 9.16, so I must be finite. □

Remark 9.18. This is false for general compactly generated triangulated categories. For example, let $R = k[x_1, x_2, \ldots]$ be the infinite polynomial ring. Then this fails in $\mathsf{D}(R)$: we have $D(k) = \mathrm{Hom}_R(k, R) = 0$ is compact, but k is not compact. (We showed that k is not even proxy-small in Example 5.9.)

Proposition 9.19. *Let $f^* \colon \mathsf{T} \to \mathsf{U}$ be a geometric functor, and suppose that U is pure semisimple and f_* is conservative. Let $M \in \mathsf{T}$. Then M is a Matlis dualising object if and only if it is an external dualising object.*

Proof. By Proposition 9.12 it suffices to show that if M is an external dualising object, then $f^{(1)}M$ is rigid. Since M is an external dualising object, we have $\mathbb{1}_{\mathsf{U}} \simeq \underline{\mathrm{Hom}}(f^{(1)}M, f^{(1)}M)$. We may write $f^{(1)}M = \oplus_{i \in I} C_i$ for non-zero compacts C_i by pure semisimplicity. Therefore $\mathbb{1}_{\mathsf{U}} \simeq \prod_{i \in I} \oplus_{j \in I} \underline{\mathrm{Hom}}(C_i, C_j)$. If I were infinite, then for any n we may choose a partition of $I = \coprod_{r=1}^n I_r$ and thus have that $\mathbb{1}_{\mathsf{U}} \simeq \oplus_{r=1}^n \prod_{i \in I_r} \oplus_{j \in I} \underline{\mathrm{Hom}}(C_i, C_j)$. This contradicts Lemma 9.16 and hence I is finite showing that $f^{(1)}M$ is compact. □

Taking $M = \mathbb{1}_{\mathsf{T}}$ in the previous we obtain the following corollary:

Corollary 9.20. *Let $f^* \colon \mathsf{T} \to \mathsf{U}$ be a geometric functor, and suppose that U is pure semisimple and f_* is conservative. Then f^* is Gorenstein if and only if $f_*\mathbb{1}_{\mathsf{U}}$ is reflexive.*

Recall the trichotomy theorem from Theorem 8.19 showing that a geometric functor fits into either an adjoint triple, an adjoint quintuple, or an infinite tower of adjoints. The previous shows that this collapses to a dichotomy theorem under the pure semisimplicity and conservativity assumptions:

Corollary 9.21. *Let $f^* \colon \mathsf{T} \to \mathsf{U}$ be a geometric functor and suppose that U is pure semisimple and f_* is conservative. Then exactly one of the following holds:*

(1) f^* fits into an adjoint triple $f^* \dashv f_* \dashv f^{(1)}$ with no further adjunctions;
(2) f^* fits into an infinite tower of adjunctions.

Proof. By Theorem 8.19 it suffices to show that if f^* has Grothendieck–Neeman duality then f^* is a Wirthmüller context. If f^* has Grothendieck–Neeman duality then f_* preserves compacts since $f^{(1)}$ preserves coproducts. Therefore $f_*\mathbb{1}_{\mathsf{U}}$ is compact and in particular reflexive. We conclude by Corollary 9.20. □

Remark 9.22. This gives a general, abstract justification for the regular implies Gorenstein argument used to see that Grothendieck–Neeman duality automatically gives a Wirthmüller context for $f^* = k \otimes^{\mathsf{L}}_R -$ for a commutative Noetherian local ring $(R, \mathfrak{m}, k)$. It also shows that the Gorenstein local rings are characterised by the property that k is reflexive in $\mathsf{D}(R)$.

9.C. **Classifying Matlis dualising objects by proxy-smallness.** In this section we give a characterisation of Matlis dualising objects in terms of the behaviour of torsion and completion functors. The results of the previous subsections classified Matlis dualising objects in terms of their duality features so it was 'backwards'. The point should instead be that we want to produce Matlis dualising objects in order to prove duality statements, rather than the converse. The classification in this section does this.

We first have the following analogue of Theorem 7.4.

Theorem 9.23. *Let $f^* \colon \mathsf{T} \to \mathsf{U}$ be a proxy-small geometric functor and let $M \in \mathsf{T}$. The following are equivalent:*

(1) the evaluation map $\mathrm{ev}_M \colon M \otimes DM \to \mathbb{1}_{\mathcal{C}}$ is an f^*-isomorphism, i.e., $f^*(\mathrm{ev}_M)$ is an isomorphism;
(2) $\Gamma_f M \in \Gamma_f \mathsf{T}$ is invertible;
(3) $\Lambda_f M \in \Lambda_f \mathsf{T}$ is invertible;
(4) $f^* M \in \mathsf{U}$ is invertible.

Proof. Recall that an object M is invertible if and only if $\mathrm{ev}_M \colon DM \otimes M \to \mathbb{1}$ is an isomorphism. The proof is then analogous to Theorem 7.4.

(1) ⇔ (3): We have a commutative diagram

$$\begin{tikzcd}
\Lambda_f(DM) \widehat{\otimes} \Lambda_f M \arrow[r, "\simeq"] \arrow[d, "\alpha_{M,\mathbb{1}} \widehat{\otimes} \Lambda_f M"'] & \Lambda_f(DM \otimes M) \arrow[d, "\Lambda_f \mathrm{ev}_M"] \\
D_{\Lambda_f}(\Lambda_f M) \widehat{\otimes} \Lambda_f M \arrow[r, "\mathrm{ev}_{\Lambda_f M}"'] & \Lambda_f \mathbb{1}_{\mathcal{C}}.
\end{tikzcd}$$

We note that $\Lambda_f \colon \mathsf{T} \to \Lambda_f \mathsf{T}$ is closed monoidal, and so the left vertical arrow in the diagram is an isomorphism. Therefore, $\mathrm{ev}_{\Lambda_f M}$ is an isomorphism if and only if ev_M is a Λ_f-isomorphism, which is the same as an f^*-isomorphism by Proposition 6.26.

(2) ⇔ (3): Since the MGM equivalence is monoidal it preserves invertible objects.

(2) ⇒ (4): The restriction of f^* to $\Gamma_f \mathsf{T}$ is strong symmetric monoidal, and thus preserves invertible objects. Since $f^* \simeq f^* \Gamma_f$, we therefore see that if $\Gamma_f M$ is invertible in $\Gamma_f \mathsf{T}$, then $f^* M \simeq f^* \Gamma_f M$ is invertible in U.

(4) $\Rightarrow$ (1): We have a commutative diagram

$$\begin{tikzcd} f^*(DM) \otimes f^*M \arrow[r, "\simeq"] \arrow[d, "\alpha_{M,\mathbb{1}} \otimes f^*M"'] & f^*(DM \otimes M) \arrow[d, "f^*\mathrm{ev}_M"] \\ D(f^*M) \otimes f^*M \arrow[r, "\mathrm{ev}_{f^*M}"'] & f^*\mathbb{1}_{\mathsf{T}} = \mathbb{1}_{\mathsf{U}}. \end{tikzcd}$$

Since f^*M is assumed to be invertible and hence rigid, by (7.14) the left hand vertical is an isomorphism. Therefore we see that ev_{f^*M} being an isomorphism implies $f^*(\mathrm{ev}_M)$ is one too. $\square$

Remark 9.24. Let us immediately dispel a misunderstanding. We always have a factorisation

$$\begin{tikzcd} \mathrm{Pic}(\mathsf{T}) \arrow[d] \arrow[dr] & \\ \mathrm{Pic}(\Gamma_f \mathsf{T}) \arrow[r] & \mathrm{Pic}(\mathsf{U}). \end{tikzcd}$$

However, the previous theorem does not say that $f^* \colon \mathrm{Pic}(\Gamma_f \mathsf{T}) \to \mathrm{Pic}(\mathsf{U})$ is an isomorphism. In fact, it is even possible for both T and U to have trivial Picard group and $\mathrm{Pic}(\Gamma_f \mathsf{T})$ to not have trivial Picard group. The injectivity of the map $\mathrm{Pic}(\Gamma_f \mathsf{T}) \to \mathrm{Pic}(\mathsf{U})$ is important in the study of Gorenstein duality as we will see in the next section.

Example 9.25. Let $(R, \mathfrak{m}, k)$ be a commutative Noetherian local ring. The Picard group of $\Gamma_{\mathfrak{m}} \mathsf{D}(R)$ is trivial. Indeed, if $M \in \mathrm{Pic}(\Gamma_{\mathfrak{m}} \mathsf{D}(R))$ then $\Lambda_{\mathfrak{m}} M \in \mathrm{Pic}(\Lambda_{\mathfrak{m}} \mathsf{D}(R))$ and hence by Theorem 9.23 we have $f^*M = k \otimes^{\mathsf{L}}_R M \in \mathrm{Pic}(\mathsf{D}(k))$. This means that $k \otimes^{\mathsf{L}}_R M \simeq \Sigma^a k$ for some $a \in \mathbb{Z}$. By Corollary 7.9, this means that $\Lambda_{\mathfrak{m}} M \simeq \Sigma^a \Lambda_{\mathfrak{m}} R$. We conclude by the MGM equivalence. Note that Corollary 7.9 actually gives a stronger statement than Theorem 9.23 in this specific example.

As such it remains to relate the invertibility of f^*M with the invertibility of $f^{(1)}M$. We achieve this through Grothendieck duality to obtain the following:

Theorem 9.26. *Let $f^* \colon \mathsf{T} \to \mathsf{U}$ be a proxy-small geometric functor with U pure semisimple. If f^* is Gorenstein then the following are equivalent for $M \in \mathsf{T}$:*

(1) M is Matlis dualising (i.e., $f^{(1)}M \in \mathrm{Pic}(\mathsf{U})$);
(2) $f^*M \in \mathrm{Pic}(\mathsf{U})$;
(3) $\Gamma_f M \in \mathrm{Pic}(\Gamma_f \mathsf{T})$;
(4) $\Lambda_f M \in \mathrm{Pic}(\Lambda_f \mathsf{T})$.

Proof. Conditions (2)-(4) are equivalent by Theorem 9.23, so it remains to prove the equivalence with (1).

(2) $\Rightarrow$ (1): Since $f^*M \in \mathrm{Pic}(\mathsf{U})$, we have that f^*M is rigid. Therefore Theorem 7.15 applies, so we have the Grothendieck duality isomorphism $\omega_f \otimes f^*M \simeq f^{(1)}M$. By assumption, both ω_f and f^*M are invertible, and hence so is $f^{(1)}M$.

(1) $\Rightarrow$ (2): We first show that $\varphi^M_{\mathbb{1}_{\mathsf{T}}} \colon \mathbb{1}_{\mathsf{T}} \to D^2_M(\mathbb{1}_{\mathsf{T}}) = \underline{\mathrm{Hom}}(M, M)$ is an $f^{(1)}$-isomorphism. By Proposition 6.26, this is equivalent to checking that $W \otimes \varphi^M_{\mathbb{1}_{\mathsf{T}}}$ is an isomorphism for all $W \in \mathcal{W} \subseteq \mathrm{thick}_{\otimes}(f_* \mathbb{1}_{\mathsf{U}})$. By Proposition 9.9, we have that φ^M_W is an isomorphism. Since W is rigid, $\varphi^M_W = W \otimes \varphi^M_{\mathbb{1}_{\mathsf{T}}}$ so we conclude that

$$\omega_f = f^{(1)}(\mathbb{1}_{\mathsf{T}}) \xrightarrow{\simeq} f^{(1)} \underline{\mathrm{Hom}}(M, M)$$

is an isomorphism. Using (3.10) and the fact that $f^{(1)}M$ is invertible (hence rigid), we see that $\omega_f \simeq f^{(1)}\underline{\mathrm{Hom}}(M,M) \simeq \underline{\mathrm{Hom}}(f^*M, f^{(1)}M) \simeq D(f^*M) \otimes f^{(1)}M$. Therefore by the 2-out-of-3 property of invertible objects, we see that $D(f^*M) \in \mathrm{Pic}(\mathsf{U})$ and hence $D(f^*M)$ is rigid. By pure semisimplicity it follows that f^*M is rigid (see Lemma 9.17), and hence Grothendieck duality $\omega_f \otimes f^*M \simeq f^{(1)}M$ holds. Since both ω_f and $f^{(1)}M$ are invertible, we deduce that f^*M is also invertible as required. □

10. Orientability and Gorenstein duality

In this section we explain a duality phenomenon experienced by some Gorenstein geometric functors. Recall that a commutative Noetherian local ring $(R, \mathfrak{m}, k)$ is Gorenstein if and only if $\mathrm{Ext}^*_R(k, R)$ is k in a single degree if and only if the local cohomology satisfies

$$H^*_{\mathfrak{m}}(R) = \begin{cases} E(k) & * = \dim(R) \\ 0 & \text{otherwise.} \end{cases}$$

Our definition of Gorenstein geometric functors was modelled on the former characterisation of Gorenstein rings. We call this latter characterisation *Gorenstein duality*. Although these conditions are equivalent for a commutative Noetherian local ring, they are not equivalent in general for geometric functors. Let us first formulate this duality feature in the setting of geometric functors.

We need to understand two things: what the local cohomology is, and what the injective hull is. The first is straightforward: Γ_f is the appropriate analogue of local cohomology. But without additional assumptions, it is not clear what the correct analogue of $E(k)$ is for a general geometric functor $f^*\colon \mathsf{T} \to \mathsf{U}$. However, recall that if $(R, \mathfrak{m}, k)$ is moreover a k-algebra, then $E(k) = \mathrm{Hom}_k(R, k)$. This can be easily generalised into our setting:

Definition 10.1. We say that $f^*\colon \mathsf{T} \to \mathsf{U}$ *admits a geometric section* if there is a geometric functor $\eta^*\colon \mathsf{U} \to \mathsf{T}$ such that $f^*\eta^* \simeq \mathrm{id}_{\mathsf{U}}$. Therefore diagrammatically we have

$$\begin{tikzcd} \mathsf{U} \arrow[r, shift left=2, "\eta^*"] \arrow[r, shift right=2, "\eta^{(1)}"'] \arrow[rr, bend left, "\mathrm{id}_{\mathsf{U}}"] & \mathsf{T} \arrow[l, "\eta_*"] \arrow[r, shift left=2, "f^*"] \arrow[r, shift right=2, "f^{(1)}"'] & \mathsf{U} \arrow[l, "f_*"] \end{tikzcd}$$

and we note that it follows that $\eta_* f_* \simeq \mathrm{id}_{\mathsf{U}}$ and $f^{(1)}\eta^{(1)} \simeq \mathrm{id}_{\mathsf{U}}$. In analogy with $\omega_f := f^{(1)}(\mathbb{1}_{\mathsf{T}})$, we set $\omega_\eta := \eta^{(1)}(\mathbb{1}_{\mathsf{U}})$. Notice that if f^* has a geometric section η^*, then both f_* and η^* are conservative: $f_*X \simeq 0$ implies $X \simeq \eta_* f_* X \simeq 0$ and similarly for η^*.

Examples 10.2.

(i) Let $\mathcal{C}$ be an enhanced rigidly-compactly generated tt-category, and let $f\colon R \to S$ be a map of commutative algebras in $\mathcal{C}$. If R is an S-algebra with unit map $\eta\colon S \to R$, then extension of scalars along η is a geometric section for extension of scalars along the augmentation f.

(ii) Let G be a compact Lie group, and let $f^* = \mathrm{res}^G_1\colon \mathrm{Sp}_G \to \mathrm{Sp}$ be the restriction functor to the trivial group. This has a geometric section given by the inflation functor $\eta^* = \mathrm{inf}^G_1$.

This suggests the following analogue of the Gorenstein duality statement in our generality:

$$\omega_\eta = \Sigma^a \Gamma_f \mathbb{1}_{\mathsf{T}}.$$

However this has some shortcomings:

(1) In the algebraic setting, $\mathrm{Hom}_k(R, k)$ is automatically in the image of Γ_f (since $k \in \mathrm{Loc}(R)$ so $\mathrm{Hom}_k(R, k) \in \mathrm{Loc}(\mathrm{Hom}_k(k, k)) = \mathrm{Loc}(k)$). However, in general there is no reason for

ω_η to be in the image of Γ_f. This would immediate rule out the definition holding since the right hand side is in the image of Γ_f.

(2) It is convenient to introduce a little more flexibility in the right hand side too. One of the messages of previous sections is that restricting to suspensions of the unit, it often unnecessary, and we might as well assume that it is just some invertible object instead. On the other hand, for a bit more control, rather than any invertible object, we ask that it comes from an invertible object in U, i.e., is of the form $\eta^*(X)$ for some $X \in \mathrm{Pic}(\mathsf{U})$.

This leads us to the definition:

Definition 10.3. A geometric functor $f^*\colon \mathsf{T} \to \mathsf{U}$ with a geometric section $\eta^*\colon \mathsf{U} \to \mathsf{T}$ is said to have *Gorenstein duality* if there is an $X \in \mathrm{Pic}(\mathsf{U})$ such that $\Gamma_f\omega_\eta \simeq \Gamma_f\eta^*(X)$.

Remark 10.4. To give some idea of what the definition is doing beyond the setting of local rings, consider a finite CW-complex X and a self-injective local Noetherian ring R. Then the geometric functor given by extension of scalars along the map of commutative ring spectra $C^*(X;R) \to R$ has Gorenstein duality if and only if X has Poincaré duality with respect to R, see Proposition 11.4.

We observe that Gorenstein duality implies the Gorenstein property itself.

Proposition 10.5. *Suppose $f^*\colon \mathsf{T} \to \mathsf{U}$ is a proxy-small geometric functor with geometric section η^*. If f^* has Gorenstein duality, then f^* is Gorenstein.*

Proof. Applying the MGM-equivalence to the Gorenstein duality statement we have that $\Lambda_f\omega_\eta \simeq \Lambda_f\eta^*(X)$ for some $X \in \mathrm{Pic}(\mathsf{U})$. Applying $f^{(1)}$ to this together with Proposition 6.25, we obtain that $f^{(1)}\omega_\eta \simeq f^{(1)}\eta^*(X)$. Therefore

$$\mathbb{1}_{\mathsf{U}} \simeq f^{(1)}\eta^{(1)}(\mathbb{1}_{\mathsf{U}}) \simeq f^{(1)}(\omega_\eta) \simeq f^{(1)}\eta^*(X).$$

Since $f^*\eta^*(X) \simeq X$ is rigid, by Grothendieck duality (7.17) it follows that $\mathbb{1}_{\mathcal{D}} \simeq \omega_f \otimes X$. Therefore ω_f is invertible with inverse X. □

Remark 10.6. The proof of the previous result shows that Gorenstein duality is in fact equivalent to $\Gamma_f\omega_\eta \simeq \Gamma_f\eta^*(\omega_f^{-1})$, that is, one may always take $X = \omega_f^{-1}$ in Definition 10.3.

Unlike in algebra, in the generality of geometric functors, Gorenstein duality is not equivalent to being Gorenstein. We will give explicit examples justifying this in the next section. For now though, we deal with the theoretical aspects. In order to state a precise converse, we introduce the following terminology.

Definition 10.7. Let $f^*\colon \mathsf{T} \to \mathsf{U}$ be a geometric functor, and let $Y \in \mathsf{U}$. We say that an object $X \in \mathsf{T}$ is a *Matlis lift* of Y if $f^{(1)}X \simeq Y$.

The object ω_f plays a distinguished role (e.g., through Grothendieck duality), so we are particularly interested in Matlis lifts of ω_f. Moreover, if f^* is Gorenstein then any Matlis lift of ω_f is a Matlis dualising object. The following definition singles out a particular class of these Matlis lifts of interest.

Definition 10.8. Let $f^*\colon \mathsf{T} \to \mathsf{U}$ be a proxy-small geometric functor, and let M be a Matlis lift of ω_f. We say that M is an *orientable Matlis lift of* ω_f if $\Gamma_f M \simeq \Gamma_f\mathbb{1}_{\mathsf{T}}$. We say that f^* is *automatically orientable* if every Matlis lift of ω_f is orientable.

Remark 10.9. We warn the reader that Proposition 6.26 does *not* imply that any f^* is automatically orientable. Proposition 6.26 ensures that a morphism $X \to Y$ is an $f^{(1)}$-isomorphism if and only if it is a Γ_f-isomorphism, however here we only have an abstract isomorphism $f^{(1)}M \simeq \omega_f$ which need not be the image under $f^{(1)}$ of a morphism in T.

Proposition 10.10. *Let $f^*\colon \mathsf{T} \to \mathsf{U}$ be a proxy-small, Gorenstein geometric functor with a geometric section η^*. Then f^* has Gorenstein duality if and only if $\eta^{(1)}(\omega_f)$ is an orientable Matlis lift of ω_f.*

Proof. We note that since $\omega_f \in \mathcal{D}$ is invertible, it is rigid, and so Grothendieck duality shows that $\omega_\eta \otimes \eta^*(\omega_f) \simeq \eta^{(1)}(\omega_f)$. (Note that Grothendieck duality holds here without the proxy-smallness assumption on η^* since ω_f is rigid and not just η^*-rigid, see Lemma 4.3) Therefore the statement of orientability of $\eta^{(1)}(\omega_f)$ is precisely

$$\Gamma_f(\eta^*(\omega_f) \otimes \omega_\eta) \simeq \Gamma_f \mathbb{1}_{\mathcal{C}}$$

which by invertibility of ω_f is equivalent to the Gorenstein duality statement

$$\Gamma_f \omega_\eta \simeq \Gamma_f \eta^*(\omega_f^{-1}).$$

Since the X in the statement of Gorenstein duality must always be the inverse to ω_f (Remark 10.6), this shows the two conditions are always equivalent. □

This is the primary usage of automatic orientability: it ensures that the Gorenstein condition and Gorenstein duality are always equivalent:

Corollary 10.11. *Let $f^*\colon \mathcal{C} \to \mathcal{D}$ be a proxy-small geometric functor with a geometric section η^*, and suppose that f^* is automatically orientable. Then f^* is Gorenstein if and only if it has Gorenstein duality.*

Proof. If f^* has Gorenstein duality, then it is Gorenstein by Proposition 10.5. Conversely, automatic orientability applies to $\eta^!(\omega_f)$ and so the Gorenstein property implies Gorenstein duality by Proposition 10.10. □

Remark 10.12. We summarise the implications between the Gorenstein property, Gorenstein duality, and automatic orientability in the following diagram:

$$\begin{array}{c} \text{Gorenstein and} \\ \text{automatically orientable} \end{array} \underset{\not\Longleftarrow}{\overset{(10.11)}{\Longrightarrow}} \begin{array}{c} \text{Gorenstein} \\ \text{duality} \end{array} \underset{\not\Longleftarrow}{\overset{(10.5)}{\Longrightarrow}} \text{Gorenstein}$$

As such, it is convenient to recast automatic orientabilty in a form which can be easier to check in practice.

Proposition 10.13. *Let $f^*\colon \mathsf{T} \to \mathsf{U}$ be a proxy-small, Gorenstein, geometric functor. If f^* is automatically orientable then the group homomorphism*

$$f^*\colon \mathrm{Pic}(\Gamma_f \mathsf{T}) \to \mathrm{Pic}(\mathsf{U})$$

is injective. If U is moreover pure semisimple then the converse implication also holds.

Proof. If $f^*M \simeq \mathbb{1}_{\mathsf{U}}$ then by Grothendieck duality (7.17), it follows that M is a Matlis lift of ω_f so we conclude that the group homomorphism $f^*\colon \mathrm{Pic}(\Gamma_f \mathsf{T}) \to \mathrm{Pic}(\mathsf{U})$ is injective by the definition of orientability. For the converse, we now assume that U is pure semisimple. Suppose that M is a Matlis lift of ω_f. Since ω_f is invertible as f^* is Gorenstein, M is therefore a Matlis dualising object, so by Theorem 9.26, we have that f^*M is invertible and hence rigid. Therefore by Grothendieck duality (7.17) and the invertibility of ω_f, we have that $f^*M \simeq \mathbb{1}_{\mathsf{U}}$. Since $f^*\colon \mathsf{T} \to \mathsf{U}$ factors over $\Gamma_f \mathsf{T}$ by Proposition 6.25 the claim follows by injectivity. □

Remark 10.14. One can give an alternative characterisation of automatic orientability in terms of Morita theory by studying the endomorphism ring $\underline{\mathrm{Hom}}(f_*\mathbb{1}_{\mathsf{U}}, f_*\mathbb{1}_{\mathsf{U}})$. This gets a bit fiddly so we omit it, but in practice it is the most effective method for verifying automatic orientability.

Example 10.15. Let $(R, \mathfrak{m}, k)$ be a commutative Noetherian local ring, and consider the geometric functor $f^* = k \otimes^{\mathsf{L}}_R - : \mathsf{D}(R) \to \mathsf{D}(k)$. This is automatically orientable, and this explains why the Gorenstein property and Gorenstein duality are equivalent in this setting. Indeed, by Proposition 10.13 it suffices to show that $f^* \colon \mathrm{Pic}(\Gamma_{\mathfrak{m}}\mathsf{D}(R)) \to \mathrm{Pic}(\mathsf{D}(k))$ is injective, so suppose that $M \in \Gamma_{\mathfrak{m}}\mathsf{D}(R)$ has $k \otimes^{\mathsf{L}}_R M \simeq k$. Then since $k \simeq \Gamma_{\mathfrak{m}} k$, by the MGM equivalence we have $k \otimes^{\mathsf{L}}_R \Lambda_{\mathfrak{m}} M \simeq k$. By Corollary 7.9 we must have $\Lambda_{\mathfrak{m}} M \simeq \Lambda_{\mathfrak{m}} R$. Therefore by the MGM equivalence again, $M \simeq \Gamma_{\mathfrak{m}} R$ proving injectivity.

11. Examples of Gorenstein duality

In this section we focus on the Gorenstein duality property for cochain spectra. Rather than give complete details we instead focus on giving an idea of how one generally argues about Gorenstein geometric functors and what consequences one can draw from this.

Let A be a commutative ring and X be a based topological space. Let $C^*(X; A)$ be the function spectrum $\underline{\mathrm{Hom}}(\Sigma^\infty X_+, A)$, which is a commutative ring spectrum via the cocommutative diagonal on X. There is a map of commutative ring spectra $f \colon C^*(X; A) \to A$ which has a section $\eta \colon A \to C^*(X; A)$. These are induced by maps of based spaces $X_+ \to S^0 \to X_+$ and hence by the obvious maps of unbased spaces $X \to * \to X$. Writing f^* for the extension of scalars along f and likewise for η, we see that f^* is a geometric functor with a geometric section. Note that $\pi_* C^*(X; A) = H^{-*}(X; A)$ as

$$\pi_n C^*(X; A) = \mathrm{Hom}(S^n, C^*(X; A)) = \mathrm{Hom}(S^n, \underline{\mathrm{Hom}}(X_+, A)) = \mathrm{Hom}(\Sigma^n X_+, A) = H^{-n}(X; A).$$

We explain how Gorenstein duality for f^* relates to the behaviour of $H^*(X; A)$. For brevity, we will say that $f \colon C^*(X; A) \to A$ has Gorenstein duality when f^* does.

Let's suppose for simplicity that A is a commutative Noetherian local ring, so that $\mathrm{Pic}(\mathsf{D}(A))$ consists only of shifts of A by Proposition 8.14. The Gorenstein duality statement for $f \colon C^*(X; A) \to A$ then amounts to the statement that

$$\Gamma_f \mathrm{Hom}_R(C^*(X; A), A) \simeq \Sigma^a \Gamma_f C^*(X; A) \tag{11.1}$$

for some $a \in \mathbb{Z}$.

In order to probe this, we need to understand what the torsion functor is doing. In many cases we can be extremely explicit about this:

Lemma 11.2. *We have* $\Gamma_f \mathrm{Hom}_R(C^*(X; A), A) \simeq \mathrm{Hom}_A(C^*(X; A), A)$.

Proof. By definition of Γ_f it suffices to show that $\mathrm{Hom}_A(C^*(X; A), A) \in \mathrm{Loc}_\otimes(A)$ as a $C^*(X; A)$-module. Notice that $\mathrm{Hom}_A(C^*(X; A), A)$ is bounded below (in π_*) since $C^*(X; A)$ is bounded above.

Let M be a bounded below module. We claim that M is in $\mathrm{Loc}(A)$. By truncating we have a triangle

$$F_n \to M \to \tau_{\geq n} M$$

for each n, and M is the homotopy colimit of the F_n. As such it suffices to show that each F_n is built from A. But by construction, F_n has only finitely many non-trivial homotopy groups, each of which is an A module, so we conclude that F_n is finitely built from A as required. □

This lets us refine (11.1) a little: Gorenstein duality holds for $f \colon C^*(X; A) \to A$ if and only if

$$\mathrm{Hom}_A(C^*(X; A), A) \simeq \Sigma^a \Gamma_f C^*(X; A) \tag{11.3}$$

for some $a \in \mathbb{Z}$.

11.A. **Poincaré duality.** We now deal with our first explicit example. For this, we restrict our attention to when X is a finite CW-complex and A is self-injective.

Proposition 11.4. *Let X be a finite CW-complex and A be a self-injective commutative Noetherian local ring. Then $f\colon C^*(X;A) \to A$ has Gorenstein duality if and only if X has Poincaré duality with respect to A.*

Proof. We first claim that $\Gamma_f C^*(X;A) = C^*(X;A)$. To see this is suffices to show that $C^*(X;A) \in \mathrm{Loc}_{\otimes}(A)$. This is believable: since X is a finite CW-complex, $X \in \mathrm{thick}(S^0)$ in spectra, and applying $\underline{\mathrm{Hom}}(-,A)$ we see that $C^*(X;A) \in \mathrm{thick}(A)$ in spectra. However, one must argue that this happens in $C^*(X;A)$-modules rather than in spectra. We omit this argument (which uses the Rothenberg–Steenrod equivalence $C^*(X;A) \simeq \mathrm{Hom}_{C_*(\Omega X;A)}(A,A)$).

By (11.3), $f\colon C^*(X;A) \to A$ therefore has Gorenstein duality if and only if

$$\mathrm{Hom}_A(C^*(X;A),A) \simeq \Sigma^a C^*(X;A)$$

for some $a \in \mathbb{Z}$.

There is a spectral sequence

$$\mathrm{Ext}^*_A(\pi_* C^*(X;A),A) \Rightarrow \pi_*\mathrm{Hom}_A(C^*(X;A),A).$$

Since A is assumed to be self-injective, there are no higher Ext groups so this collapses to give an isomorphism $\mathrm{Hom}_A(\pi_* C^*(X;A),A) = \pi_*\mathrm{Hom}_A(C^*(X;A),A)$.

Suppose that f^* has Gorenstein duality. Then

$$\begin{aligned}
H^*(X;A) &= \pi_{-*}C^*(X;A) && \text{by definition}\\
&= \pi_{-*+a}\mathrm{Hom}_R(C^*(X;A),A) && \text{by Gorenstein duality}\\
&= \mathrm{Hom}_A(\pi_{-*+a}C^*(X;A),A) && \text{by ss argument}\\
&= \mathrm{Hom}_A(H^{*-a}(X;A),A) && \text{by definition}\\
&= H_{*-a}(X;A) && \text{by uni coeff as } A \text{ self-injective.}
\end{aligned}$$

The same argument shows the converse. □

Example 11.5. Consider $X = \mathbb{RP}^2$ and take coefficients in the self-injective ring $A = \mathbb{Z}/4$. Since X is not orientable, Poincaré duality fails and hence $C^*(\mathbb{RP}^2;\mathbb{Z}/4) \to \mathbb{Z}/4$ does not have Gorenstein duality. However, one can show that it is Gorenstein so this shows that Gorenstein duality is indeed a stronger condition than the Gorenstein property in general.

11.B. **Modular representation theory.** Before turning to the key examples of group cohomology, we set the scene.

Let $f\colon R \to k$ be a map of commutative ring spectra, and assume that $\pi_* R$ is Noetherian and local (in the graded sense) and that f induces an isomorphism $\pi_* R/\mathfrak{m} \simeq k$. Note that this holds if R is connective or coconnective since then $\mathfrak{m}$ consists of the elements of positive (co)degree. We write f^* for the extension of scalars functor along f^* and say that $f\colon R \to k$ is Gorenstein if f^* is so.

Lemma 11.6. *If $\pi_* R$ is a Gorenstein ring, then $f\colon R \to k$ is Gorenstein.*

Proof. There is a spectral sequence

$$\mathrm{Ext}^*_{\pi_* R}(k,\pi_* R) \Rightarrow \pi_*\mathrm{Hom}_R(k,R).$$

If $\pi_* R$ is Gorenstein, then the Ext is concentrated in a single degree where it is k so that the spectral sequence collapses to give an isomorphism $k = \mathrm{Ext}^a_{\pi_* R}(k,\pi_* R) = \pi_*\mathrm{Hom}_R(k,R)$. This

means that $\mathrm{Hom}_R(k, R) \simeq \Sigma^{-a}k$ which is invertible in $\mathsf{D}(k)$, so that $f\colon R \to k$ is Gorenstein as claimed. □

A key point is that the converse is false. In fact, one key use of the Gorenstein property for $R \to k$ is to deduce features of $\pi_* R$ even when it is not Gorenstein.

In this setting, one can describe the torsion functor in terms of stable Koszul complexes in spectra just as in algebra: for an element $x \in \pi_* R$ define $K(x)$ to be the cone of $R \xrightarrow{x} R$ and then $K_\infty(x)$ to be the homotopy colimit of the tower $K(x) \to K(x^2) \to \dots$. This gives a filtration on Γ_f coming from the unstable Koszul complexes, which gives rise to a spectral sequence:

$$H^{-s}_{\mathfrak{m}}(\pi_* R)_t \Rightarrow \pi_{t-s}\Gamma_f R. \tag{11.7}$$

Let us now assume moreover that $f\colon R \to k$ has a section η, so that R is a k-algebra. It then follows that f^* has a geometric section given by η^* (the extension of scalars along $\eta\colon k \to R$). As such we are in a context where Gorenstein duality makes sense. As above, we say that $f\colon R \to k$ has Gorenstein duality if f^* does.

We now hone in on the case of interest. Namely, we consider the case $R = C^*(BG; k)$ for a finite group G and a field k. Then $\pi_* R = H^*(G; k)$ is the group cohomology ring. This is local and is Noetherian by Venkov's theorem.

We will first argue that $f\colon C^*(BG; k) \to k$ is Gorenstein. There are various ways to do this, but for our purposes the easiest way is by making use of the following descent result:

Proposition 11.8. *Let $g^*\colon \mathcal{B} \to \mathcal{C}$ and $f^*\colon \mathcal{C} \to \mathcal{D}$ be geometric functors and suppose that f^* is proxy-small. Let $M \in \mathcal{B}$ be a Matlis dualising object for g^*. Then f^* is Gorenstein if and only if M is Matlis dualising for f^*g^*.*

Proof. Since M is Matlis dualising for g^*, we have that $g^!M$ is invertible and hence $f^*g^!M$ is invertible (thus rigid). Therefore by Grothendieck duality (Lemma 4.3), we have an isomorphism

$$f^!g^!M \simeq \omega_f \otimes f^*g^!M \tag{11.9}$$

showing that $f^!g^!M$ is invertible if and only if ω_f is invertible. □

We write $\omega_{gf} \coloneqq f^!g^!(\mathbb{1}_{\mathcal{B}})$ in line with our geometric notation.

Corollary 11.10. *Let $g^*\colon \mathcal{B} \to \mathcal{C}$ and $f^*\colon \mathcal{C} \to \mathcal{D}$ be geometric functors and suppose that f^* is proxy-small. Suppose that g^* is Gorenstein. Then there is an isomorphism*

$$\omega_{gf} \simeq \omega_f \otimes f^*\omega_g$$

and thus f^* is Gorenstein if and only if f^*g^* is Gorenstein.

Proof. This is the case $M = \mathbb{1}_{\mathcal{B}}$ of Proposition 11.8 together with (11.9). □

Theorem 11.11. *Let G be a finite group and k be a field. Then $f\colon C^*(BG; k) \to k$ is Gorenstein.*

Proof. To be added.

□

We take as a fact that f^* is automatically orientable, and hence by Corollary 10.11 has Gorenstein duality. Therefore by (11.3) we have

$$\mathrm{Hom}_k(C^*(BG; k), k) \simeq \Gamma_f C^*(BG' k).$$

Combining this with the spectral sequence (11.7) we obtain a spectral sequence

$$H^{-s}_{\mathfrak{m}}(\pi_* R)_t \Rightarrow \pi_{t-s}\mathrm{Hom}_k(C^*(BG; k), k). \tag{11.12}$$

Let us record a consequence of this for the group cohomology ring. Recall that a commutative Noetherian local ring $(R, \mathfrak{m}, k)$ is Cohen–Macaulay if and only if $H^*_{\mathfrak{m}}(R)$ is concentrated in a single degree. (Equivalently, if $\operatorname{depth}(R) = \dim(R)$.)

Theorem 11.13 (Benson–Carlson duality)**.** *Let G be a finite group and k be a field. Then $H^*(G;k)$ is Cohen–Macaulay if and only if it is Gorenstein.*

Proof. The reverse implication is true for any commutative ring, so let us assume that $H^*(G;k)$ is Cohen–Macaulay. Then the local cohomology $H^{-s}_{\mathfrak{m}}(H^*(G;k))$ is zero unless $-s = d$ (where d is the Krull dimension) so the spectral sequence (11.12) collapses to an isomorphism $H^d_{\mathfrak{m}}(\pi_* R) = \pi_{-*}\operatorname{Hom}_k(C^*(BG;k), k) = \operatorname{Hom}_k(H^*(G;k), k)$. Since $H^*(G;k)$ is a k-algebra, the injective hull $E(k) = \operatorname{Hom}_k(H^*(G;k), k)$, and hence

$$H^i_{\mathfrak{m}}(H^*(G;k)) = \begin{cases} E(k) & i = d \\ 0 & i \neq d. \end{cases}$$

This is a characterisation of Gorenstein rings and hence $H^*(G;k)$ is Gorenstein. □

Remark 11.14. There was nothing particularly special about the group cohomology ring here. Consider $f\colon R \to k$ where $\pi_* R$ is a Noetherian local ring with residue field k. Then if $f\colon R \to k$ has Gorenstein duality and $\operatorname{Hom}_k(R, k)$ is torsion, one obtains the analogue of the spectral sequence (11.12) and can draw the conclusion that $\pi_* R$ is Cohen-Macaulay if and only if it is Gorenstein.

Appendix A. Exercises

Exercise A.1. Prove Proposition 2.11.

Exercise A.2. Prove that for any ring R, the compact objects in $\mathsf{D}(R)$ are the perfect complexes.

Exercise A.3. Prove that if $M \in \mathsf{stmod}kG$ (that is, M is finitely generated), then M is compact in $\mathsf{StMod}(kG)$.

Exercise A.4. Let T be a tt-category and let $Y \in \mathsf{T}$ be rigid. Prove that the natural map

$$F(X, \mathbb{1}) \otimes Y \to F(X, Y)$$

is an isomorphism for all $X \in \mathsf{T}$.

Exercise A.5. Let T be a tensor-triangulated category and $X \in \mathsf{T}$ be rigid. Prove that if Y is internally compact (resp., rigid), then $X \otimes Y$ is internally compact (resp., rigid). (See Proposition 2.31.)

Exercise A.6. Let $F\colon \mathsf{T} \to \mathsf{U}$ be a tt-functor between tt-categories. Let $X \in \mathsf{T}$ be rigid, and $Y \in \mathsf{T}$ be arbitrary. Prove that there is a natural isomorphism $F\underline{\mathrm{Hom}}(X, Y) \simeq \underline{\mathrm{Hom}}(FX, FY)$.

Exercise A.7. Let $\mathcal{K}$ be a set of compact objects in a rigidly-compactly generated tt-category T. Consider the localising $\otimes$-ideal $\mathsf{L} = \mathrm{Loc}_{\otimes}(\mathcal{K})$. Explain how one can view this as a tt-category (note that it is *not* a tt-subcategory of T). Which of the following are true in general?:

(1) if X is compact in L then X is compact in T;
(2) if $X \in \mathsf{L}$ is compact in T then X is compact in L;
(3) if X is rigid in L then X is rigid in T;
(4) if $X \in \mathsf{L}$ is rigid in T then X is rigid in L;
(5) $\{\text{compacts in } \mathsf{L}\} \subset \{\text{rigids in } \mathsf{L}\}$;
(6) $\{\text{compacts in } \mathsf{L}\} \supset \{\text{rigids in } \mathsf{L}\}$;
(7) $\{\text{compacts in } \mathsf{L}\} = \{\text{rigids in } \mathsf{L}\}$.

Exercise A.8. Let T be a tt-category and $X \in \mathsf{T}$ be rigid. Prove that X is a retract of $X \otimes DX \otimes X$. Hence deduce that for any rigid X we have $\mathrm{thick}_{\otimes}(X) = \mathrm{thick}_{\otimes}(DX)$.

Exercise A.9. Prove Lemma 3.1.

Exercise A.10. Prove implication (5) $\Rightarrow$ (2) in Theorem 4.8. (Hint: use Lemma 4.6.)

Exercise A.11. Which of the isomorphisms listed in Corollary 4.10 imply Grothendieck–Neeman duality?

Exercise A.12. Which of the following are true? Give proofs or counterexamples accordingly:

(1) If X is proxy-small, then ΣX is proxy-small.
(2) If $\{X_i\}_{i \in I}$ is a set of proxy-small objects, then $\oplus X_i$ is proxy-small.
(3) $\mathbb{Q}$ is proxy-small in $\mathsf{D}(\mathbb{Z})$.
(4) Let R be a commutative ring, and $M \in \mathsf{D}(R)$ be arbitrary. Then $R \oplus M$ is proxy-small in $\mathsf{D}(R)$.
(5) If $X \oplus Y$ is proxy-small, then X and Y are proxy-small.
(6) Let $f\colon X \to Y$ be a map between proxy-small objects. Complete it to a triangle $X \to Y \to Z$. Then Z is proxy-small.

Exercise A.13. Let T be a tensor-triangulated category. Prove that the one-to-one correspondence between localizations and colocalizations in Proposition 6.5 restricts to a one-to-one correspondence between monoidal localizations and monoidal colocalizations.

Exercise A.14. Let T be a tensor-triangulated category, and L be a monoidal localization. Prove that the natural map $\alpha_X \colon L\mathbb{1} \otimes X \to LX$ from Lemma 6.10 is an isomorphism if X is rigid.

Exercise A.15. Let T be a tensor-triangulated category, L be a monoidal localization, and write $\mathcal{S}$ for the full subcategory of L-local objects.

(1) Prove Proposition 6.11.
(2) Give an example of a smashing localization.
(3) Prove that $L \colon \mathsf{T} \to \mathcal{S}$ preserve compacts if L is smashing. Does this still hold if L is not smashing?

Exercise A.16. Let T be a rigidly-compactly generated tt-category and $L \colon \mathsf{T} \to \mathsf{T}$ be a monoidal localisation functor. Prove that $L\mathsf{T}$ is rigidly-compactly generated if and only if L is smashing.

Exercise A.17. Let $(\mathsf{T}, \mathcal{K})$ be a local duality context, and assume that $\mathcal{K} = \{K\}$ is a singleton. An object $X \in \mathsf{T}$ is said to be *homologically K-local* if $\mathrm{Hom}(Y, X) \simeq 0$ whenever $K \otimes Y \simeq 0$. A map $f \colon X \to Y$ in T is said to be a *K-equivalence* if $K \otimes f$ is an equivalence. This question is about showing that completion may be equivalently described as homological K-localization.

(1) Show that X is homologically K-local if and only if for all $f \colon Y \to Z$ such that $K \otimes f$ is an equivalence, the induced map $\mathrm{Hom}(f, X)$ is an equivalence.
(2) Prove that the map $X \to \Lambda X$ is a K-equivalence.
(3) Prove that X is complete if and only if X is homologically K-local.